# Online Social Network Access at Labor Market Entry and Long-Run Career Success*

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**Abstract**

How does access to online social networks at labor market entry affect workers' careers? I study this question by exploiting the staggered rollout of Facebook across U.S. colleges in a difference-in-differences design, tracking the career trajectories of nearly 900,000 business and economics students through their public LinkedIn profiles. I show that early access to Facebook leads to more successful careers: affected students work in higher-paying jobs, are 5.1% more likely to receive early-career promotions, and are 3% more likely to reach senior leadership roles in their late 30s. Investigating potential mechanisms, I find that these effects can be attributed to (i) a more successful college-to-work transition, driven by help from college peers in accessing higher-quality early-career jobs, (ii) more stable and productive early-career worker-firm matches, and (iii) intensified early-career sorting into higher-quality and career-supporting firms. The results highlight how access to online social networks during a sensitive period of a young worker's career can reduce information frictions, facilitate the college-to-work transition, and generate long-lasting career benefits.

**Keywords:** online social networks, Facebook, college graduates, labor market entry, career development, leadership, firms, networking

**JEL codes:** J24, J30, J62, I26, D83, L86

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# 1 Introduction

How do online social networks affect workers' careers? Platforms such as Facebook and LinkedIn have become an essential part of modern labor markets, with over 3 billion people worldwide using them for work-related purposes (Kemp 2025). Compared to prior technologies, they increase the flow of information and reduce the cost of adding and maintaining social connections (e.g., Bakshy *et al.* 2012; Armona 2025; Armona and Bailey 2025). Given the well-established evidence that social connections are valuable for job search and career advancement (e.g., Simon and Warner 1992; Granovetter 1995; Calvó-Armengol and Jackson 2004; Topa 2011; Schmutte 2015; Pallais and Sands 2016; Rajkumar *et al.* 2022; Cullen and Perez-Truglia 2023), providing workers with access to these platforms could reduce information frictions and improve labor market prospects. This may be especially relevant for younger workers, who have less experience with the job search process and face greater information frictions (e.g., Altonji and Pierret 2001; Pallais 2014). Yet we know surprisingly little about whether online social network access improves young workers' transition into the labor market and how it shapes their careers in the long run.

In this paper, I provide novel evidence on the causal impact of access to online social networking platforms on the initial job placement and career trajectories of young workers. To isolate plausibly exogenous variation, I exploit a natural experiment: the staggered introduction of Facebook across U.S. colleges in the mid-2000s, before it became publicly available in September 2006. As some college cohorts during the rollout phase graduated before their college gained access, the variation allows me to capture the effect of having access to Facebook right before vs. after labor market entry, a sensitive period of a young worker's career that can have long-lasting consequences for labor market outcomes (e.g., von Wachter 2020).[1]

I use two main datasets for the analysis. The first comprises the introduction dates of Facebook across the first 842 colleges (Armona 2025), which allows me to identify the academic year in which each U.S. college received access. The second dataset, provided by Revelio Labs, contains detailed CV data from public LinkedIn profiles. The LinkedIn data include two critical pieces of information that administrative data do not combine at this scale. First, the education histories provide the name of the postsecondary institution as well as start and end dates, allowing me to identify which students were directly affected by Facebook's staggered introduction. Second, the employment histories contain information on job titles,

[1]The labor market entry phase is also the point in time when many universities and career services recommend starting to use online social networking platforms for work-related purposes (e.g., School of Information, San Jose State University 2018).

job-based earnings, and company names, allowing me to track the career trajectories of nearly 900,000 U.S. business and economics students in great detail.[2] I use the information available on the profiles to measure job mobility and career development (e.g., promotions and the attainment of leadership roles), to construct networking indicators, and to characterize job and firm choices. This allows me to track how the effects of early access to online social networks propagate through a worker's career from labor market entry up to 19 years after graduation.

To estimate the causal impact of Facebook access at labor market entry on career outcomes, I use a generalized difference-in-differences design. Founded at Harvard in February 2004, Facebook was gradually introduced to other U.S. colleges before opening to the public in September 2006, creating sharp differences in availability across colleges and cohorts. Due to factors such as limited server capacity (Kirkpatrick 2011) and the desire to cultivate a sense of exclusivity (Braghieri *et al.* 2022), the rollout of Facebook was not random but targeted elite and highly selective colleges first. In the main specification, I therefore exclusively rely on the plausibly exogenous variation within each selectivity-by-region cell, narrowing down the comparison to treated and not-yet-treated college cohorts within the same selectivity and region. This strategy accounts for various endogeneity concerns such as spurious correlations with the non-random timing of the Facebook rollout, time-invariant college-specific characteristics (e.g., prestige), and cohort-specific shocks within the same region at colleges of similar selectivity (e.g., fluctuations in labor market entry conditions and returns to education). To address potential biases in staggered difference-in-differences estimation using two-way fixed effects (Goodman-Bacon 2021), I apply the heterogeneity-robust two-stage estimator by Gardner (2022).[3]

The main finding of the paper is that access to Facebook at labor market entry generates long-lasting career benefits. Students with early access to Facebook are 5.1% more likely to receive early-career promotions and 3% more likely to reach senior leadership positions in their late 30s. They also tend to work in higher-paying jobs, resulting in an increase in cumulative job-based earnings of around 1.2% over the first 19 years after graduation, which corresponds to nearly half of the earnings loss college graduates can expect when graduating in a large recession (e.g., von Wachter 2020).

Investigating potential mechanisms behind the long-run career effects, I show that treated students have a more successful college-to-work transition (the first two years after expected labor market entry) at

[2] I focus on business and economics students because their LinkedIn coverage is by far the highest among all fields for the mid-2000s graduation cohorts.

[3] Appendix F presents the main results using the heterogeneity-robust difference-in-differences estimators of Callaway and Sant'Anna (2021) and de Chaisemartin and D'Haultfœuille (2020), which differ slightly in their identifying assumptions. The results are very similar.

the employment, firm quality, occupation, and job margins. Students are, on average, 1.5% more likely to be employed, 6.8% more likely to begin their careers in high-wage firms, and 8.8% more likely to obtain jobs at firms offering high career support (as indicated by employer ratings on Glassdoor). In addition, treated students are 4.8% more likely to start their careers in high-paying occupations and 7% more likely to find early-career jobs with high chances of reaching a senior leadership position. The college-to-work transition effects are not explained by graduation timing or postgraduate education choices and are not apparent for lower-quality early-career jobs.

I further show that the improved college-to-work transition largely reflects help from college peers in accessing higher-quality early-career jobs. Using the job help proxy by Gee *et al.* (2017), which is based on relational firm mobility patterns, I find that students with early access to Facebook are 10.2% more likely to obtain an early-career job through their college network and 17.1% (16.3%) more likely to receive job help from their college peers to obtain an early-career job at a high-wage (career-supporting) firm.

Looking at potential longer-run treatment dynamics, I do not find evidence that the career benefits are attributable to a long-run information advantage in the search for higher-paying jobs. While early Facebook adopters have larger social networks in the long run (Armona and Bailey 2025), the estimated impact on between-firm mobility and promotion patterns is close to zero. This holds even when combining the between-firm mobility outcomes with job help indicators for the college network. In the spirit of traditional job search models (e.g., Burdett 1978), this finding suggests that early access to Facebook did not lead to a permanent increase in the rate of job offer arrivals. This is consistent with students in the control group creating a Facebook account shortly after it became publicly available, reducing differences in the flow of work-related information in the long run.[4]

Instead, I find evidence that part of the career benefits are explained by improved stability and productivity of early-career worker-firm matches, enabling students to climb up the career ladder within their organizations. In the first six years after graduation, treated students are 1% more likely to be in stable worker-firm matches and accumulate 0.37 months more firm tenure on average. The firm tenure effect persists in the longer run. Consistent with more stable and productive early-career worker-firm matches, treated students are 13.6% more likely to receive early-career promotions within their firm, while there is no impact on within-firm demotion patterns.

[4]Consistent with this finding, Armona and Bailey (2025) show that the positive network size effects decay over time, Facebook was also highly popular among the control group students, and there are no detectable long-run differences in terms of the intensity of Facebook usage between the treatment and control groups.

Part of the documented career benefits can also be explained by intensified sorting into firms that provide better learning opportunities and career support. Consistent with the college-to-work transition effects at the firm quality margin, students with early access to Facebook accumulate 5.4% more work experience at high-wage firms in the first six years after graduation on average. They also accumulate more work experience in firms that offer better career support, with the effect persisting in the long run. Given the recent evidence by Arellano-Bover (2022), Gregory (2023), and Arellano-Bover and Saltiel (2026) that firms differ substantially in the learning opportunities they provide and that heterogeneous human capital accumulation across firms can explain a considerable part of life-cycle differences in career outcomes, the findings suggest that faster on-the-job skill accumulation at higher-quality firms explains part of the career benefits. There is no evidence that a long-run reduction in unemployment risk drives the career benefits or that the career effects differ markedly between men and women.

The findings are robust to a series of sensitivity checks. One key concern with the identification strategy is that the selectivity-by-region cells may not be sufficient to account for the non-random rollout strategy of Facebook and may pick up spurious correlations within those cells. To test this, I randomly reshuffle the Facebook introduction dates across colleges within each selectivity-by-region cell. The placebo estimates are tightly clustered around zero, with all statistically significant treatment effects above the 95$^{th}$ percentile of the placebo distribution. Another concern is that the results may be driven by treated students taking away jobs from students in the control group, thereby upward-biasing the treatment effects. To test this, I predict the control cohorts' exposure to spillovers from the historical career overlap of students across colleges and drop every college with at least one highly exposed control cohort. The results are very similar to the baseline estimates, suggesting that the documented career benefits of early access to Facebook do not simply reflect a reallocation of high-quality jobs but are instead consistent with a reduction in information frictions.

The results of this paper contribute to the large literature on the individual and societal impacts of online social networking platforms. Several studies have documented negative consequences, such as increased mental health problems (Braghieri *et al.* 2022), greater political polarization (Zhuravskaya *et al.* 2020), the spread of misinformation (Allcott and Gentzkow 2017), and the amplification of hate speech (Müller and Schwarz 2021). Other papers highlight potential benefits, such as increased collective action and protest participation (Fergusson and Molina 2019; Enikolopov *et al.* 2020). This paper contributes by showing that access to online social networks can generate economically meaningful and long-lasting career benefits. There are three more closely related papers in the labor market context. Wheeler *et al.*

(2022) show that providing young job seekers with training to use LinkedIn improves their short-run job-finding rate. Armona (2025) exploits the same variation as this paper to show that access to Facebook at labor market entry increases the earnings rank of affected college cohorts in 2014, providing indirect evidence that the effect stems from stronger social ties among college peers and improved horizontal matching between major and occupation. Armona and Bailey (2025) use the same variation coupled with Facebook data to show that students with early access to Facebook have larger and more diverse networks in the long run, are more likely to sort into areas with higher economic opportunity, and work in higher-paying occupations. I contribute to this literature by demonstrating that online social network access at labor market entry also improves workers' initial job placement and shapes their long-run career progression. I further improve the understanding of the *dynamics* and *mechanisms* behind the labor market effects of early access to Facebook documented so far.

The findings of this paper also speak to the literature on the role of college networks in labor market outcomes. Social connections among college peers have been shown to affect early-career job placement (Marmaros and Sacerdote 2002; Zhu 2025; Gompers *et al.* 2026), channel students into top jobs (Kramarz and Thesmar 2013; Zimmerman 2019; Michelman *et al.* 2022; Fischer *et al.* 2025; Hampole *et al.* 2026), mitigate the negative consequences of job loss due to layoffs (Ost *et al.* 2026), and affect managerial decision-making (Lerner and Malmendier 2013; Shue 2013). This paper highlights that access to online social networks for college students increases the *value* of the college network as an information system for accessing early-career jobs and *shifts* their early-career job search strategy toward it.

Finally, this paper contributes to the literature on the role of labor market entry in long-run career success. Kahn (2010), Oreopoulos *et al.* (2012), Altonji *et al.* (2016), Schwandt and von Wachter (2019), and von Wachter (2020) show that graduating in a recession hinders a successful transition to the labor market, reducing the likelihood of working in high-quality jobs and harming workers' careers in the long run. Arellano-Bover (2024) shows that starting a career at a large firm benefits workers' long-run labor market success. This paper shows that online social network access at labor market entry can facilitate the transition into the labor market and improve initial job placement, with lasting effects on career progression.

The rest of the paper is organized as follows. Section 2 provides a short history of Facebook. Section 3 describes the data. Section 4 discusses the empirical strategy. Section 5 presents the results. Section 6 concludes.

## 2 Background

**The early expansion of Facebook.** Founded in February 2004 at Harvard, Facebook started as an exclusive social networking platform for college students, requiring a verified *.edu* email address to create an account and connect with other students from the same university. Over the following years, Facebook was gradually rolled out to other U.S. colleges and to universities abroad until it became publicly available in September 2006. Appendix Figure A1 shows the rollout of Facebook across all selective U.S. colleges offering four-year business and economics degrees by academic year and college selectivity. The rollout was not random but initially focused on highly selective colleges, driven by limited server capacity (Kirkpatrick 2011), a deliberate strategy to launch in college networks where students already knew each other well offline (Aral 2021), and the desire to cultivate a sense of exclusivity (Braghieri *et al.* 2022).

During the expansion period, Facebook was highly popular among college students, with around 85% of undergraduates creating an account (Arrington 2005) and nearly three-quarters of users logging in daily (Hass 2006). Compared to prior technologies, such as email and instant messaging, Facebook increased the flow of information and reduced the cost of adding and maintaining social connections (e.g., Ellison *et al.* 2007; Bakshy *et al.* 2012). Consistent with this, students used it primarily to keep in touch with friends and learn more about classmates (Lampe *et al.* 2006, 2008), with each additional year of early access increasing the number of Facebook connections within the college network by 13.6% (Armona 2025). Since alumni could sign up with their *.edu* email address, these networks also extended to former students. Early access to Facebook thus primarily enabled students to improve their connectedness within the college network before entering the labor market.

**Facebook in a work-related context.** While Facebook never fully specialized as a work-related online social networking platform, its features and immense popularity supported work-related activities, such as job search, professional networking, and recruiting, from the very beginning. As early as April 2004, Facebook allowed a small set of employers to advertise jobs and products to specific colleges (Grynbaum 2004), and in April 2006, it enabled students to list their current workplace and introduced workplace networks (Facebook 2006), which around 20–25% of early-adopter U.S. students used to report their employment history (Armona and Bailey 2025). Features addressing job seekers and employers more directly only followed in 2007: *Facebook Marketplace*, allowing users to post job ads within their networks (Facebook 2008), and *Facebook Pages*, enabling employers to create public company profiles (Facebook 2007). Over time, Facebook became the most important online social networking platform for job-seeking activities: in 2011, 48% of U.S. job seekers used Facebook for job search and 16% obtained

a job referral from a Facebook friend (Jobvite 2011).[5] On the firm side, among U.S. employers using or planning to use online social networking platforms for recruiting, 36% used Facebook in 2008 (Jobvite 2009), a number that increased to 68% in 2021 (Jobvite 2021).

Facebook's rollout and timeline have two important implications for students with early access. First, when they entered the labor market, Facebook's work-related features were still in their infancy, with the majority of employers not yet actively using the platform for recruiting or screening. For their early-career job search, they therefore primarily had access to a larger, more closely connected college network and the supply-side-focused work-related platform features, but were not necessarily exposed to vastly different demand-side information channels.[6] Second, as Facebook became publicly available in September 2006, students from the control group also received access to the technology shortly after their labor market entry. From then on, students with early access differed from the control group mainly in their head start: the larger online network they had built before labor market entry (Armona and Bailey 2025), their longer experience with the platform when the later work-related features were introduced, and the jobs they had obtained at labor market entry. I examine the relevance of these channels and dynamics in Section 5.2.

# 3 Data

To assess the career effects of access to Facebook at labor market entry, this paper primarily combines four data sources: LinkedIn profile data provided by the company Revelio Labs, the introduction dates of Facebook across the first 842 colleges, the number of awards and demographic composition of college graduation classes from the National Center for Education Statistics, and institutional college characteristics including Barron's selectivity index. This section provides an overview of the data sources, key variables, and descriptive statistics.

## 3.1 LinkedIn Data

Information on the career trajectories of U.S. students and firm characteristics comes from a snapshot of public LinkedIn data from November 2024. With more than 1 billion members and 67 million companies, LinkedIn is currently the world's largest professional networking site (LinkedIn 2024). In addition to its extensive global coverage, LinkedIn is particularly popular among U.S. college graduates (Auxier

[5]The second-largest platform during this period was LinkedIn, with substantially smaller usage numbers (26% and 9%) due to its lower adoption rate.

[6]The only exception at this stage was targeted employer ads. Section 5.2.1 discusses their relevance in greater detail.

and Anderson 2021). Users create profiles that resemble CVs and contain detailed information about employment history and educational background. Firms also create profiles that include information about firm characteristics to advertise open positions and increase their visibility.

The LinkedIn data comes from the company Revelio Labs. Revelio Labs is a workforce intelligence company that provides an extensive human resource database from various online sources. In the case of LinkedIn, Revelio Labs scrapes information from public user and company profiles and converts it into a structured database using proprietary algorithms.

The employment history section of public LinkedIn profiles contains information on job titles, start and end dates of employment, employer names, and job locations. The educational background section contains information on the postsecondary institution, degree, field of study, and start and end dates. The key advantage of using public LinkedIn profile data is that most of these variables are typically unavailable in public or administrative datasets at this level of detail and scale. The start and end dates of each stage in the CV allow me to construct a retrospective yearly career panel for each student up to 19 years after graduation.[7] At the end of the career panel, students are typically in their early 40s, when career trajectories have largely stabilized (Chetty *et al.* 2020). To ensure adequate coverage of the underlying student population, I focus on business and economics graduates, for whom LinkedIn coverage is by far the highest among all fields in the cohorts graduating around Facebook's introduction. Below, I describe the key variables derived from the public LinkedIn data.

**Job-based earnings.** A frequently used measure of workers' career trajectories is earnings (e.g., Bronson and Thoursie 2019; von Wachter 2020). While LinkedIn profiles do not report individual-level earnings, I can use Revelio Labs' job-based earnings imputation to track whether students work in typically higher- or lower-paying jobs over their careers. To generate job-based earnings, Revelio Labs employs a machine learning model that predicts the annual salary of each job stage using information on job title, company name, job location, tenure, and calendar year. The model is trained on publicly available online job postings and visa application data. Dorn *et al.* (2025) shows that the imputed earnings align closely with official earnings data from the American Community Survey and serve as a reliable proxy for job-based earnings. To make the job-based earnings measure comparable across years and cohorts, I convert it into real 2024 dollars using the U.S. Consumer Price Index (CPI).

**Job-based promotions.** To assess career-ladder steps more formally, I use the job-based earnings measure to construct a proxy for promotions. Following Bronson and Thoursie (2019), I define promotions

[7] To construct the yearly career panel, I take a snapshot on June 30 of each year, similar to Armona and Bailey (2025).

as large relative real wage growth jumps of at least 10 log points. The measure compares the real annual wage growth of a worker to the predicted real annual wage growth the worker would have obtained had they remained in their previous position.[8] Since I do not observe within-job earnings differentials, this measure of promotions essentially captures job-based promotions identified through changes in job titles, firms, or geographic location. Appendix Table A1 and Figure A2 investigate whether the job-based earnings measure by Revelio Labs, combined with the "relative wage growth jump" methodology in Bronson and Thoursie (2019), is capable of detecting promotion patterns in the LinkedIn data. Appendix Table A1 exploits the availability of job titles on LinkedIn profiles to assess the most frequent job title changes coinciding with a promotion. The most common job title transitions closely reflect standard moves up the career ladder. For example, the most common promotion event in the early-career stage reflects a change in job title from "Staff Accountant" to "Senior Accountant," in the mid-career stage from "Financial Analyst" to "Senior Financial Analyst," and in the later-career stage from "Project Manager" to "Senior Project Manager." Appendix Figure A2 further supports the suitability of the promotion measure by showing that the distribution of the imputed real annual wage growth (in log points) closely mirrors the individual-level wage growth distribution in Bronson and Thoursie (2019). The majority of the real annual wage growth distribution is concentrated around zero, with a slight left skewness. Consistent with their evidence, the distribution has a pronounced right tail, indicative of promotion events, with one-tenth of the observations having imputed real annual wage growth above 10%.

**Leadership positions.** One key advantage of the employment history on LinkedIn profiles is the availability of job titles, which I can use to identify leadership positions as an additional measure of workers' career development that does not rely on the job-based earnings imputation.[9] To do so, I use the keyword-based approach from Hampole *et al.* (2026), which maps job titles into the following managerial career pipeline: individual contributor, manager, director, vice president, senior vice president, and c-suite. Following Hampole *et al.* (2026), I define senior leadership positions as director-level to c-suite-level positions. To validate the mapping, Appendix Table A2 presents basic descriptive statistics for each category in the managerial career pipeline, while Appendix Table A3 lists the most commonly mapped job titles. Average job-based annual earnings increase monotonically along the career pipeline, from an average of $66,543 for individual contributor roles to $170,116 for director-level jobs and up to $250,123 for C-suite positions. Consistent with the literature (Lean In and McKinsey & Company 2020;

[8] Since the job-based earnings measure predicts the same earnings trajectory for *all* workers in a given job, this definition coincides with their methodology, which uses coworkers' earnings to predict counterfactual wage growth.

[9] Leadership positions are considered major career steps for workers, coming with a significant increase in responsibility, earnings, and prestige (e.g., Gibbons and Waldman 1999).

Hampole *et al.* 2026), the share of female U.S. business graduates gradually decreases with each step on the career ladder. In addition, Appendix Table A3 demonstrates that the most commonly mapped job titles closely align with each concept in the managerial pipeline and show no sign of misclassification.[10]

**Firm choice.** Job stages on LinkedIn profiles also contain information on the company name, with Revelio Labs linking each company to the corresponding LinkedIn company profile and Glassdoor amenity ratings. This unique feature of the data allows me to investigate the role of firm choice across various dimensions, which has been shown to be an important driver of workers' career trajectories (e.g., Oreopoulos *et al.* 2012; von Wachter 2020; Arellano-Bover 2024). To analyze firm choice, I focus on several measures of firm quality. Starting with firm pay as the first quality dimension, I combine information on company names with the job-based earnings imputation to classify high-wage and non-high-wage firms following the approach in Oreopoulos *et al.* (2012); Haltiwanger *et al.* (2018); von Wachter (2020). To do so, I use information from *all* LinkedIn profiles and companies that employ college students in my estimation sample and compute the average median log earnings at the firm level, net of year fixed effects. I then split the employment-weighted firm-earnings distribution into quartiles and define high-wage firms as all firms in the upper quartile of the distribution and non-high-wage firms as all firms below this threshold.[11]

As workers also highly value non-pecuniary aspects of work (e.g., Mas and Pallais 2017; Sorkin 2018; Sockin 2022; Maestas *et al.* 2023; Caldwell *et al.* 2026), I extend the firm quality measures to non-wage attributes using the information on Glassdoor employer ratings.[12] To do so, I use the same employment-weighted quartile split procedure after calculating the average employer rating, net of year fixed effects, to create firm quality proxies for the following dimensions: overall rating, compensation and benefits, career opportunities, culture and values, senior leadership, and work-life balance. Appendix Table A4 shows the coverage, distribution, and relation to the firm pay quality proxy in more detail. I merge the firm amenity characteristics to slightly more than two-thirds of the observed firm choices in the career panel. The upper quartile threshold for the different dimensions is around 3.5 to 4 on the Glassdoor rating scale from 1 to 5.[13] Looking at the correlation between the different firm quality margins provides

[10]For example, the most frequent terms for the C-suite category are "President" and "CEO," the most common terms for director-level positions are "Partner" and "Controller," and the most common terms for individual contributor positions are "Account Executive" and "Senior Accountant."

[11]Appendix Table A4 shows the coverage and distribution of the firm pay variable in more detail. Using alternative thresholds produces similar results.

[12]For a detailed discussion on the coverage and validation of the Glassdoor amenity ratings in the U.S., see Sockin (2022).

[13]Also here, using alternative thresholds for the classification of high-quality firms in the various amenity dimensions produces similar results.

several important insights. First, the firm pay and amenity variables are weakly positively correlated (correlations of 0.04–0.1), in line with recent evidence by Sockin (2022) and Arellano-Bover and Saltiel (2026). Second, there is a strong positive correlation between the different Glassdoor amenity rating dimensions (0.63–0.87), indicating that firms tend to offer amenities in a bundle rather than focusing on a single dimension.

**Occupational choice.** While LinkedIn profiles do not contain occupational codes, Revelio Labs infers 2018 SOC classifications using proprietary algorithms trained primarily on job titles, reported skills, and job descriptions. This measure allows me to analyze occupational choice as another important determinant of career trajectories (e.g., Keane and Wolpin 1997; Speer 2017). Similar to the firm quality proxies, I use job-based earnings imputation to construct a measure of occupational quality. To do so, I compute the average median log earnings for each 5-digit SOC occupation in the Revelio Labs data between 2000 and 2024 and perform an employment-weighted quartile split of the occupation-earnings distribution. I then classify the upper quartile of the occupation-earnings distribution as high-wage occupations, mainly reflecting white-collar professional and managerial jobs (e.g., top executives, financial managers, engineers, lawyers, and management analysts).

**Steep career ladder early-career jobs.** The detailed employment history on LinkedIn profiles also enables me to assess whether students sort into early-career jobs with steep or flat career prospects. To do this, I follow the idea in Almar *et al.* (2025) and predict the long-run wage growth and likelihood of attaining a senior leadership position for each early-career job. To construct the measures, I aggregate early-career jobs at the occupation-by-firm level and perform a leave-one-cohort-out prediction of both measures 15 years later.[14] I then split both occupation-by-firm-level career prospect distributions at the upper quartile to define early-career jobs with steep career prospects.

**Job help in the college network.** LinkedIn profiles do not contain explicit information on whether students shared work-related information or helped one another to access jobs. Nevertheless, the panel structure and the availability of firm names allow me to construct a proxy for the use of social ties in the job search process based on relational firm mobility patterns, following Gee *et al.* (2017). Suppose I observe two college peers in the data. If one of them starts working at a firm where the other has already worked for at least one year, this is taken as evidence that the first has received help from the college peer to access the new job at the connected firm. This measure has been validated by Gee *et al.* (2017) and is considered the state-of-the-art approach to measuring relational job mobility (Rajkumar *et al.* 2022).

[14]For occupation-by-firm cells with fewer than 30 observations, I collapse the career cells at the 5-digit SOC level to avoid small-sample bias. I define an early-career job as any job in the first two years after expected graduation.

**Degree, field of study, and postsecondary institution.** The educational background section on LinkedIn profiles contains information on degree, field of study, and educational institution, enabling me to merge the Facebook introduction dates with each student's career outcomes. To convert the raw text strings into standardized educational background characteristics, I follow the keyword approach in Dorn *et al.* (2025). For educational degrees, I map the raw degree input into six education levels using a keyword approach based on several online sources on typical degree abbreviations (YourDictionary 2022; StudentNews Group 2024; Best Universities 2024). The six levels are as follows: high school, associate, bachelor, master, professional degree, and doctorate. To classify field of study, I use the Classification of Instructional Programs (CIP) taxonomy provided by National Center for Education Statistics (2024a). As field-of-study entries on LinkedIn profiles are closely related to the CIP classification, I can map most of them to the corresponding CIP codes using regular expressions. For the most common non-mapped entries, I apply manual mapping. To standardize information about educational institutions, I use the clustering algorithm provided by Revelio Labs, which also accounts for common abbreviations. To further improve the clustering algorithm's accuracy, I refine the mapping by manually removing the most frequent false-positive cases from the estimation sample.

## 3.2 Further Data Sources

**Facebook introduction dates.** The second primary data source comes from Armona (2025) and contains information on Facebook's introduction dates for the first 842 postsecondary institutions. While the exact introduction date of Facebook is only known for all selective U.S. colleges up to August 2005, I can use the remaining selective U.S. colleges as a control group. They did not have access to Facebook before the academic year 2005–2006. Using this information, I can determine for all selective U.S. colleges the academic year in which they received access to Facebook and merge the treatment information with the LinkedIn profiles.

**IPEDS data.** The third data source comes from the Integrated Postsecondary Education Data System (IPEDS) and contains detailed information on the demographic composition and degree counts of each graduation class at a U.S. postsecondary institution (National Center for Education Statistics 2024b). I use this information to investigate the representativeness of the LinkedIn sample and to merge the number of business and economics bachelor's degrees awarded to each graduation cohort.

**Mobility Report Cards data.** To merge institutional characteristics of all four-year colleges to the LinkedIn profiles, I utilize the Mobility Report Cards (MRC) data from the Equality of Opportunity Project

(Chetty *et al.* 2020). The MRC dataset provides information on the location of postsecondary institutions and their selectivity, measured by Barron's selectivity index (Barron's Educational Series 2008).

### 3.3 Data Merge, Sample Selection, and Representativeness of LinkedIn Profiles

To construct the final estimation sample, I start by linking the Facebook introduction dates, IPEDS data, and MRC data using the unitid identifier. Since I focus on undergraduate business and economics students at selective U.S. colleges, I limit the sample to selective four-year colleges that offered an undergraduate business or economics degree between 1999 and 2005, resulting in 1,242 potential colleges.[15] In the next step, I merge the college-level information into the LinkedIn profiles using the information on college names. Appendix Table A5 documents the sampling procedure in detail. I can match 1,235 colleges (99.4%) to at least one LinkedIn profile reporting a bachelor's degree in business and economics, resulting in 1,071,848 profiles. To ensure that the estimation sample does not contain any incomplete or stale profiles, I drop 23,271 observations reporting an invalid undergraduate graduation date (2.2%).[16] I also drop 133,601 observations (12.7%) that do not report any employment for more than half of the career panel.[17] Following Overgoor *et al.* (2020), I further restrict the sample to colleges with average LinkedIn coverage rates between 0.25 and 1.25 to maintain representative coverage across college cohorts.[18] This refinement leaves a final estimation sample with 1,154 colleges and 890,951 profiles. The sample selection steps ensure that only sufficiently complete LinkedIn profiles and sufficiently representative college cohorts enter the final estimation sample. Appendix Table A5 shows that this refinement has little overall impact on the sample's composition in terms of age, gender, or college selectivity.

To assess the representativeness and coverage of the final estimation sample, I compare basic descriptive statistics with information from the IPEDS data. Appendix Table A6 shows that the final estimation sample covers 46.7% of all U.S. business and economics students during that time period. Consistent with other studies that use a LinkedIn sample of U.S. college graduates (e.g., Dorn *et al.* 2025), females are slightly underrepresented, with 41.6% of the LinkedIn sample compared to 47.2% in

[15] I define business and economics majors following the methodology used by U.S. News & World Report. Business majors are defined as any major with the two-digit CIP code 52, excluding 52.06. Economics majors are classified as any major with the CIP codes 45.06, 01.103, 03.0204, 51.2007, 52.06, 30.3901, and 30.4001 (Brooks 2025).

[16] I define a valid undergraduate graduation date as any end date of the education spell that is not missing, strictly after the indicated start date, and implies a study duration of under 10 years.

[17] The employment filter ensures that LinkedIn profiles that did not indicate an early-career employment history and did not update their LinkedIn profiles in recent years are not included in the estimation sample.

[18] Alternative thresholds do not affect the results.

IPEDS. The shares of students coming from colleges located in different U.S. regions and from colleges in different selectivity tiers are closely aligned across both data sets. Very selective colleges are slightly overrepresented in the LinkedIn data (51.9%) compared to IPEDS (48.2%).[19]

## 3.4 Descriptive Statistics

Table 1 presents descriptive statistics on the career trajectories of U.S. business and economics students, measured at the person-by-year level. In the first six years after expected graduation, 77.5% of students report being employed, with average job-based real annual earnings of around $76,000 (2024 USD). Consistent with the well-documented evidence on life cycle employment and earnings dynamics (e.g., Rubinstein and Weiss 2006), the employment share of U.S. business and economics students quickly rises and stabilizes at around 94% with the average real annual earnings reaching $122,000, 13–19 years after expected graduation. Taking a closer look at job mobility patterns, Table 1 shows that the annual promotion rate in the sample is around 8%, with roughly two-thirds of the promotions occurring between firms and one-third within firms. Consistent with previously documented evidence on mobility patterns within- and across-firms (e.g., Belzil and Bognanno 2008; Frederiksen *et al.* 2016), promotions occur more frequently than lateral job moves (3.5% to 4.1%) and demotions (4.4% to 5.9%), but are not the only type of job transition in the data. Investigating firm choice, Table 1 indicates that the share of students reporting to work at high-wage firms starts at 25.7% in the first six years after graduation and rises to 34.7% in the later career stage. Looking at top jobs, the share of U.S. business and economics students holding a senior leadership position starts at 10% in the first six years of their careers and increases up to 27.2% 13–19 years after expected graduation.[20]

How important are college peers for job finding and career development? I define college peers as students between three classes below and three classes above the respective cohort. The relational firm mobility job help proxy by Gee *et al.* (2017) shows that in the first 12 years, roughly 14% of U.S. business and economics students received job-related information from their college network to obtain their current job. While the share declines to 11.6% in the later career stage, the evidence suggests that more than 12.5% of the reported jobs in the career panel were obtained with the help of college peers. This underscores their relevance in the job search process. Interacting the job help proxy with firm quality and career

[19]While the main analysis utilizes the LinkedIn sample composition for estimation, Appendix G shows that the results are very similar when reweighting the sample in terms of demographic and college characteristics using the official statistics from the IPEDS data.

[20]This pattern closely aligns with the findings in Hampole *et al.* (2026) on the dynamics of holding senior leadership positions for MBA graduates, with the levels shifted downwards as the sample here comprises undergraduate students.

development outcomes suggests that college peers are also an important source for career advancement. Between 4.3% and 5.4% of students used their college network to obtain a job at a high-wage firm, 0.7% to 1.3% to obtain a promotion at a connected firm, and around 0.2% to transition into a senior leadership position. The relevance of college peers for career advancement declines slowly over time.

Table 1: Career Outcomes Over Time

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Career Development* | | | |
| Employment | 0.775 | 0.938 | 0.943 |
| Job-Based Real Annual Earnings (2024, USD) | 76,300 | 107,068 | 121,926 |
| Job-Based Promotion (≥10 log pts.) | 0.081 | 0.088 | 0.074 |
| Between-Firm Promotions (≥10 log pts.) | 0.057 | 0.056 | 0.045 |
| Within-Firm Promotions (≥10 log pts.) | 0.024 | 0.031 | 0.030 |
| Lateral Moves | 0.035 | 0.041 | 0.037 |
| Job-Based Demotion (≤ −10 log pts.) | 0.044 | 0.058 | 0.059 |
| High-Wage Firm | 0.257 | 0.329 | 0.347 |
| Senior Leadership Positions | 0.100 | 0.199 | 0.272 |
| *Panel B: College Network Outcomes* | | | |
| Job via College Network | 0.137 | 0.144 | 0.116 |
| High-Wage Firm Job via College Network | 0.053 | 0.054 | 0.043 |
| Promotion via College Network (≥10 log pts.) | 0.013 | 0.011 | 0.007 |
| Senior Leadership Transition via College Network | 0.002 | 0.002 | 0.002 |
| Observations | 5,345,706 | 5,345,706 | 6,236,657 |

*Notes:* Descriptive statistics of career outcomes over time reported at the person*year level. Each value represents the mean outcome pooled over one of the following time intervals after expected graduation: 1–6 Years, 7–12 Years, 13–19 Years. Sample: all business and economics undergraduates expected to graduate from 1999 to 2005 at selective 4-year colleges with valid LinkedIn profile and reasonable college-level coverage (1,154 colleges). Job-based real annual earnings are set to zero in years without employment. Job mobility outcomes start at year 2 after expected graduation. *Data source:* Revelio Labs.

# 4 Empirical Strategy

## 4.1 Identification

To isolate the causal effect of access to Facebook at labor market entry on career outcomes, I exploit the sharp and staggered rollout of Facebook across U.S. colleges in a generalized difference-in-differences design. The variation in treatment timing allows me to compare the career outcomes of business and economics students from the same college across treated and untreated cohorts. I account for common

cohort differences and spurious correlations with the Facebook rollout using not-yet-treated cohorts from colleges in the same region and selectivity tier.

The baseline model in a two-way fixed effects (TWFE) setting can be specified as follows:

$$Y_{fs,c,t} = \alpha_1 Treat_{fs,c} + \delta_{fs,\text{tier,region}} + \phi_c + \epsilon_{fs,c,t}\,, \tag{1}$$

where $Y_{fs,c,t}$ represents various career outcome measures for freshman cohort $fs$ at college $c$, $t$ years after expected graduation. $Treat_{fs,c}$ is a binary variable indicating whether freshman cohort $fs$ from college $c$ had access to Facebook before the *expected* graduation date, avoiding a potentially endogenous relation between treatment and the actual graduation date.[21] Since Facebook became public in 2006, the treatment indicator essentially captures the intent-to-treat (ITT) effect of having access to Facebook before vs. after labor market entry. $\phi_c$ denotes college fixed effects, accounting for time-invariant differences across colleges, such as prestige. $\delta_{fs,\text{tier,region}}$ represents cohort-by-college-tier-by-region fixed effects, narrowing down the comparison in the difference-in-differences setting to college cohorts within the same region and selectivity cell.[22] As the rollout of Facebook was not random, the key idea behind $\delta_{fs,\text{tier,region}}$ is to absorb all systematic differences across college cohorts that might be spuriously correlated with the treatment timing and exclusively rely on the plausibly exogenous variation within these cells. Examples of potential spurious correlations absorbed by $\delta_{fs,\text{tier,region}}$ are cohort-specific shocks in career outcomes within the same region at colleges of similar selectivity (e.g., fluctuations in labor market entry conditions and returns to education) or differential cohort-specific trends across those cells.

Consistent with the interpretation that the remaining variation is plausibly exogenous, Appendix Table A7 shows that the correlation between the Facebook rollout and various potential confounders, such as college characteristics, student and major composition, as well as local labor market entry conditions, vanishes after incorporating the fixed-effects structure.[23]

A growing body of literature emphasizes that in staggered adoption designs, the conventional two-way fixed effects (TWFE) model requires not only the parallel-trends assumption but also treatment-effect

[21] I follow Armona and Bailey (2025) and define access to Facebook before the *expected* graduation date as having access to Facebook at the respective U.S. college before June 30 of the expected graduation year (fourth year of the undergraduate).

[22] Since all but one elite college received early access to Facebook, I group elite and highly competitive institutions (Barron's tiers 1 and 2) into a single category for the fixed-effects estimation. To still allow for differential trends across cohorts within the broader group of highly selective colleges, I include linear cohort trends for Ivy Plus colleges, other elite colleges, and all highly competitive-by-region interactions. Appendix H shows that the results are robust to dropping Ivy Plus and other elite colleges.

[23] The main results in Table 2 are very similar when including these characteristics as controls in the regression (not shown here).

homogeneity to produce consistent estimates (Goodman-Bacon 2021; Sun and Abraham 2021; Roth *et al.* 2023). Specifically, Goodman-Bacon (2021) shows that the estimate of the TWFE model is a weighted average of all possible 2×2 difference-in-differences comparisons, including comparisons of already-treated units to newly-treated units. If the treatment effect varies over time or by adoption cohort, this biases the treatment effect obtained with TWFE. Given that Facebook initially targeted more selective universities, it is possible that earlier adoption cohorts experienced different treatment effects than later cohorts, calling into question the strong assumption of treatment effect homogeneity.

To ensure that the results are not driven by this bias, I estimate the treatment effect in the spirit of equation (1) using the two-stage heterogeneity-robust estimator proposed by Gardner (2022). The idea behind the two-stage approach is to first regress the outcome $Y_{fs,c,t}$ on the group $\phi_c$ and period $\delta_{fs,\text{tier,region}}$ fixed effects using the not-yet-treated observations only. The second stage regresses the residualized outcome for all observations on the treatment indicator $Treat_{fs,c}$. This procedure effectively avoids using already-treated units as controls for newly-treated ones and is robust to heterogeneous treatment effects.[24] To account for potential correlations in the error term across cohorts within the same college, I cluster standard errors at the college level (Athey and Imbens 2022). For computational efficiency, I estimate the regressions at the college-by-cohort-by-year level (as noted in equation (1)), weighting each cell by the corresponding number of LinkedIn profiles.

## 4.2 Threats to Identification

The core identifying assumption is that, in the absence of treatment, the career outcomes of students in treated colleges would have evolved in parallel to those in not-yet-treated colleges of similar selectivity located in the same U.S. region. To assess the plausibility of the parallel trends assumption for each career outcome, Appendix B shows estimates of an event-study version of equation (1) (Figures B1 to B18). The event refers to the first freshman cohort expected to be exposed to the introduction of Facebook at the respective college. Reassuringly, the pre-treatment coefficients are precisely estimated to be close to zero, and almost all coefficients are individually and jointly insignificant at the 5% level, supporting the parallel trends assumption in this setting.[25]

[24] Note that the two-stage heterogeneity-robust estimator is numerically equivalent to the imputation estimator of Borusyak *et al.* (2024). Appendix F presents the main results using two further heterogeneity-robust estimators: Callaway and Sant'Anna (2021) and de Chaisemartin and D'Haultfœuille (2020). While these differ from the baseline in their identifying assumptions and precision, the results are very similar.

[25] In the few cases where the null hypothesis of zero pre-treatment coefficients is rejected, the event study figures clearly indicate that the pre-treatment dynamics are precisely estimated, close to zero, and cannot explain the estimated treatment effects (e.g., Figure B6, Panel (c)). A future version of this paper will integrate honest DiD estimates following the

Another key concern is that access to Facebook may affect the probability of having a LinkedIn account or may change the composition of the sample. In this case, estimated treatment effects might simply reflect sample selection bias. Using official IPEDS data, Appendix Table A8 shows that Facebook access at labor market entry does not affect the likelihood of having a LinkedIn account. Facebook access also does not alter the sample composition with respect to baseline demographic characteristics such as gender or age.

Lastly, early access to Facebook might affect the completeness of LinkedIn profiles, even if the propensity to have a LinkedIn account remains unaffected. Since the sample restrictions exclude incomplete profiles from the baseline sample, this could challenge a causal interpretation of the results. To assess the severity of this concern, Appendix Table A9 estimates the treatment effect on each incomplete-profile restriction, starting with all LinkedIn profiles at the colleges in the estimation sample. Reassuringly, early Facebook access affects neither the LinkedIn coverage before any sample restriction nor the share of profiles that pass each incomplete-profile restriction (individually or jointly), indicating that treated and untreated students have LinkedIn profiles of similar completeness.[26] I present further evidence supporting the identifying assumptions in Section 5.3.

# 5 Results

## 5.1 Career Development

Does access to Facebook at labor market entry affect the career development of U.S. students? Table 2 investigates this question by tracking their career trajectories up to 19 years after graduation. Panel A shows the results using the job-based earnings measure. In the first six years after expected graduation, students with early access to Facebook earn on average $1,286 more than the control group, with the earnings effect increasing to $1,343 in the years 7–12.[27] Comparing the treatment effect to the average job-based annual earnings of the control group indicates a relative effect size of 1.7% ($75,876) and 1.3% ($106,240), respectively, suggesting a moderate but economically meaningful shift in average annual earnings. While the earnings effect is fading over time (see Column 3), the results in Panel A provide clear evidence that the initial career trajectories of students with early access to Facebook have been positively

recommended practice by Rambachan and Roth (2023).

[26]The main results are also robust to estimating them on the broader LinkedIn sample without the restrictions on the employment history (not shown here).

[27]The estimated earnings effect 7–12 years after expected graduation is well in line with Armona (2025), who finds that each year of Facebook access during college increases average earnings in 2014 by $972 ($1,280 in 2024 dollars).

affected. Appendix Table A10 shows that the positive earnings effects in the first six years are primarily driven by an increase in labor supply, while the effect in years 7–12 mainly comes from students working in higher-paying jobs.[28]

Table 2, Panel B, uses the relative wage growth jump methodology by Bronson and Thoursie (2019) to analyze how early access to Facebook affects promotion patterns. Consistent with the positive earnings effect, Column 1 shows that students with early access to Facebook are 0.4 percentage points more likely to receive promotions in the first six years after graduation. Comparing the effect size to the control group's average annual promotion rate (7.8%) indicates an economically meaningful relative effect size of 5.1%. While the positive effect on promotions fades out over time (see Columns 2 and 3), the point estimate remains positive though statistically insignificant in the years 7–12, suggesting a *cumulative* edge in the number of promotions in the long run.[29] There is no impact on lateral job mobility and demotions (see Appendix Table A12), suggesting that the positive earnings effects are not explained by treated students being less likely to receive a demotion or by frequent small positive wage-growth jumps due to increased lateral job mobility.

The positive cumulative promotion effect raises the question of whether access to Facebook enabled some students to reach top jobs in the economy. Panel C investigates this by examining the senior leadership measure initially proposed in Hampole *et al.* (2026). Consistent with the intuition that reaching a senior leadership position takes time, there is no impact in the first six years after expected graduation. However, Columns 2 and 3 show that Facebook access at labor market entry increases the likelihood of attaining a senior leadership position in the longer run. 13–19 years after graduation, students with early access to Facebook are 0.8 percentage points more likely to reach senior leadership positions, a 3% increase relative to the control group mean. This suggests that while the positive average earnings effects are fading out, early access to Facebook enabled some students to reach top jobs in the economy through a persistent effect on their career progression.[30] Consistent with this interpretation, Appendix Table A14 shows that early access to Facebook also has a positive impact on the likelihood of reaching high-paying jobs 7–19 years after expected graduation, with the effect being concentrated around the top quartile of the earnings distribution. Overall, the results in Table 2 indicate that access to Facebook at labor market entry

[28]The labor supply and earnings dynamics are very similar to the documented career patterns in the literature on graduating in a recession for college graduates (e.g., Oreopoulos *et al.* 2012; Altonji *et al.* 2016; von Wachter 2020).

[29]Appendix Table A11 shows that the results are robust to alternative specifications of the promotion threshold.

[30]Appendix Table A13 repeats the estimation using management occupations (SOC 11), top executives (SOC 11-1), and board members as alternative definitions of top jobs. While the concepts differ slightly in the kinds of jobs they capture, the results for management and top executive occupations are very similar. There is no impact on the likelihood of becoming a board member.

improved the long-run career trajectories of U.S. business and economics students in an economically meaningful way. Appendix C shows that the career benefits are shared equally between men and women.

Table 2: Facebook Access and Career Development

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Real Annual Earnings (2024, USD)* | | | |
| FB Access | 1,286** | 1,343* | 768 |
| | (479) | (556) | (649) |
| Dep. Var. Mean | 75,876 | 106,240 | 121,452 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel B: Job-Based Promotion (≥10 log pts.)* | | | |
| FB Access | 0.004** | 0.002 | -0.0010 |
| | (0.001) | (0.001) | (0.0010) |
| Dep. Var. Mean | 0.078 | 0.085 | 0.075 |
| Observations | 40,210 | 48,252 | 56,294 |
| *Panel C: Senior Leadership Positions* | | | |
| FB Access | 0.002 | 0.006* | 0.008* |
| | (0.002) | (0.003) | (0.004) |
| Dep. Var. Mean | 0.101 | 0.199 | 0.270 |
| Observations | 48,252 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 Years after expected graduation. Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

**Cumulative earnings.** What does having early access to Facebook imply for the cumulative earnings of U.S. students? Appendix Table A15 investigates this question using the cumulative job-based earnings in the first 19 years after graduation. Assuming a 5% discount rate, the estimated increase in cumulative job-based earnings from early access to Facebook is around $14,976, equivalent to a 1.2% increase relative to the control group. This corresponds to 44% of the reduction in the present discounted value of earnings that college graduates can expect when graduating in a large recession (e.g., von Wachter 2020), suggesting that early Facebook access had an economically meaningful impact on the cumulative earnings of the affected U.S. business and economics students.

## 5.2 Mechanisms

Having shown that access to Facebook at labor market entry has a long-run positive impact on students' career trajectories, I next examine potential mechanisms. To do so, I split the treatment dynamics into two stages: the college-to-work transition and the period after it.

### 5.2.1 The College-to-Work Transition

During the college-to-work transition, students may rely on improved information flows from their larger and more connected online social networks or the early work-related platform features to support the early-career job search process.[31] As an extensive literature emphasizes that a successful college-to-work transition and better initial job placement are key to long-run career development (e.g., Kahn 2010; Oreopoulos *et al.* 2012; Altonji *et al.* 2016; Schwandt and von Wachter 2019; von Wachter 2020; Arellano-Bover 2024), this mechanism could explain part of the documented career benefits.

**Early employment and quality of firms and jobs.** Table 3 investigates this channel by examining early-career employment and job-placement outcomes in the first two years after expected labor market entry.[32] Consistent with the estimated employment patterns in Appendix Table A10, Column 1 shows that early Facebook access increases the probability of finding a job during the expected labor market entry phase by 1 percentage point (1.5%). This provides the first evidence of an improvement in the college-to-work transition.

To assess whether the *job quality* margin was also affected, Columns 2–6 focus on several indicators of high-quality entry-level jobs. Starting with firm quality, Columns 2 and 3 show that students with early access to Facebook are 1.4 percentage points more likely to start their careers at high-wage firms and at firms highly rated for career support. Compared to the control group, this implies a relative effect size of 6.8% and 8.8%, respectively, suggesting an economically meaningful shift toward high-quality firms. Consistent with the positive earnings effects documented in Table 2, students with early access to Facebook are also more likely to sort into firms that have a high rating in terms of compensation and benefits, while Appendix Table A16 shows no effect on the other Glassdoor dimensions (overall rating, culture and values, senior leadership, and work-life balance).[33]

Column 4 shows that treated students are also 1.2 percentage points (4.8%) more likely to start working

[31] As discussed in Section 2, the majority of employers were not yet actively using Facebook during the labor market entry phase of the first exposed college cohorts, so that most demand-side information channels (e.g., employer screening, recruiters) were not at play. The main exception, targeted employer ads, is discussed below.

[32] Varying the definition of the labor market entry phase does not affect the results.

[33] Appendix Table A17 shows that the results are not driven by firms with missing information on amenities.

in a high-paying occupation. This suggests that, next to the firm quality margin, early access to Facebook also improved early-career outcomes at the occupational margin.

Table 3: Facebook Access and the College-to-Work Transition

| | | Indicators of High-Quality Early-Career Job Placement | | | | |
|---|---|---|---|---|---|---|
| | Employment | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
| FB Access | 0.010** | 0.014*** | 0.014*** | 0.012** | -0.001 | 0.014*** |
| | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.653 | 0.206 | 0.159 | 0.249 | 0.159 | 0.201 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-wage firm: an indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: an indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after, above the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

To also consider the long-run perspective of early-career jobs on career development, Columns 5 and 6 investigate sorting into early-career jobs associated with *steep career prospects* in terms of wage growth and the likelihood of reaching a senior leadership position. Column 5 indicates that early access to Facebook did not increase the likelihood of obtaining an early-career job with a steep wage-growth profile. Column 6 shows that students are 1.4 percentage points (7%) more likely to access entry-level jobs with a high chance of reaching a senior leadership position later in their careers.

Appendix Table A18 investigates whether the college-to-work transition effect is also apparent for *lower-quality* early-career jobs. The coefficients are close to zero and slightly negative, with non-steep wage growth jobs being the only outcome that is positively affected. Taken together, the findings suggest that early access to Facebook primarily enabled students to find *high-quality* early-career jobs after labor market entry, improving their initial job placement and shifting their career prospects in the short as well as long run.

**Graduation timing and postgraduate education choice.** Appendix Table A19 investigates to what extent the college-to-work transition effects are driven by graduation timing and postgraduate education

choices. Looking at the reported undergraduate study duration and likelihood of studying in the first two years after expected graduation, the effects are close to zero and insignificant. This suggests that students without early access to Facebook did not strategically delay their labor market entry by staying in college. I also find no impact on the likelihood of obtaining a postgraduate degree or on the type of postgraduate degree, suggesting that the documented treatment effects are not mediated through postgraduate education decisions. However, students with early access to Facebook report their first stable job after expected graduation two-thirds of a month earlier than the control group. While small in economic magnitude, this suggests that part of the positive college-to-work transition effect might be explained by students finding their first stable job at a faster rate.[34]

**Early-career job help in the college network.** While the previous results indicate that Facebook access at labor market entry facilitated the college-to-work transition, it remains unclear where this effect comes from. As described in Section 2, treated students could increase their connectedness within the college network, maintain those connections after exiting college at a lower cost, and use the early-stage platform features of Facebook to quickly exchange information. The improved transition may therefore stem from students sharing work-related information and helping one another in the early-career job search process. Consistent with this hypothesis, early survey evidence by Ellison *et al.* (2007) suggests that U.S. students expected Facebook to be useful for obtaining job referrals and learning more about jobs from their social network. To examine this channel, Table 4 decomposes the college-to-work transition effect in Table 3 into early-career jobs at firms with and without a college peer already working there, using the relational firm mobility job-help proxy by Gee *et al.* (2017).

Panel A, Column 1, investigates whether early access to Facebook increased the likelihood that students find an early-career job at a firm connected to their college network. The treatment effect is positive and statistically significant at 1.1 percentage points (10.2%). At the same time, Panel B, Column 1, indicates that there is no treatment effect on the likelihood that students find an early-career job *without* such a connection. This suggests that the entire early-career job finding effect in Table 3, Column 1, is concentrated at firms connected to the college network, consistent with improved access to work-related information and job help from college peers.

[34]This would be consistent with the idea that early access to Facebook increases the arrival of job offers at the college-to-work transition stage, enabling students to find a job that matches their expectations at a faster rate.

Table 4: Facebook Access and Early-Career Job Help in the College Network

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Early-Career Job via College Network* | | | | | | |
| FB Access | 0.011*** | 0.007*** | 0.008*** | 0.004* | 0.004** | 0.007*** |
| | (0.002) | (0.002) | (0.002) | (0.002) | (0.001) | (0.002) |
| Dep. Var. Mean | 0.107 | 0.041 | 0.049 | 0.042 | 0.027 | 0.047 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| *Panel B: Early-Career Job without College Network* | | | | | | |
| FB Access | -0.0004 | 0.008* | 0.006 | 0.008* | -0.005 | 0.007 |
| | (0.004) | (0.003) | (0.003) | (0.004) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.546 | 0.165 | 0.110 | 0.207 | 0.132 | 0.154 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Panel A: early-career job at a firm with a college peer already working there. Panel B: early-career job at a firm without a college peer already working there. The two panels partition the corresponding outcome in Table 3, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. Column 1 (Any Job) is the partition of the Employment outcome shown in Column 1 of that table. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

To investigate whether students also used the college network more intensively to find high-quality entry-level jobs, Columns 2–6 combine the job help proxy with the high-quality early-career job indicators, so that each outcome indicates a job of the respective type at a firm where a college peer already worked. Columns 2 and 3, Panel A, show that treated students are 0.7 percentage points (17.1%) more likely to obtain a job at a connected high-wage firm and 0.8 percentage points (16.3%) more likely to obtain a job at a connected highly career-supporting firm. Comparing the coefficients to the estimated effect sizes in Columns 2 and 3 of Panel B (0.8 and 0.6 percentage points) suggests that about half of the positive impact of early access to Facebook on the likelihood of obtaining a job at a high-quality firm comes from jobs at firms connected to the college network. Columns 4 and 6, Panels A and B, show that the college network is also relevant for accessing high-paying occupations (0.4 vs. 0.8 percentage points) and jobs associated with a high chance of reaching senior leadership positions (0.7 vs. 0.7 percentage points) at connected firms. Notably, while Column 5 indicates that there is no aggregate treatment effect on the likelihood

of attaining a steep wage growth job, treated students are more likely to obtain these jobs through their college network (0.4 percentage points) than without (−0.5 percentage points). This suggests that early access to Facebook ultimately *shifted* students' early-career job search toward their college network, a pattern also evident at the lower-quality early-career job margin (see Appendix Table A20).

Which type of college peer is responsible for the improved college-to-work transition of treated students? Appendix Table A21 investigates this question, decomposing the early-career job help effect into below-cohort, same-cohort, and above-cohort college peer ties. The results indicate that the effects are primarily driven by above-cohort college peer ties, consistent with the idea that they can provide the most useful work-related information without competing for the same jobs.[35]

**Employer ads.** Another potential reason behind the improved college-to-work transition is that students with early access to Facebook may have engaged more actively with employer ads, which were introduced as early as April 2004 (Grynbaum 2004). If targeted ads increased awareness of jobs and companies among treated students, this could also have facilitated their search for early-career jobs. However, there are two reasons to believe this channel is unlikely to explain a meaningful part of the college-to-work transition effects in Table 3. First, targeted ads were only temporarily available to a very small set of firms during the college-to-work transition phase of the early Facebook adopters.[36] Second, survey evidence from the late 2000s indicates that only 12% of U.S. students actively engaged with employer ads during their first job search (National Association of Colleges and Employers 2009), suggesting that student engagement with this feature was low.

**Geographic mobility.** Lastly, Armona and Bailey (2025) show that students with early access to Facebook were more likely to move away from their hometown and to live in denser areas with higher economic opportunity. As more urban areas provide access to higher-paying jobs and higher-quality firms (e.g., Glaeser and Maré 2001; Dauth *et al.* 2022; Card *et al.* 2025), increased sorting into stronger labor markets might also explain part of the positive college-to-work transition effect. Appendix Table A22 investigates the relevance of this channel using the job location information reported on LinkedIn profiles.[37] Consistent with Armona and Bailey (2025), I find some evidence that treated students are more likely to find jobs in large metropolitan areas outside the MSA of their college (marginally

[35]The slightly positive below-cohort effect likely reflects peers who held a job at the firm during college (e.g., internships). The zero same-cohort effect is consistent with competition for the same entry-level jobs.

[36]To be precise, targeted ads were available to a small set of firms during April 2004 (Grynbaum 2004). The importance and adoption of targeted ads changed with the official launch of *Facebook Ads* in November 2007 (Facebook 2007), shortly after the college-to-work transition phase of the early Facebook adopters, which is here defined as the period until June 2007.

[37]I use geographic information at the metropolitan statistical area (MSA) level of the job, which is the most fine-grained harmonized location measure available in the Revelio Labs data.

significant at the 10% level). However, accounting for geographic job sorting only explains 1% to 14% of the documented college-to-work transition effects on the high-quality firm, occupation, and job margins, suggesting that most of the improvement in initial job placement occurred within local labor markets.

### 5.2.2 After the College-to-Work Transition

The evidence so far indicates that part of the documented career benefits stem from a more successful college-to-work transition in which the college network played a central role. Another part may reflect longer-run treatment dynamics that unfold after labor market entry.

**Long-run job search, firm mobility, and job help in the college network.** All U.S. business and economics students received access to Facebook in the long run. Part of the documented career benefits for the early adopters could nevertheless be explained by a long-run increase in access to information about job opportunities, facilitating their search for higher-paying jobs.[38] There are three reasons why this might be the case. First, treated students may continue to receive more information about job opportunities as they had more time to build their social network and were less likely to lose their college ties when transitioning into the labor market. Relatedly, Armona and Bailey (2025) show that students with early access to Facebook have larger online social networks and are more closely connected to their college peers in the long run. As workers rely heavily on their social networks for information about job opportunities (e.g., Caldwell and Harmon 2019) and larger networks predict a greater arrival rate of job offers from other firms, this channel could explain part of the long-run career benefits. Second, students with early access to Facebook may be using the later-introduced work-related platform features, such as Facebook Marketplace, Pages, or Ads, more intensively to acquire job-related information. Third, due to a potentially higher visibility on the platform, students with early access to Facebook might receive more job-related information from employers and recruiters, who were increasingly using the platform.

Panel A in Table 5 investigates the joint significance of these channels by examining the likelihood of receiving a between-firm promotion. If early access to Facebook increases access to information about job opportunities in the long run, job search models would predict an increase in the annual rate of between-firm promotions. However, Columns 1 to 3 in Panel A show that the estimated treatment effect on between-firm promotions is close to zero and insignificant across the entire career panel. This suggests that a long-run increase in the job offer arrival rate is not driving the documented career benefits. Treatment effects are also close to zero for any between-firm switches and for between-firm switches with

[38] In traditional job search models (e.g., Burdett 1978), this channel would reflect a permanent (or slowly decaying) increase in the job offer arrival rate, allowing workers to draw from better job offers in the wage distribution.

no immediate promotion (see Appendix Table A23). Panels B and C test more directly the hypothesis that early access to Facebook leads to a long-run improvement in access to job opportunities through the more closely connected college network by interacting the between-firm promotion measure with the job help proxy for college peers. Similar to the findings in Panel A, the results are close to zero and insignificant.[39]

Table 5: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Between-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.0005 | 0.0003 | -0.001 |
| | (0.001) | (0.0009) | (0.0008) |
| Dep. Var. Mean | 0.055 | 0.055 | 0.045 |
| Observations | 40,210 | 48,249 | 56,294 |
| *Panel B: Promotion via College Network (≥10 log pts.)* | | | |
| FB Access | -0.00009 | 0.0005 | -0.00004 |
| | (0.0004) | (0.0004) | (0.0003) |
| Dep. Var. Mean | 0.012 | 0.010 | 0.007 |
| Observations | 40,210 | 48,252 | 56,294 |
| *Panel C: Senior Leadership Transition via College Network* | | | |
| FB Access | -0.00007 | 0.0002 | 0.000006 |
| | (0.0002) | (0.0002) | (0.0001) |
| Dep. Var. Mean | 0.002 | 0.002 | 0.002 |
| Observations | 40,210 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior Leadership Transition via College Network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

While the results in Table 5 may appear surprising at first, they are in line with evidence by Armona and Bailey (2025) that the social network size advantage of the early adopters of Facebook decreased over time, that Facebook was also quickly adopted by the control group after it became publicly available, and that there was no long-run difference in the overall platform activity between treatment and control group.

[39] Appendix Table A23 also shows that the results for moves to connected firms with no promotion are close to zero or negative, providing further evidence that long-run information sharing within the college network does not explain the documented career benefits.

Overall, the results are consistent with differences in the job offer arrival rate decaying quickly once the control group received access to Facebook.

**Match quality, skill accumulation, and long-run firm sorting.** While the previous results show no impact on between-firm promotions, another potential reason behind the documented career benefits could be that treated students use their initial information advantage during the college-to-work transition to sort into higher-quality worker-firm matches. This would enable them to accelerate their human capital accumulation and move up the career ladder within their organizations. A higher-quality worker-firm match in this context could reflect two things. First, students with early access to Facebook might have an easier time finding *stable* and *productive* worker-firm matches. Second, they might increasingly *sort* into firms that provide better learning opportunities and career support, even if the stability and productivity of the match remains unchanged.

Starting with the stable and productive worker-firm match channel, Table 6, Panel A, analyzes the fraction of students reporting to work at their current employer for at least 6 months as a measure of early-career worker-firm match stability.[40] Column 1 shows that in the first six years after expected graduation, students with early access to Facebook are 0.7 percentage points (1%) more likely to report working at their employer for at least 6 months, while there is no impact on the likelihood of working in an unstable worker-firm match (see Appendix Table A24). This provides the first evidence that early access to Facebook improved worker-firm match stability. While Columns 2 and 3 show that the positive effect is fading over time, this could be driven by the well-documented fact that firm tenure increases with experience (e.g., Topel and Ward 1992).

To investigate whether worker-firm match stability also improved in the longer run, Panel B therefore shows the results for current firm tenure. Consistent with the evidence in Panel A, Column 1 indicates a positive effect on firm tenure in the first six years after expected graduation. Compared to the average firm tenure of the control group (1.64 years or 19.68 months), students with early access to Facebook stay on average 0.031 (0.37) more years (months) with their current employer. Column 2 shows that the firm tenure effect persists in the longer run and increases over time, reaching 0.07 (0.84) years (months) 7–12 years after graduation. While the firm tenure effect becomes insignificant in the long run, the findings in Panels A and B provide evidence that treated students were more likely to find stable early-career worker-firm matches that persisted in the medium run.

[40]The results for alternative early-career worker-firm match stability thresholds are very similar (not shown here).

Table 6: Facebook Access and Worker-Firm Match Stability and Productivity

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Stable Worker-Firm Match (≥6 Months)* | | | |
| FB Access | 0.007* | 0.001 | -0.0004 |
| | (0.003) | (0.002) | (0.002) |
| Dep. Var. Mean | 0.688 | 0.866 | 0.881 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel B: Current Firm Tenure (Years)* | | | |
| FB Access | 0.031** | 0.070* | 0.083 |
| | (0.011) | (0.028) | (0.047) |
| Dep. Var. Mean | 1.64 | 4.02 | 6.04 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel C: Within-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.003*** | 0.002* | 0.0002 |
| | (0.0008) | (0.0008) | (0.0006) |
| Dep. Var. Mean | 0.022 | 0.030 | 0.029 |
| Observations | 40,210 | 48,249 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 6 months. Current firm tenure: years at current employer. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

As a more stable worker-firm match does not automatically imply that treated students benefited in terms of career development, Panel C examines whether early access to Facebook also improved the productivity of the worker-firm match, using the probability of within-firm promotions as the outcome. Column 1 shows that in the first six years after graduation, students with early access to Facebook are 0.3 percentage points more likely to obtain an early-career within-firm promotion, a 13.6% increase compared to the control group. The positive effect also persists 7–12 years after graduation (0.2 percentage points) and only fades at the later career stage, while there is no positive impact on within-firm demotions and only a slight positive impact on within-firm lateral moves (see Appendix Table A24). Taken together, the results suggest that early access to Facebook enabled students to secure stable, productive worker-firm matches, allowing them to accumulate firm-specific human capital and advance within their organizations.

While the previous results provide evidence of improved stability and productivity of worker-firm

matches, another reason for the positive career benefits could be intensified sorting into firms that offer better learning opportunities and career support. Theoretical work by Rosen (1972) and Gibbons and Waldman (2006), as well as more recent empirical evidence by Arellano-Bover (2022), Gregory (2023), and Arellano-Bover and Saltiel (2026) emphasizes that firms differ substantially in the learning opportunities they provide, rationalizing a sizable fraction of life-cycle differences in career outcomes. The college-to-work transition results already indicate that early access to Facebook enabled students to start their careers in higher-quality firms and jobs. Longer-run sorting into training-intensive and career-supporting firms could therefore be a complementary mechanism that explains the documented career effects. Table 7, Panel A, uses the accumulated work experience at high-wage firms as a first proxy for sorting into firms that provide better learning opportunities.[41] Consistent with the college-to-work transition results, Column 1 shows that in the first six years after graduation, treated students accumulated on average 0.08 years (0.96 months) more work experience at high-wage firms, a 5.4% increase compared to the control group (1.48 years). The positive effect fades out over time, as students from the control group are more likely to switch into high-wage firms later in their career (see Appendix Table A25).[42]

To capture the effect of sorting into firms that provide better learning opportunities and career support more directly, Panel B analyzes accumulated work experience in high-career-support firms, using Glassdoor's career opportunity rating dimension. Consistent with the work experience effects for high-wage firms, Column 1 shows that in the first six years after graduation, treated students also accumulate, on average, 0.054 years more work experience in high-career-support firms (6.4% relative to the baseline). Contrary to the fading effect in Panel A, the results in Panel B (Columns 2 and 3) indicate that the effect on accumulated work experience in career-supporting firms increases over time. The results in Panels A and B are thus consistent with the idea that an acceleration of on-the-job skill accumulation due to intensified early-career sorting into firms that provide better learning opportunities and career support explains part of the documented career benefits.

**General labor market experience and long-run unemployment risk.** Another potential reason behind the documented career benefits could be that students with early access to Facebook accumulated more general labor market experience and faced a lower risk of unemployment in the long run. Table 7, Panel C, investigates this channel using cumulative work experience. Consistent with the documented

[41] While the findings of the literature on heterogeneous skill accumulation across firms suggest that it is difficult to predict learning opportunities based on observable firm characteristics, Arellano-Bover and Saltiel (2026) provides some evidence that there is a weak positive correlation between firm pay and learning opportunities.

[42] The increased switching rate into high-wage firms from the control group is very similar to the behavior of students graduating in a recession (e.g., Oreopoulos *et al.* 2012; Schwandt and von Wachter 2019).

positive early-career employment effects, students with early access to Facebook accumulate 0.048 years (0.58 months) more work experience in the first six years after expected graduation. The effect remains positive and significant 12 years after graduation (0.054 years) before fading over the long run. However, the relative effect sizes are small (1% and 0.5%, respectively), suggesting that general labor market experience contributes little to the documented career effects.[43] Consistent with these results, Appendix Table A10 shows that the positive employment effects quickly fade out after the first six years, suggesting that a long-run reduction in unemployment risk does not explain the documented career effects.

Table 7: Facebook Access, Firm Sorting, and Human Capital Accumulation

| | 6Y | 12Y | 19Y |
|---|---|---|---|
| *Panel A: Work Experience at High-Wage Firms (Years)* | | | |
| FB Access | 0.080*** | 0.076 | 0.044 |
| | (0.024) | (0.049) | (0.080) |
| Dep. Var. Mean | 1.48 | 3.45 | 5.88 |
| Observations | 8,042 | 8,042 | 8,042 |
| *Panel B: Work Experience at High-Career-Support Firms (Years)* | | | |
| FB Access | 0.054** | 0.105** | 0.145** |
| | (0.017) | (0.035) | (0.054) |
| Dep. Var. Mean | 0.846 | 1.80 | 2.97 |
| Observations | 8,029 | 8,027 | 8,029 |
| *Panel C: Cumulative Work Experience (Years)* | | | |
| FB Access | 0.048** | 0.054** | 0.043 |
| | (0.017) | (0.021) | (0.022) |
| Dep. Var. Mean | 4.49 | 10.12 | 16.78 |
| Observations | 8,042 | 8,042 | 8,042 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, estimated at 6, 12, and 19 years after expected graduation. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Work experience at high-career-support firms: cumulative years employed at a high-career-support firm. Cumulative work experience: total years of employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

[43] This is especially true in light of the well-documented evidence of decreasing returns to general labor market experience over a worker's career (e.g., Mincer 1974; Rubinstein and Weiss 2006).

## 5.3 Robustness

This section presents a battery of robustness checks addressing concerns related to remaining spurious correlations with the Facebook rollout, spillover effects, multiple hypothesis testing, the choice of heterogeneity-robust estimator, the sample composition on LinkedIn, and empirical design choices for elite colleges.

**Randomization inference.** A key concern with the main empirical strategy is that even within narrow selectivity-by-region comparisons, there might be remaining unobservable spurious correlations with the timing of the Facebook rollout, biasing the main estimates. For example, Facebook may have (deliberately or not) expanded early to colleges *within* the same selectivity and U.S. region where career outcomes of U.S. business and economics cohorts were already on a differential trend. To formally test this, I implement a randomization inference test in which I randomly reshuffle the introduction year of Facebook across colleges within each selectivity-by-region cell. The test preserves the total number of treated colleges per cell and the overall distribution of treatment timing. Appendix D presents the results. The placebo estimates are densely distributed around zero, and virtually all statistically significant effects from the main specification fall above the 95$^{th}$ percentile of the placebo distribution. The remaining variation in the Facebook rollout is therefore unlikely to reflect spurious correlations.

**Spillover analysis.** A further concern is that the results are driven by spillovers. In particular, treated students might displace students in the control group from jobs, biasing the treatment effect upward and violating the stable-unit treatment value assumption (SUTVA). To assess the severity of this concern, I predict each potential control cohort's exposure to spillovers based on the historical career overlap of U.S. business and economics students across colleges prior to the Facebook rollout. I then drop all colleges that have at least one potential control cohort with the highest exposure score from the estimation. Reassuringly, Appendix E shows that the main patterns are very similar to the baseline estimates. Taken together with the previous findings, the results suggest that the treatment effects do not simply reflect a reallocation of jobs within the economy. They are rather consistent with the idea that access to Facebook reduces information frictions about job opportunities.

**Multiple hypothesis testing.** Given the large number of main results, some statistically significant findings could simply arise by chance. Appendix Table A26 reports adjusted p-values controlling for the false discovery rate following Benjamini and Hochberg (1995). The main findings remain statistically significant after the correction, suggesting that the key results are unlikely to reflect false positives.

**Further robustness.** Appendix F examines whether the main effects are sensitive to the choice

of heterogeneity-robust estimator, presenting results using the estimators of Callaway and Sant'Anna (2021) and de Chaisemartin and D'Haultfœuille (2020) as two alternatives. While a few estimated coefficients tend to be lower and standard errors are slightly larger, the main patterns remain very similar, indicating that the results are not driven by the choice of estimator. Appendix G reweights the analysis sample by gender and cohort size using IPEDS enrollment data to ensure representativeness in baseline demographics and college characteristics. The results are nearly identical to the baseline estimates, indicating that the main findings are not driven by LinkedIn's sample composition. Appendix H excludes Ivy Plus and other elite colleges, which are among the earliest adopters and can only be modeled with linear cohort trends. The estimates remain similar, suggesting that the main findings are not driven by empirical design choices for elite colleges and that the career benefits also accrue to non-elite colleges.

# 6 Conclusion

More than 3 billion people worldwide report using online social networks such as Facebook or LinkedIn for work-related purposes (Kemp 2025), yet evidence on the labor market impacts of these platforms remains scarce. This paper investigates how providing young workers with access to Facebook during a sensitive period in their careers – the labor market entry phase – affects initial job placement and long-run career progression. To do so, I exploit the staggered introduction of Facebook across U.S. colleges as a natural experiment and trace the careers of affected students through their public LinkedIn profiles. I document that access to Facebook at labor market entry generates long-lasting career benefits. U.S. business and economics students with early access to Facebook are 5.1% more likely to receive early-career promotions and 3% more likely to reach senior leadership positions in their late 30s. They also tend to work in higher-paying jobs, resulting in an economically meaningful increase in cumulative job-based earnings of around 1.2% over the first 19 years after graduation. The career benefits do not differ markedly between men and women.

A considerable part of the long-run career benefits can be explained by a more successful college-to-work transition, in which the college network played a central role in accessing higher-quality early-career jobs. Looking at longer-run treatment dynamics, I further show that part of the career benefits can be explained by improved stability and productivity of early-career worker-firm matches, enabling students to climb the career ladder within their organizations, and by intensified early-career sorting into higher-quality and career-supporting firms. I do not find evidence that the career benefits

are attributable to a long-run information advantage in the search for higher-paying jobs or a long-run reduction in unemployment risk. Overall, the findings of this paper highlight that access to online social networks during a sensitive period of a young worker's career can reduce information frictions, facilitate the transition into the labor market, and generate long-lasting career benefits.

What can we learn from the findings about the role of online social networking platforms in job search and career development today? The use of online social networking platforms, such as Facebook, has increased considerably in recent decades, spreading from a small subset of college students to over 70% of U.S. adults by 2025 (Pew Research Center 2025). With a higher job-offer arrival rate across the entire population, traditional job search models (e.g., Burdett 1978) would predict that the career benefits of using online social networking platforms diminish as individuals increasingly hear about and compete for the same jobs.[44] Nevertheless, three pieces of evidence suggest that some of these benefits persist today: (i) the finding of this paper that the career benefits do not simply reflect a reallocation of high-quality jobs but are consistent with reduced information frictions and improved worker-firm match quality, (ii) the well-documented evidence of persistent information frictions in the labor market (e.g., Conlon *et al.* 2018; Heller and Kessler 2024), with job seekers continuing to be inaccurately informed about potential job opportunities (e.g., Belot *et al.* 2019; Jäger *et al.* 2024; Caldwell *et al.* 2026), and (iii) empirical evidence by Wheeler *et al.* (2022) on positive short-run labor market effects of LinkedIn training for young job seekers in more recent years. Therefore, career services and unemployment agencies should actively encourage the use of online social networking platforms as a low-cost tool for job search and career development that can mitigate search frictions and improve prospects for workers in sensitive periods of their careers.

While the results in this paper improve our understanding of the relevance of online social networks for job finding and career development of young workers, several important questions for future research remain. In particular, examining (i) how the career effects differ for less-educated workers and workers at older ages, (ii) the general equilibrium effects, and (iii) the relative importance of the different channels (e.g., referrals, recruiters, job ads, company profiles) and platforms (e.g., Facebook, LinkedIn, X) is key to fully understanding the labor market impacts of online social networking platforms today.

[44] This is especially true for a single online social networking platform, as more social networking platforms have nowadays gained popularity and provide work-related information (e.g., Facebook, LinkedIn, X).

# References


Allcott, H. and Gentzkow, M. (2017) Social media and fake news in the 2016 election, *Journal of Economic Perspectives*, **31**, 211–236.

Almar, F., Friedrich, B., Reynoso, A., Schulz, B. and Vejlin, R. M. (2025) Families' career investments and firms' promotion decisions, NBER Working Paper, National Bureau of Economic Research.

Altonji, J. G., Kahn, L. B. and Speer, J. D. (2016) Cashier or consultant? Entry labor market conditions, field of study, and career success, *Journal of Labor Economics*, **34**, 361–401.

Altonji, J. G. and Pierret, C. R. (2001) Employer learning and statistical discrimination, *Quarterly Journal of Economics*, **116**, 313–350.

Aral, S. (2021) *The Hype Machine: How Social Media Disrupts Our Elections, Our Economy, and Our Health – and How We Must Adapt*, Currency, New York.

Arellano-Bover, J. (2022) The effect of labor market conditions at entry on workers' long-term skills, *Review of Economics and Statistics*, **104**, 1028–1045.

Arellano-Bover, J. (2024) Career consequences of firm heterogeneity for young workers: First job and firm size, *Journal of Labor Economics*, **42**, 549–589.

Arellano-Bover, J. and Saltiel, F. (2026) Differences in on-the-job learning across firms, *Journal of Labor Economics*, **44**, 149–188.

Armona, L. (2025) Online social network effects in labor markets: Evidence from Facebook's entry to college campuses, *Review of Economics and Statistics*, **107**, 873–888.

Armona, L. and Bailey, M. (2025) The long-term effects of expanding social capital via social media, working paper.

Arrington, M. (2005) 85% of college students use Facebook, *TechCrunch*, https://techcrunch.com/2005/09/07/85-of-college-students-use-facebook/, accessed March 28, 2025.

Athey, S. and Imbens, G. W. (2022) Design-based analysis in difference-in-differences settings with staggered adoption, *Journal of Econometrics*, **226**, 62–79.

Auxier, B. and Anderson, M. (2021) Social media use in 2021, https://www.pewresearch.org/internet/2021/04/07/social-media-use-in-2021/, accessed November 22, 2024.

Bakshy, E., Rosenn, I., Marlow, C. and Adamic, L. (2012) The role of social networks in information diffusion, in *Proceedings of the 21st International Conference on World Wide Web (WWW '12)*, ACM, pp. 519–528.

Barron's Educational Series (2008) *Barron's Profiles of American Colleges 2009*, Barron's Educational Series, Inc., Hauppauge, NY.

Beaman, L., Keleher, N. and Magruder, J. (2018) Do job networks disadvantage women? Evidence from a recruitment experiment in Malawi, *Journal of Labor Economics*, **36**, 121–157.

Belot, M., Kircher, P. and Muller, P. (2019) Providing advice to jobseekers at low cost: An experimental study on online advice, *Review of Economic Studies*, **86**, 1411–1447.

Belzil, C. and Bognanno, M. (2008) Promotions, demotions, halo effects, and the earnings dynamics of American executives, *Journal of Labor Economics*, **26**, 287–310.

Benjamini, Y. and Hochberg, Y. (1995) Controlling the false discovery rate: A practical and powerful approach to multiple testing, *Journal of the Royal Statistical Society: Series B (Methodological)*, **57**, 289–300.

Bertrand, M., Goldin, C. and Katz, L. F. (2010) Dynamics of the gender gap for young professionals in the financial and corporate sectors, *American Economic Journal: Applied Economics*, **2**, 228–255.

Best Universities (2024) Abbreviations for degrees and other academic distinctions, https://best-universities.net/degrees/degree-abbreviations/, accessed January 8, 2024.

Borusyak, K., Jaravel, X. and Spiess, J. (2024) Revisiting event-study designs: Robust and efficient estimation, *Review of Economic Studies*, **91**, 3253–3285.

Braghieri, L., Levy, R. and Makarin, A. (2022) Social media and mental health, *American Economic Review*, **112**, 3660–3693.

Bronson, M. A. and Thoursie, P. S. (2019) The wage growth and within-firm mobility of men and women: New evidence and theory, working paper.

Brooks, E. (2025) Undergraduate ranking methodologies, *U.S. News & World Report*, https://www.usnews.com/education/best-colleges/articles/undergraduate-ranking-methodologies, accessed January 11, 2026.

Burdett, K. (1978) A theory of employee search and quits, *American Economic Review*, **68**, 212–220.

Buser, T., Niederle, M. and Oosterbeek, H. (2014) Gender, competitiveness, and career choices, *Quarterly Journal of Economics*, **129**, 1409–1447.

Caldwell, S., Haegele, I. and Heining, J. (2026) Firm pay and worker search, working paper.

Caldwell, S. and Harmon, N. (2019) Outside options, bargaining, and wages: Evidence from coworker networks, working paper.

Callaway, B. and Sant'Anna, P. H. C. (2021) Difference-in-differences with multiple time periods, *Journal of Econometrics*, **225**, 200–230.

Calvó-Armengol, A. and Jackson, M. O. (2004) The effects of social networks on employment and inequality, *American Economic Review*, **94**, 426–454.

Card, D., Rothstein, J. and Yi, M. (2025) Location, location, location, *American Economic Journal: Applied Economics*, **17**, 297–336.

Chetty, R., Friedman, J. N., Saez, E., Turner, N. and Yagan, D. (2020) Income segregation and intergenerational mobility across colleges in the United States, *Quarterly Journal of Economics*, **135**, 1567–1633.

Conlon, J., Pilossoph, L., Wiswall, M. and Zafar, B. (2018) Labor market search with imperfect information and learning, NBER Working Paper 24988, National Bureau of Economic Research.

Cullen, Z. and Perez-Truglia, R. (2023) The old boys' club: Schmoozing and the gender gap, *American Economic Review*, **113**, 1703–1740.

Dauth, W., Findeisen, S., Moretti, E. and Suedekum, J. (2022) Matching in cities, *Journal of the European Economic Association*, **20**, 1478–1521.

de Chaisemartin, C. and D'Haultfœuille, X. (2020) Two-way fixed effects estimators with heterogeneous treatment effects, *American Economic Review*, **110**, 2964–2996.

Dorn, D., Schoner, F., Seebacher, M., Simon, L. and Woessmann, L. (2025) Multidimensional skills on LinkedIn profiles: Measuring human capital and the gender skill gap, IZA Discussion Paper 17896, IZA – Institute of Labor Economics.

Ellison, N. B., Steinfield, C. and Lampe, C. (2007) The benefits of Facebook "friends": Social capital and college students' use of online social network sites, *Journal of Computer-Mediated Communication*, **12**, 1143–1168.

Enikolopov, R., Makarin, A. and Petrova, M. (2020) Social media and protest participation: Evidence from Russia, *Econometrica*, **88**, 1479–1514.

Facebook (2006) Facebook expands to include work networks, https://about.fb.com/news/2006/05/facebook-expands-to-include-work-networks-2/, accessed January 10, 2026.

Facebook (2007) Facebook unveils Facebook ads, https://about.fb.com/news/2007/11/facebook-unveils-facebook-ads/, accessed January 10, 2026.

Facebook (2008) Facebook chooses Oodle to power its marketplace application, https://about.fb.com/news/2008/12/facebook-chooses-oodle-to-power-its-marketplace-application/, accessed January 10, 2026.

Fergusson, L. and Molina, C. (2019) Facebook causes protests, Documento CEDE 41.

Fischer, A., Gorshkov, A., Sandoy, T. M. and Walldorf, J. (2025) Peers, heirs, and careers: Labor market effects of alumni networks, working paper.

Frederiksen, A., Halliday, T. and Koch, A. K. (2016) Within- and cross-firm mobility and earnings growth, *ILR Review*, **69**, 320–353.

Gardner, J. (2022) Two-stage differences in differences, arXiv:2207.05943.

Gee, L. K., Jones, J. and Burke, M. (2017) Social networks and labor markets: How strong ties relate to job finding on Facebook's social network, *Journal of Labor Economics*, **35**, 485–518.

Gibbons, R. and Waldman, M. (1999) Careers in organizations: Theory and evidence, *Handbook of Labor Economics*, **3**, 2373–2437.

Gibbons, R. and Waldman, M. (2006) Enriching a theory of wage and promotion dynamics inside firms, *Journal of Labor Economics*, **24**, 59–107.

Glaeser, E. L. and Maré, D. C. (2001) Cities and skills, *Journal of Labor Economics*, **19**, 316–342.

Goldin, C. (2014) A grand gender convergence: Its last chapter, *American Economic Review*, **104**, 1091–1119.

Gompers, P. A., Hu, G., Levinson, W. and Srivastava, S. (2026) From field to firm: College sports and early-stage career choice, NBER Working Paper 35527, National Bureau of Economic Research, Cambridge, MA.

Goodman-Bacon, A. (2021) Difference-in-differences with variation in treatment timing, *Journal of Econometrics*, **225**, 254–277.

Granovetter, M. (1995) *Getting a Job: A Study of Contacts and Careers*, University of Chicago Press, Chicago.

Gregory, V. (2023) Firms as learning environments: Implications for earnings dynamics and job search, unpublished manuscript, Federal Reserve Bank of St. Louis.

Grynbaum, M. M. (2004) Online Facebook solicits new ads, *The Harvard Crimson*, https://www.thecrimson.com/article/2004/5/7/online-facebook-solicits-new-ads-thefacebookcom/, accessed January 10, 2026.

Haltiwanger, J. C., Hyatt, H. R., Kahn, L. B. and McEntarfer, E. (2018) Cyclical job ladders by firm size and firm wage, *American Economic Journal: Macroeconomics*, **10**, 52–85.

Hampole, M., Papanikolaou, D., Schmidt, L. D. W. and Seegmiller, B. (2025) Artificial intelligence and the labor market, NBER Working Paper 33509, National Bureau of Economic Research.

Hampole, M., Truffa, F. and Wong, A. (2026) Peer effects and the gender gap in corporate leadership: Evidence from MBA students, *Quarterly Journal of Economics*, **141**, 2499–2554.

Hass, N. (2006) In your Facebook.com, https://www.nytimes.com/2006/01/08/education/edlife/in-your-facebookcom.html, accessed March 28, 2025.

Heller, S. B. and Kessler, J. B. (2024) Information frictions and skill signaling in the youth labor market, *American Economic Journal: Economic Policy*, **16**, 1–33.

Hindlian, A., Lawson, S., Banerjee, S., Shan, H., Mirabal, D. and Campbell-Mohn, E. (2018) Closing the gender gaps: Advancing women in corporate America, Global Markets Institute Report, Goldman Sachs.

Ingram, D. D. and Franco, S. J. (2014) 2013 NCHS urban–rural classification scheme for counties, Vital and Health Statistics Series 2, No. 166, National Center for Health Statistics.

Jäger, S., Roth, C., Roussille, N. and Schoefer, B. (2024) Worker beliefs about outside options, *Quarterly Journal of Economics*, **139**, 1505–1556.

Jobvite (2009) Social recruitment survey: Hiring during down economy, https://www.jobvite.com/news_item/jobvite-social-recruitment-survey-companies-turn-employee-referrals-social-networks-hiring-economy/, accessed January 10, 2026.

Jobvite (2011) One in six workers successfully used social networks to get hired, https://www.jobvite.com/news_item/social-job-seekers-getting-ahead-jobvite-survey-reveals-one-six-workers-successfully-used-social-networks-get-hired/, accessed January 10, 2026.

Jobvite (2021) Jobvite's 2021 recruiter nation report reveals new levels of agility in recruiting, https://www.jobvite.com/news_item/jobvites-2021-recruiter-nation-report-reveals-new-levels-of-agility-in-recruiting/, accessed January 10, 2026.

Kahn, L. B. (2010) The long-term labor market consequences of graduating from college in a bad economy, *Labour Economics*, **17**, 303–316.

Keane, M. P. and Wolpin, K. I. (1997) The career decisions of young men, *Journal of Political Economy*, **105**, 473–522.

Kemp, S. (2025) Digital 2025: The state of social media, https://datareportal.com/reports/digital-2025-sub-section-state-of-social, accessed March 26, 2025.

Kirkpatrick, D. (2011) *The Facebook Effect: The Inside Story of the Company That Is Connecting the World*, Simon & Schuster, New York.

Kleven, H. J., Landais, C. and Søgaard, J. E. (2019) Children and gender inequality: Evidence from Denmark, *American Economic Journal: Applied Economics*, **11**, 181–209.

Kramarz, F. and Skans, O. N. (2014) When strong ties are strong: Networks and youth labour market entry, *Review of Economic Studies*, **81**, 1164–1200.

Kramarz, F. and Thesmar, D. (2013) Social networks in the boardroom, *Journal of the European Economic Association*, **11**, 780–807.

Lalanne, M. and Seabright, P. (2016) The old boy network: The impact of professional networks on remuneration in top executive jobs, SSRN Working Paper 2717988.

Lampe, C., Ellison, N. and Steinfield, C. (2006) A face(book) in the crowd: Social searching vs. social browsing, in *Proceedings of the 2006 20th Anniversary Conference on Computer Supported Cooperative Work (CSCW '06)*, ACM Press, New York, NY, USA, pp. 167–170.

Lampe, C., Ellison, N. B. and Steinfield, C. (2008) Changes in use and perception of Facebook, in *Proceedings of the 2008 ACM Conference on Computer Supported Cooperative Work (CSCW '08)*, ACM Press, San Diego, CA, USA, pp. 721–730.

Lean In and McKinsey & Company (2020) Women in the workplace, https://womenintheworkplace.com/, accessed May 8, 2025.

Lerner, J. and Malmendier, U. (2013) With a little help from my (random) friends: Success and failure in post-business school entrepreneurship, *Review of Financial Studies*, **26**, 2411–2452.

LinkedIn (2024) About us, https://web.archive.org/web/20241029035736/https:/news.linkedin.com/about-us/, accessed November 22, 2024.

Maestas, N., Mullen, K. J., Powell, D., von Wachter, T. and Wenger, J. B. (2023) The value of working conditions in the United States and implications for the structure of wages, *American Economic Review*, **113**, 2007–2047.

Marmaros, D. and Sacerdote, B. (2002) Peer and social networks in job search, *European Economic Review*, **46**, 870–879.

Mas, A. and Pallais, A. (2017) Valuing alternative work arrangements, *American Economic Review*, **107**, 3722–3759.

Mengel, F. (2020) Gender differences in networking, *Economic Journal*, **130**, 1842–1873.

Michelman, V., Price, J. and Zimmerman, S. D. (2022) Old boys' clubs and upward mobility among the educational elite, *Quarterly Journal of Economics*, **137**, 845–909.

Mincer, J. (1974) *Schooling, Experience and Earnings*, Columbia University Press, New York.

Müller, K. and Schwarz, C. (2021) Fanning the flames of hate: Social media and hate crime, *Journal of the European Economic Association*, **19**, 2131–2167.

National Association of Colleges and Employers (2009) 2009 student survey, https://www.k-state.edu/careercenter/conference/documents/nacemovingonreportjan2010.pdf, accessed January 10, 2026.

National Center for Education Statistics (2024a) The classification of instructional programs (CIP), https://nces.ed.gov/ipeds/cipcode/default.aspx?y=56, accessed January 8, 2024.

National Center for Education Statistics (2024b) Integrated postsecondary education data system (IPEDS), U.S. Department of Education, Institute of Education Sciences, https://nces.ed.gov/ipeds/, accessed October 10, 2024.

Niederle, M. and Vesterlund, L. (2007) Do women shy away from competition? Do men compete too much?, *Quarterly Journal of Economics*, **122**, 1067–1101.

Olivetti, C., Pan, J. and Petrongolo, B. (2024) The evolution of gender in the labor market, in *Handbook of Labor Economics*, Elsevier, vol. 5, pp. 619–677.

Oreopoulos, P., von Wachter, T. and Heisz, A. (2012) The short- and long-term career effects of graduating in a recession, *American Economic Journal: Applied Economics*, **4**, 1–29.

Ost, B., Pan, W. and Webber, D. A. (2026) College networks: The importance of employer connections, *Journal of Labor Economics*, **44**, 351–381.

Overgoor, J., Adamic, L. A. *et al.* (2020) The structure of US college networks on Facebook, in *Proceedings of the International AAAI Conference on Web and Social Media*, vol. 14, pp. 499–510.

Pallais, A. (2014) Inefficient hiring in entry-level labor markets, *American Economic Review*, **104**, 3565–3599.

Pallais, A. and Sands, E. G. (2016) Why the referential treatment? Evidence from field experiments on referrals, *Journal of Political Economy*, **124**, 1793–1828.

Pew Research Center (2025) Americans' social media use 2025, https://www.pewresearch.org/internet/2025/11/20/americans-social-media-use-2025/, accessed May 8, 2026.

Rajkumar, K., Saint-Jacques, G., Bojinov, I., Brynjolfsson, E. and Aral, S. (2022) A causal test of the strength of weak ties, *Science*, **377**, 1304–1310.

Rambachan, A. and Roth, J. (2023) A more credible approach to parallel trends, *Review of Economic Studies*, **90**, 2555–2591.

Rosen, S. (1972) Learning and experience in the labor market, *Journal of Human Resources*, **7**, 326–342.

Roth, J., Sant'Anna, P. H. C., Bilinski, A. and Poe, J. (2023) What's trending in difference-in-differences? A synthesis of the recent econometrics literature, *Journal of Econometrics*, **235**, 2218–2244.

Rubinstein, Y. and Weiss, Y. (2006) Post schooling wage growth: Investment, search and learning, in *Handbook of the Economics of Education* (Eds.) E. A. Hanushek and F. Welch, North-Holland, Amsterdam, vol. 1, chap. 1, pp. 1–67.

Schmidheiny, K. and Siegloch, S. (2023) On event studies and distributed lags in two-way fixed effects models: Identification, equivalence, and generalization, *Journal of Applied Econometrics*, **38**, 695–713.

Schmutte, I. M. (2015) Job referral networks and the determination of earnings in local labor markets, *Journal of Labor Economics*, **33**, 1–32.

School of Information, San Jose State University (2018) Social media job search, https://ischool.sjsu.edu/social-media-job-search, accessed May 2026.

Schwandt, H. and von Wachter, T. (2019) Unlucky cohorts: Estimating the long-term effects of entering the labor market in a recession in large cross-sectional data sets, *Journal of Labor Economics*, **37**, 161–198.

Shue, K. (2013) Executive networks and firm policies: Evidence from the random assignment of MBA peers, *Review of Financial Studies*, **26**, 1401–1442.

Simon, C. J. and Warner, J. T. (1992) Matchmaker, matchmaker: The effect of old boy networks on job match quality, earnings, and tenure, *Journal of Labor Economics*, **10**, 306–330.

Sockin, J. (2022) Show me the amenity: Are higher-paying firms better all around?, CESifo Working Paper 9842, CESifo.

Sorkin, I. (2018) Ranking firms using revealed preference, *Quarterly Journal of Economics*, **133**, 1331–1393.

Speer, J. D. (2017) Pre-market skills, occupational choice, and career progression, *Journal of Human Resources*, **52**, 187–246.

StudentNews Group (2024) Degree abbreviations, https://degree.studentnews.eu/, accessed January 8, 2024.

Sun, L. and Abraham, S. (2021) Estimating dynamic treatment effects in event studies with heterogeneous treatment effects, *Journal of Econometrics*, **225**, 175–199.

Topa, G. (2011) Labor markets and referrals, in *Handbook of Social Economics*, Elsevier, vol. 1, pp. 1193–1221.

Topel, R. H. and Ward, M. P. (1992) Job mobility and the careers of young men, *Quarterly Journal of Economics*, **107**, 439–479.

von Essen, E. and Smith, N. (2023) Network connections and board seats: Are female networks less valuable?, *Journal of Labor Economics*, **41**, 323–360.

von Wachter, T. (2020) The persistent effects of initial labor market conditions for young adults and their sources, *Journal of Economic Perspectives*, **34**, 168–194.

Wheeler, L., Garlick, R., Johnson, E., Shaw, P. and Gargano, M. (2022) LinkedIn (to) job opportunities: Experimental evidence from job readiness training, *American Economic Journal: Applied Economics*, **14**, 101–125.

Wiswall, M. and Zafar, B. (2018) Preference for the workplace, investment in human capital, and gender, *Quarterly Journal of Economics*, **133**, 457–507.

YourDictionary (2022) Degree abbreviations, https://www.yourdictionary.com/articles/degree-abbreviations, accessed January 8, 2024.

Zeltzer, D. (2020) Gender homophily in referral networks: Consequences for the Medicare physician earnings gap, *American Economic Journal: Applied Economics*, **12**, 169–197.

Zhu, M. (2025) Beyond the "Old Boys' Network": Social networks and job finding at community colleges, *Journal of Human Resources*, **60**, 1405–1434.

Zhuravskaya, E., Petrova, M. and Enikolopov, R. (2020) Political effects of the internet and social media, *Annual Review of Economics*, **12**, 415–438.

Zimmerman, S. D. (2019) Elite colleges and upward mobility to top jobs and top incomes, *American Economic Review*, **109**, 1–47.

# Appendix A: Additional Tables and Figures

Table A1: Most Common Job Title Transitions for Promotion Events

| Years Since Graduation | Top 10 Job Title Transitions (Promotion Events) |
|---|---|
| 2-6 Years | Staff Accountant -> Senior Accountant, Financial Analyst -> Senior Financial Analyst, Analyst -> Associate, Associate -> Vice President, Consultant -> Senior Consultant, Accountant -> Senior Accountant, Account Executive -> Senior Account Executive, Account Executive -> Account Executive, Associate -> Associate, Associate -> Senior Associate |
| 7-12 Years | Financial Analyst -> Senior Financial Analyst, Associate -> Vice President, Manager -> Senior Manager, Attorney -> Attorney, Associate -> Partner, Associate -> Associate, Staff Accountant -> Senior Accountant, Assistant Vice President -> Vice President, Project Manager -> Senior Project Manager, Senior Accountant -> Accounting Manager |
| 13-19 Years | Project Manager -> Senior Project Manager, Vice President -> Senior Vice President, Associate -> Partner, Controller -> Chief Financial Officer, Assistant Controller -> Controller, Manager -> Senior Manager, Vice President -> President, Vice President -> Director, Attorney -> Partner, General Manager -> General Manager |

*Notes:* Most common job title transitions for identified promotion events by years since graduation. Promotions are identified by large relative real annual wage growth jumps of at least 10 log points following Bronson and Thoursie (2019). *Data source:* Revelio Labs.

Table A2: Managerial Pipeline Validation – Earnings

| Variable | Individual Contributor | Manager | Director | Vice president | Senior vice president | C-Suite |
|---|---|---|---|---|---|---|
| Job-based annual earnings | 66,543 | 102,596 | 170,116 | 190,486 | 209,820 | 250,123 |
| Share female | 0.450 | 0.414 | 0.409 | 0.303 | 0.257 | 0.246 |
| Observations | 8,765,845 | 4,053,126 | 1,633,676 | 756,400 | 462,375 | 442,071 |

*Notes:* Descriptive statistics at the person–year level by managerial pipeline position. Sample: all U.S. business & economics undergraduates expected to graduate from 1999 to 2005 at selective 4-year colleges with a valid LinkedIn profile and reasonable college-level coverage (1,154 colleges). *Data source:* Revelio Labs

## Table A3: Most Common Job Titles – Leadership Pipeline

| Category | Top 10 job titles |
|---|---|
| Individual Contributor | Account Executive, Senior Accountant, Accountant, Associate, Financial Advisor, Attorney, Financial Analyst, Sales Representative, Staff Accountant, Senior Financial Analyst |
| Manager | Project Manager, Manager, Account Manager, Operations Manager, Sales Manager, Branch Manager, Store Manager, Accounting Manager, Senior Manager, Marketing Manager |
| Director | Partner, Controller, Director, Director of Operations, Assistant Vice President, Marketing Director, Director of Finance, Director of Marketing, Director of Sales, Director Of Operations |
| Vice President | Vice President, Principal, Regional Vice President, Vice President of Sales, Vice President of Operations, Senior Director, VP, Vice President Operations, Vice President of Finance, VP of Operations |
| Senior Vice President | General Manager, Managing Director, Executive Director, Managing Partner, Senior Vice President, Executive Vice President, Assistant General Manager, Senior Managing Director, Managing Principal, GM |
| C-Level | President, CEO, Chief Financial Officer, Chief Executive Officer, CFO, Chief Operating Officer, COO, President & CEO, President/CEO, General Counsel |

*Notes:* Most common job titles by leadership pipeline category. *Data source:* Revelio Labs.

## Table A4: Coverage, Distribution, and Correlation of Permanent Firm Characteristics

| | Coverage (%) | | Distribution (emp.-wtd.) | | | Correlation matrix | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | LinkedIn | Est. sample | $p_{25}$ | Median | $p_{75}$ | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| (1) Firm pay | 100.0 | 99.8 | 10.70 | 11.00 | 11.26 | 1.00 | 0.04 | 0.05 | 0.05 | 0.10 | 0.07 | 0.10 |
| (2) Overall rating | 43.8 | 69.4 | 3.34 | 3.65 | 3.94 | 0.04 | 1.00 | 0.77 | 0.84 | 0.73 | 0.84 | 0.69 |
| (3) Career opportunities | 43.2 | 68.8 | 3.01 | 3.35 | 3.63 | 0.05 | 0.77 | 1.00 | 0.77 | 0.65 | 0.79 | 0.73 |
| (4) Culture & values | 43.1 | 68.7 | 3.13 | 3.51 | 3.87 | 0.05 | 0.84 | 0.77 | 1.00 | 0.76 | 0.87 | 0.68 |
| (5) Work-life balance | 43.2 | 68.8 | 3.14 | 3.50 | 3.85 | 0.10 | 0.73 | 0.65 | 0.76 | 1.00 | 0.75 | 0.63 |
| (6) Senior leadership | 43.2 | 68.8 | 2.87 | 3.17 | 3.48 | 0.07 | 0.84 | 0.79 | 0.87 | 0.75 | 1.00 | 0.69 |
| (7) Comp. & benefits | 43.2 | 68.8 | 3.02 | 3.41 | 3.73 | 0.10 | 0.69 | 0.73 | 0.68 | 0.63 | 0.69 | 1.00 |

*Notes:* Descriptive statistics on the coverage, distribution, and correlation of all permanent firm characteristics. LinkedIn: coverage of the respective firm characteristic among all companies on LinkedIn that employ college students in the main estimation sample weighted by the average LinkedIn employee size. Est. sample: coverage of the respective firm characteristic among all person-year observations indicating an employer in the main estimation sample. Distribution columns: $p_{25}$, median, and $p_{75}$ of each permanent firm characteristic among all companies on LinkedIn that employ college students in the main estimation sample weighted by the average LinkedIn employee size. Firm pay: average median log annual earnings, net of year fixed effects. Glassdoor amenity ratings are measured on a scale from 1–5: overall rating, career opportunities, culture & values, work-life balance, senior leadership, comp. & benefits. All Glassdoor amenity ratings are averaged across years, net of year fixed effects. Correlation matrix: Pearson correlations at the firm level between the different firm quality measures. *Data source:* Revelio Labs and Glassdoor.

## Table A5: Sample Selection

| Filtering step | Observations | Number of Colleges | Cumulative | Age | Female | At Least Very Selective College |
|---|---|---|---|---|---|---|
| All LinkedIn profiles, U.S. business and economics undergraduates | 1,071,848 | 1,235 | 1.00 | 18.42 | 0.434 | 0.503 |
| Valid undergraduate graduation date | 1,048,577 | 1,235 | 0.978 | 18.41 | 0.433 | 0.504 |
| Complete employment history | 914,976 | 1,234 | 0.854 | 18.41 | 0.424 | 0.511 |
| Reasonable college-level coverage | 890,951 | 1,154 | 0.831 | 18.39 | 0.423 | 0.519 |

*Notes:* Each row reports the number of LinkedIn profiles remaining after the corresponding filtering step is applied. The final estimation sample covers all U.S. business & economics undergraduates expected to graduate from 1999 to 2005 at selective 4-year colleges with a valid LinkedIn profile and reasonable college-level coverage (1,154 colleges). *Data source:* Revelio Labs

Table A6: Comparison Table – Revelio Labs vs. IPEDS

| | Actual Graduation | Expected Graduation | IPEDS |
|---|---|---|---|
| *Selectivity* | | | |
| Elite | 0.075 | 0.076 | 0.067 |
| Highly competitive | 0.161 | 0.162 | 0.150 |
| Very competitive | 0.280 | 0.281 | 0.265 |
| Competitive | 0.394 | 0.392 | 0.415 |
| Less competitive | 0.090 | 0.089 | 0.103 |
| *College type* | | | |
| Public | 0.638 | 0.640 | 0.643 |
| *Region* | | | |
| Northeast | 0.220 | 0.220 | 0.211 |
| Midwest | 0.276 | 0.275 | 0.266 |
| South | 0.342 | 0.342 | 0.351 |
| West | 0.162 | 0.162 | 0.173 |
| *Gender* | | | |
| Female | 0.420 | 0.416 | 0.472 |
| Observations | 905,281 | 890,951 | 1,907,885 |

*Notes:* Descriptive statistics for various samples. Actual Graduation: all business & economics graduates from the 1,154 colleges in the graduation cohorts from 1999 to 2005 with a valid LinkedIn profile and reasonable college-level coverage. Expected Graduation: all business & economics students from the 1,154 colleges in the expected graduation cohorts from 1999 to 2005 with a valid LinkedIn profile and reasonable college-level coverage. IPEDS: all business & economics graduates from the 1,154 colleges in the graduation cohorts from 1999 to 2005 in the IPEDS data. *Data source:* Revelio Labs and IPEDS.

Table A7: Treatment Timing and Potential Confounders

| | Raw | Baseline |
|---|---|---|
| *College characteristics* | | |
| Bachelor's degrees awarded (log) | 0.040*** | 0.022 |
| | (0.004) | (0.040) |
| Region (ref.: West) | | |
| Midwest | 0.026* | — |
| | (0.011) | |
| Northeast | 0.069*** | — |
| | (0.014) | |
| South | 0.017 | — |
| | (0.010) | |
| Selectivity (ref.: Less competitive) | | |
| Competitive | 0.022* | — |
| | (0.009) | |
| Very competitive | 0.040*** | — |
| | (0.010) | |
| Highly competitive | 0.041** | — |
| | (0.012) | |
| Elite | 0.111*** | — |
| | (0.014) | |
| *Student and major composition* | | |
| Share female (in %) | -0.0007 | -0.0005 |
| | (0.0004) | (0.001) |
| Share White (in %) | -0.002*** | 0.0007 |
| | (0.0002) | (0.0008) |
| Share computer science (in %) | -0.015*** | -0.002 |
| | (0.002) | (0.001) |
| *Local labor market entry conditions* | | |
| College-state unemployment rate | 0.009 | 0.001 |
| | (0.005) | (0.008) |
| Joint $F$-test | 41.96 [<0.001] | 0.780 [0.564] |
| $R^2$ | 0.042 | 0.832 |
| Observations | 8,042 | 8,042 |
| College FE | | X |
| Cohort*Tier*Region FE | | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. OLS regressions of the treatment indicator on potential confounders, estimated at the college*cohort level. Raw: no fixed effects. Baseline: fixed-effects structure of equation (1). The region and selectivity indicators are absorbed by the fixed effects. Bachelor's degrees awarded (log): log number of bachelor's degrees awarded by the college in the expected graduation year. Share female, share White, and share computer science (in %): shares of these bachelor's degrees awarded to women, to White students, and in computer and information sciences (CIP 11). College-state unemployment rate: unemployment rate in the college's state in the expected graduation year. Joint $F$-test: test of joint significance of the five time-varying characteristics reported in both columns; $p$-values in brackets. Regressions are weighted by the number of LinkedIn profiles per college*cohort cell. Standard errors are clustered at the college level. *Data source:* Revelio Labs, IPEDS, BLS, and Barron's.

Table A8: Facebook Access, LinkedIn Sign-Up, and Balancing

| | Coverage | | | Age at Entry | Female |
|---|---|---|---|---|---|
| | Overall | Male | Female | | |
| FB Access | 0.006 | 0.003 | 0.011 | −0.010 | −0.0008 |
| | (0.010) | (0.013) | (0.012) | (0.020) | (0.005) |
| Dep. Var. Mean | 0.453 | 0.505 | 0.397 | 18.42 | 0.417 |
| Observations | 8,042 | 7,827 | 7,953 | 8,042 | 8,042 |
| College FE | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year or college*cohort*year*gender level. Coverage: fraction of students in the LinkedIn estimation sample compared to the official graduation counts in IPEDS. Coverage regressions are weighted using IPEDS graduation counts. Age at entry: imputed age at labor market entry. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs and IPEDS.

Table A9: Facebook Access and LinkedIn Profile Completeness

| | Coverage | Share of Profiles Meeting the Sample Restriction | | | |
|---|---|---|---|---|---|
| | (All Profiles) | Valid Graduation Date | Employment History Reported | <10 Years without Job | All Restrictions (Final Sample) |
| FB Access | 0.004 | -0.002 | -0.003 | 0.001 | 0.002 |
| | (0.012) | (0.001) | (0.002) | (0.003) | (0.003) |
| Dep. Var. Mean | 0.533 | 0.979 | 0.960 | 0.866 | 0.851 |
| Observations | 8,042 | 8,042 | 8,042 | 8,042 | 8,042 |
| College FE | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort level. Each cell reports a coefficient from a separate DiD regression. Sample: all LinkedIn profiles of business and economics students expected to graduate from 1999 to 2005 at the 1,154 colleges of the estimation sample, before any sample restriction. Coverage: number of LinkedIn profiles relative to the official graduation counts in IPEDS. Valid graduation date: share of profiles reporting a valid undergraduate graduation date. Employment history reported: share of profiles reporting any work experience in the employment history section. <10 years without job: share of profiles reporting employment in at least 10 of the 19 years after expected graduation. All restrictions (final sample): share of profiles meeting all restrictions, including the match to the IPEDS graduation counts by gender, and entering the estimation sample. Column 1 is weighted using IPEDS graduation counts. Columns 2 to 5 are weighted by the number of LinkedIn profiles in the college cohort before any sample restriction. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs and IPEDS.

Table A10: Facebook Access, Employment, and Conditional Earnings

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Employment* | | | |
| FB Access | 0.007* | 0.0001 | -0.0009 |
| | (0.003) | (0.001) | (0.002) |
| Dep. Var. Mean | 0.768 | 0.937 | 0.943 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel B: Conditional Earnings (Employed Only)* | | | |
| FB Access | 561 | 1,372* | 973 |
| | (504) | (555) | (635) |
| Dep. Var. Mean | 98,787 | 113,411 | 128,820 |
| Observations | 48,200 | 48,239 | 56,284 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Panel A: share of graduates with a recorded employment spell. Panel B: average job-based annual earnings (2024 USD) conditional on employment. Observation counts differ only slightly between panels because a college*cohort*year cell drops from Panel B only if none of its graduates is employed. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A11: Facebook Access and Alternative Promotion Thresholds

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Promotion (≥7.5 log pts.)* | | | |
| FB Access | 0.004** | 0.002 | -0.0008 |
| | (0.001) | (0.001) | (0.001) |
| Dep. Var. Mean | 0.082 | 0.089 | 0.079 |
| Observations | 40,210 | 48,252 | 56,294 |
| *Panel B: Job-Based Promotion (≥12.5 log pts.)* | | | |
| FB Access | 0.003* | 0.002 | -0.0009 |
| | (0.001) | (0.001) | (0.0010) |
| Dep. Var. Mean | 0.074 | 0.080 | 0.070 |
| Observations | 40,210 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A12: Facebook Access, Lateral Moves, and Demotion

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Lateral Moves* | | | |
| FB Access | 0.001 | 0.0008 | 0.0002 |
| | (0.001) | (0.0009) | (0.0009) |
| Dep. Var. Mean | 0.034 | 0.040 | 0.037 |
| Observations | 40,210 | 48,252 | 56,294 |
| *Panel B: Job-Based Demotion ($\leq -10$ log pts.)* | | | |
| FB Access | 0.0002 | -0.0009 | -0.0007 |
| | (0.0010) | (0.001) | (0.0010) |
| Dep. Var. Mean | 0.042 | 0.057 | 0.059 |
| Observations | 40,210 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. Panel A: share of graduates transitioning to a job with relative real annual wage growth between $-10$ and $+10$ log points. Panel B: share of graduates transitioning to a job with relative real annual wage growth below $-10$ log points. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A13: Facebook Access and Alternative Top Jobs

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Management Occupations (SOC 11)* | | | |
| FB Access | 0.002 | 0.001 | 0.010* |
| | (0.004) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.198 | 0.265 | 0.288 |
| Observations | 48,246 | 48,248 | 56,293 |
| *Panel B: Top Executives (SOC 11-1)* | | | |
| FB Access | 0.0007 | 0.003* | 0.003 |
| | (0.001) | (0.001) | (0.001) |
| Dep. Var. Mean | 0.020 | 0.034 | 0.042 |
| Observations | 48,246 | 48,248 | 56,293 |
| *Panel C: Member of the Board* | | | |
| FB Access | -0.0003 | -0.0003 | 0.00010 |
| | (0.0002) | (0.0004) | (0.0004) |
| Dep. Var. Mean | 0.0007 | 0.002 | 0.003 |
| Observations | 48,252 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Member of the board: indicator for working as a board member or member of the board of directors using a keyword search on job titles. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A14: Facebook Access and High-Paying Jobs

| | Fixed Earnings Threshold | | Relative Earnings Threshold | |
|---|---|---|---|---|
| | ≥$150,000 | ≥$200,000 | Top 25% | Top 10% |
| FB Access | 0.006* | 0.003 | 0.007* | 0.001 |
| | (0.003) | (0.002) | (0.003) | (0.002) |
| Dep. Var. Mean | 0.240 | 0.147 | 0.245 | 0.097 |
| Observations | 104,546 | 104,546 | 104,546 | 104,546 |
| College FE | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the years 7–19 after expected graduation. Fixed earnings threshold: indicator for job-based annual earnings (2024 USD) of at least $150,000 or $200,000. Relative earnings threshold: indicator for job-based annual earnings in the top 25% or top 10% of the distribution among all graduates in the sample of the same expected graduation cohort in the same year after expected graduation. Job-based annual earnings are set to zero in years without employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A15: Facebook Access and Cumulative Earnings

| | $\delta = 0\%$ | $\delta = 2.5\%$ | $\delta = 5\%$ |
|---|---|---|---|
| FB Access | 21,148* | 17,660* | 14,976** |
| | (8,989) | (7,105) | (5,771) |
| Dep. Var. Mean | 1,942,861 | 1,528,716 | 1,229,694 |
| Observations | 8,042 | 8,042 | 8,042 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort level. The present discounted value of earnings is the discounted sum of average annual earnings over the 19-year observation window: $\text{PDV}(\delta) = \sum_{t=1}^{19} \bar{w}_{j,c,t}/(1+\delta)^{t-1}$, where $\bar{w}_{j,c,t}$ is the average job-based annual earnings of college $j$, cohort $c$, at year $t$ after graduation, and $\delta$ is the discount rate. Job-based annual earnings are set to zero in years without employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A16: Facebook Access and the College-to-Work Transition – Other High-Amenity Firm Dimensions

| | Overall | Comp. & Benef. | Culture | Leadership | Work-Life |
|---|---|---|---|---|---|
| *Panel A: Employed at High-Amenity Firm* | | | | | |
| FB Access | 0.003 | 0.010** | 0.006 | 0.004 | 0.005 |
| | (0.004) | (0.004) | (0.003) | (0.004) | (0.003) |
| Dep. Var. Mean | 0.100 | 0.139 | 0.113 | 0.124 | 0.095 |
| Observations | 16,068 | 16,066 | 16,066 | 16,066 | 16,066 |
| *Panel B: Early-Career High-Amenity Firm Job via College Network* | | | | | |
| FB Access | 0.002 | 0.003* | 0.003 | 0.003* | 0.0010 |
| | (0.001) | (0.002) | (0.001) | (0.001) | (0.001) |
| Dep. Var. Mean | 0.017 | 0.028 | 0.021 | 0.030 | 0.014 |
| Observations | 16,068 | 16,066 | 16,066 | 16,066 | 16,066 |
| College FE | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-amenity firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor rating distribution for the given dimension (among firms with at least two ratings). The career opportunities dimension is shown separately in Tables 3, 4, and 7. Overall: overall rating on Glassdoor. Comp. & Benef.: compensation and benefits. Culture: company culture. Leadership: senior leadership. Work-Life: work-life balance. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A17: Facebook Access and Missing Glassdoor Rating – Balancing Check

| | Overall | Comp. & Benef. | Career Opps. | Culture | Leadership | Work-Life |
|---|---|---|---|---|---|---|
| FB Access | 0.003 | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 |
| | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.208 | 0.212 | 0.212 | 0.213 | 0.212 | 0.212 |
| Observations | 16,084 | 16,084 | 16,084 | 16,084 | 16,084 | 16,084 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Missing amenity: indicator for working at an employer with no Glassdoor rating in the respective dimension. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A18: Facebook Access and the College-to-Work Transition – Other Job Types

| | Non-High-Wage Firm | Non-High-Career-Support Firm | Non-High-Wage Occupation | Non-Steep Wage Growth | Non-Sr. Leadership Track |
|---|---|---|---|---|---|
| FB Access | -0.004 | -0.004 | -0.001 | 0.012* | -0.003 |
| | (0.005) | (0.005) | (0.005) | (0.005) | (0.005) |
| Dep. Var. Mean | 0.446 | 0.402 | 0.402 | 0.483 | 0.441 |
| Observations | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Non-High-Wage Firm: indicator for working at a firm not in the upper employment-weighted quartile of the average median log earnings distribution at the firm level. Non-High-Career-Support Firm: indicator for working at a firm not in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Non-High-Wage Occupation: indicator for working in an occupation not in the upper employment-weighted quartile of the average median log earnings distribution at the 5-digit SOC level. Non-Steep Wage Growth: indicator for an early-career job with a leave-one-cohort-out 15-year wage growth prediction at or below the 75th percentile. Non-Sr. Leadership Track: indicator for an early-career job with a leave-one-cohort-out prediction of reaching senior leadership 15 years after at or below the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A19: Facebook Access, Education Outcomes, and Time to First Stable Job

| | | | Postgraduate Degree | | | |
|---|---|---|---|---|---|---|
| | Study Duration | In College | Any | Business & Economics | MBA | Time to First Stable Job |
| FB Access | 0.007 | -0.004 | 0.0007 | 0.0007 | -0.004 | -0.055** |
| | (0.012) | (0.002) | (0.005) | (0.005) | (0.004) | (0.020) |
| Dep. Var. Mean | 3.83 | 0.055 | 0.326 | 0.259 | 0.147 | 1.49 |
| Observations | 8,042 | 16,084 | 8,042 | 8,037 | 8,042 | 8,042 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Study Duration: total years from enrollment to completion of the bachelor's degree. Time to First Stable Job: time to the first active employment spell after expected graduation lasting at least 4 months, following Kramarz and Skans (2014). In College: indicator whether students report being enrolled in the first two years after expected graduation. Postgraduate Degree: reporting a master's, PhD, or professional degree as the highest degree on the LinkedIn profile. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A20: Facebook Access and Early-Career Job Help in the College Network – Other Job Types

| | Non-High-Wage Firm | Non-High-Career-Support Firm | Non-High-Wage Occupation | Non-Steep Wage Growth | Non-Sr. Leadership Track |
|---|---|---|---|---|---|
| *Panel A: Early-Career Job via College Network* | | | | | |
| FB Access | 0.004 | 0.004 | 0.007*** | 0.006** | 0.003 |
| | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| Dep. Var. Mean | 0.066 | 0.075 | 0.064 | 0.081 | 0.061 |
| Observations | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| *Panel B: Early-Career Job without College Network* | | | | | |
| FB Access | -0.008 | -0.008 | -0.008 | 0.006 | -0.006 |
| | (0.005) | (0.005) | (0.005) | (0.005) | (0.005) |
| Dep. Var. Mean | 0.380 | 0.327 | 0.338 | 0.402 | 0.380 |
| Observations | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Complement outcomes to Table 4. Panel A: lower-quality early-career job at a firm with a college peer already working there. Panel B: the same job at a firm without a college peer already working there. The two panels partition the corresponding outcome in Table A18, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Non-High-Career-Support Firm: indicator for working at a firm not in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A21: Facebook Access and Early-Career Jobs via College Network by Peer Type

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Same-Cohort Peers* | | | | | | |
| FB Access | 0.002 | 0.001 | 0.002 | -0.0008 | 0.002 | 0.001 |
| | (0.002) | (0.0009) | (0.001) | (0.001) | (0.0010) | (0.001) |
| Dep. Var. Mean | 0.039 | 0.016 | 0.020 | 0.015 | 0.010 | 0.018 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| *Panel B: Alumni* | | | | | | |
| FB Access | 0.010*** | 0.006*** | 0.008*** | 0.004* | 0.004** | 0.007*** |
| | (0.002) | (0.001) | (0.002) | (0.002) | (0.001) | (0.002) |
| Dep. Var. Mean | 0.093 | 0.038 | 0.045 | 0.037 | 0.024 | 0.042 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| *Panel C: Below-Cohort Peers* | | | | | | |
| FB Access | 0.004* | 0.003** | 0.003 | -0.001 | 0.003*** | 0.002* |
| | (0.002) | (0.0009) | (0.001) | (0.001) | (0.0009) | (0.001) |
| Dep. Var. Mean | 0.040 | 0.014 | 0.017 | 0.015 | 0.011 | 0.015 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Alumni: students 1 to 3 classes above the expected graduation cohort. Below-cohort peers: students 1 to 3 classes below the expected graduation cohort. Same-cohort peers: students in the expected graduation cohort. Any job: indicator for joining a firm where a college peer of the respective type had already worked for at least one year, following Gee *et al.* (2017). High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distribution at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distribution at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching senior leadership 15 years after above the 75th percentile. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A22: Facebook Access, the College-to-Work Transition, and the Role of Job Location

| | Employment by Job Location | | Accounting for Geographic Job Sorting | | | |
|---|---|---|---|---|---|---|
| | Outside College's MSA | Outside College's MSA: Large Metro Area | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Sr. Leadership Track |
| FB Access | 0.005 | 0.009 | 0.012** | 0.014*** | 0.011** | 0.013*** |
| | (0.005) | (0.005) | (0.004) | (0.004) | (0.004) | (0.004) |
| Share Explained by Geographic Job Sorting | | | 14% | 1% | 6% | 6% |
| Dep. Var. Mean | 0.419 | 0.284 | 0.206 | 0.159 | 0.249 | 0.201 |
| Observations | 16,084 | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |
| Job Location FE | | | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Job location is defined at the level of metropolitan statistical areas (MSAs, 2009 delineation published by the U.S. Census Bureau), at the state level outside MSAs, and at the country level abroad. Columns 1 and 2: indicators for working outside the college's MSA (or non-metropolitan part of the college's state) and outside it in a large metropolitan area with at least one million residents in 2010 (Ingram and Franco 2014). Non-employed students are coded as zero. Columns 3–6 account for geographic job sorting by residualizing the high-quality job indicators of Table 3 that showed significant treatment effects at the 5% level (see the definitions there): I estimate job-location fixed effects among employed students in the not-yet-treated college cohorts (conditional on the baseline fixed effects and trends), subtract each employed student's job-location fixed effect net of its mean in that group, and run the regression on the residualized indicators. Share explained by geographic job sorting: one minus the ratio of the residualized estimate to the corresponding estimate in Table 3. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs, U.S. Census Bureau.

Table A23: Facebook Access and Between-Firm Job Changes

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Any Between-Firm Switch* | | | |
| FB Access | 0.001 | -0.002 | -0.001 |
| | (0.002) | (0.002) | (0.002) |
| Dep. Var. Mean | 0.112 | 0.123 | 0.111 |
| Observations | 40,210 | 48,249 | 56,294 |
| *Panel B: Between-Firm Switches (No Promotion)* | | | |
| FB Access | 0.0006 | -0.002 | -0.0002 |
| | (0.001) | (0.001) | (0.001) |
| Dep. Var. Mean | 0.057 | 0.068 | 0.065 |
| Observations | 40,210 | 48,249 | 56,294 |
| *Panel C: Job via College Network (No Promotion)* | | | |
| FB Access | 0.0004 | -0.008** | -0.001 |
| | (0.002) | (0.003) | (0.003) |
| Dep. Var. Mean | 0.108 | 0.125 | 0.106 |
| Observations | 40,210 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. Panel A: share of graduates who change employers in the given period. Panel B: share of graduates who change employers without a relative real annual wage growth jump of at least 10 log points. Panel C: share of graduates who change to a firm where a college peer had already worked, without a relative real annual wage growth jump of above 10 log points. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A24: Facebook Access, Job Stability, and Within-Firm Mobility

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Unstable Worker-Firm Match (<6 mo.)* | | | |
| FB Access | 0.0001 | -0.0010 | -0.0005 |
| | (0.001) | (0.001) | (0.001) |
| Dep. Var. Mean | 0.080 | 0.071 | 0.061 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel B: Within-Firm Lateral Move* | | | |
| FB Access | 0.0006 | 0.001* | -0.0007 |
| | (0.0005) | (0.0006) | (0.0005) |
| Dep. Var. Mean | 0.010 | 0.015 | 0.015 |
| Observations | 40,210 | 48,249 | 56,294 |
| *Panel C: Within-Firm Demotion* | | | |
| FB Access | 0.0005 | 0.0007 | 0.0003 |
| | (0.0005) | (0.0007) | (0.0005) |
| Dep. Var. Mean | 0.009 | 0.014 | 0.015 |
| Observations | 40,210 | 48,249 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Panel A: share of graduates with a current employer tenure of less than 6 months. Panels B–C start in year 2 as within-firm transitions are only defined for ongoing employment spells. Panel B: share of graduates who transition to a different role at the same employer with a relative real annual wage growth jump between -10 and +10 log points. Panel C: share of graduates who transition to a role at the same employer with relative real annual wage growth jump below -10 log points. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A25: Facebook Access and High-Wage Firm Transitions

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: High-Wage Firm* | | | |
| FB Access | 0.013*** | -0.001 | -0.005 |
| | (0.004) | (0.005) | (0.005) |
| Dep. Var. Mean | 0.251 | 0.325 | 0.342 |
| Observations | 48,252 | 48,250 | 56,294 |
| *Panel B: Entry into High-Wage Firm* | | | |
| FB Access | 0.0007 | -0.002* | 0.0004 |
| | (0.001) | (0.0008) | (0.0006) |
| Dep. Var. Mean | 0.051 | 0.037 | 0.029 |
| Observations | 40,210 | 48,250 | 56,294 |
| *Panel C: Exit from High-Wage Firm* | | | |
| FB Access | 0.0009 | 0.0007 | -0.00001 |
| | (0.0008) | (0.0006) | (0.0007) |
| Dep. Var. Mean | 0.029 | 0.031 | 0.027 |
| Observations | 40,210 | 48,250 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Panel A: share of graduates employed at a high-wage firm (upper employment-weighted quartile of the firm wage distribution). Panel B: share of graduates who transition from a non-high-wage to a high-wage firm. Panel C: share of graduates who transition from a high-wage to a non-high-wage firm. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A26: Multiple Hypothesis Testing Corrections

| | Coefficient | Unadj. $p$-value | BH (1995) $p$-value |
|---|---|---|---|
| *Panel A: Career Development (Table 2)* | | | |
| Job-Based Real Annual Earnings [1–6Y] | 1,286** | 0.007 | 0.022 |
| Job-Based Real Annual Earnings [7–12Y] | 1,343* | 0.016 | 0.037 |
| Job-Based Real Annual Earnings [13–19Y] | 768 | 0.237 | 0.328 |
| Job-Based Promotion (≥10 log pts.) [2–6Y] | 0.004** | 0.008 | 0.023 |
| Job-Based Promotion (≥10 log pts.) [7–12Y] | 0.002 | 0.092 | 0.146 |
| Job-Based Promotion (≥10 log pts.) [13–19Y] | -0.0010 | 0.326 | 0.441 |

| | Coefficient | Unadj. $p$-value | BH (1995) $p$-value |
|---|---|---|---|
| Senior Leadership Positions [1–6Y] | 0.002 | 0.373 | 0.479 |
| Senior Leadership Positions [7–12Y] | 0.006* | 0.037 | 0.077 |
| Senior Leadership Positions [13–19Y] | 0.008* | 0.021 | 0.047 |
| *Panel B: College-to-Work Transition (Table 3)* | | | |
| Employment [1–2Y] | 0.010** | 0.006 | 0.020 |
| High-Wage Firm [1–2Y] | 0.014*** | 0.0002 | 0.002 |
| High-Career-Support Firm [1–2Y] | 0.014*** | 0.0002 | 0.001 |
| High-Wage Occupation [1–2Y] | 0.012** | 0.005 | 0.020 |
| Steep Wage Growth [1–2Y] | -0.001 | 0.797 | 0.878 |
| Sr. Leadership Track [1–2Y] | 0.014*** | 0.0005 | 0.003 |
| *Panel C: Early-Career Job Help in the College Network (Table 4)* | | | |
| Job Help × Any Job [1–2Y] | 0.011*** | 0.00002 | 0.0003 |
| Job Help × High-Wage Firm [1–2Y] | 0.007*** | 0.00001 | 0.0003 |
| Job Help × High-Career-Support Firm [1–2Y] | 0.008*** | 0.00002 | 0.0003 |
| Job Help × High-Wage Occupation [1–2Y] | 0.004* | 0.042 | 0.081 |
| Job Help × Steep Wage Growth [1–2Y] | 0.004** | 0.001 | 0.005 |
| Job Help × Sr. Leadership [1–2Y] | 0.007*** | 0.000002 | 0.0001 |
| No Job Help × Any Job [1–2Y] | -0.0004 | 0.922 | 0.940 |
| No Job Help × High-Wage Firm [1–2Y] | 0.008* | 0.022 | 0.047 |
| No Job Help × High-Career-Support Firm [1–2Y] | 0.006 | 0.061 | 0.106 |
| No Job Help × High-Wage Occupation [1–2Y] | 0.008* | 0.045 | 0.084 |
| No Job Help × Steep Wage Growth [1–2Y] | -0.005 | 0.192 | 0.272 |
| No Job Help × Sr. Leadership [1–2Y] | 0.007 | 0.072 | 0.122 |
| *Panel D: Firm Mobility and Long-Run Job Help in the College Network (Table 5)* | | | |
| Between-Firm Promotion (≥10 log pts.) [2–6Y] | 0.0005 | 0.719 | 0.826 |
| Between-Firm Promotion (≥10 log pts.) [7–12Y] | 0.0003 | 0.758 | 0.852 |
| Between-Firm Promotion (≥10 log pts.) [13–19Y] | -0.001 | 0.109 | 0.168 |
| Job Help × Promotions [2–6Y] | -0.00009 | 0.825 | 0.891 |
| Job Help × Promotions [7–12Y] | 0.0005 | 0.178 | 0.260 |
| Job Help × Promotions [13–19Y] | -0.00004 | 0.883 | 0.917 |

| | Coefficient | Unadj. $p$-value | BH (1995) $p$-value |
|---|---|---|---|
| Job Help × Sr. Leadership [2–6Y] | -0.00007 | 0.676 | 0.811 |
| Job Help × Sr. Leadership [7–12Y] | 0.0002 | 0.369 | 0.479 |
| Job Help × Sr. Leadership [13–19Y] | 0.000006 | 0.963 | 0.963 |
| *Panel E: Worker-Firm Match Stability and Productivity (Table 6)* | | | |
| Stable Worker-Firm Match (≥6 Months) [1–6Y] | 0.007* | 0.013 | 0.031 |
| Stable Worker-Firm Match (≥6 Months) [7–12Y] | 0.001 | 0.584 | 0.717 |
| Stable Worker-Firm Match (≥6 Months) [13–19Y] | -0.0004 | 0.883 | 0.917 |
| Firm Tenure [1–6Y] | 0.031** | 0.003 | 0.014 |
| Firm Tenure [7–12Y] | 0.070* | 0.012 | 0.030 |
| Firm Tenure [13–19Y] | 0.083 | 0.077 | 0.126 |
| Within-Firm Promotion (≥10 log pts.) [2–6Y] | 0.003*** | 0.0001 | 0.001 |
| Within-Firm Promotion (≥10 log pts.) [7–12Y] | 0.002* | 0.041 | 0.081 |
| Within-Firm Promotion (≥10 log pts.) [13–19Y] | 0.0002 | 0.699 | 0.821 |
| *Panel F: Firm Sorting and Human Capital Accumulation (Table 7)* | | | |
| Exp. at High-Wage Firm [6Y] | 0.080*** | 0.0007 | 0.004 |
| Exp. at High-Wage Firm [12Y] | 0.076 | 0.123 | 0.184 |
| Exp. at High-Wage Firm [19Y] | 0.044 | 0.583 | 0.717 |
| Exp. at High-Career-Support Firm [6Y] | 0.054** | 0.001 | 0.005 |
| Exp. at High-Career-Support Firm [12Y] | 0.105** | 0.002 | 0.011 |
| Exp. at High-Career-Support Firm [19Y] | 0.145** | 0.007 | 0.022 |
| Work Experience [6Y] | 0.048** | 0.005 | 0.020 |
| Work Experience [12Y] | 0.054** | 0.009 | 0.024 |
| Work Experience [19Y] | 0.043 | 0.053 | 0.095 |

*Notes:* Unadjusted and adjusted p-values for each coefficient in the main results section. Adjusted p-values are computed using the Benjamini and Hochberg (1995) adjustment procedure. *Data source:* Revelio Labs.

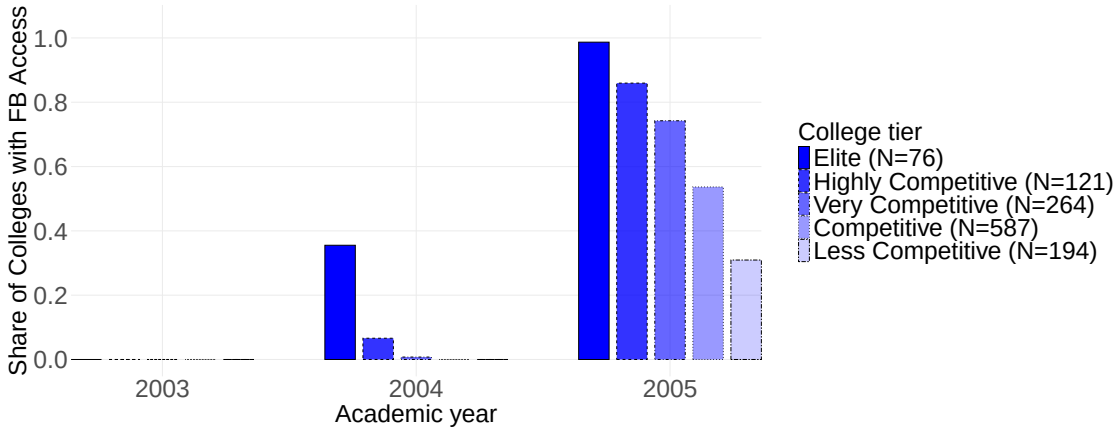



Figure A1: Facebook Rollout across Selective U.S. Colleges with a Business and Economics Undergraduate Degree by College Selectivity Tier

*Notes:* Cumulative share of colleges with access to Facebook by the end of the respective academic year, by college selectivity tier. College selectivity is measured using Barron's selectivity index (Barron's Educational Series 2008). N: number of colleges per selectivity tier. *Data source:* Facebook introduction dates from Armona (2025).

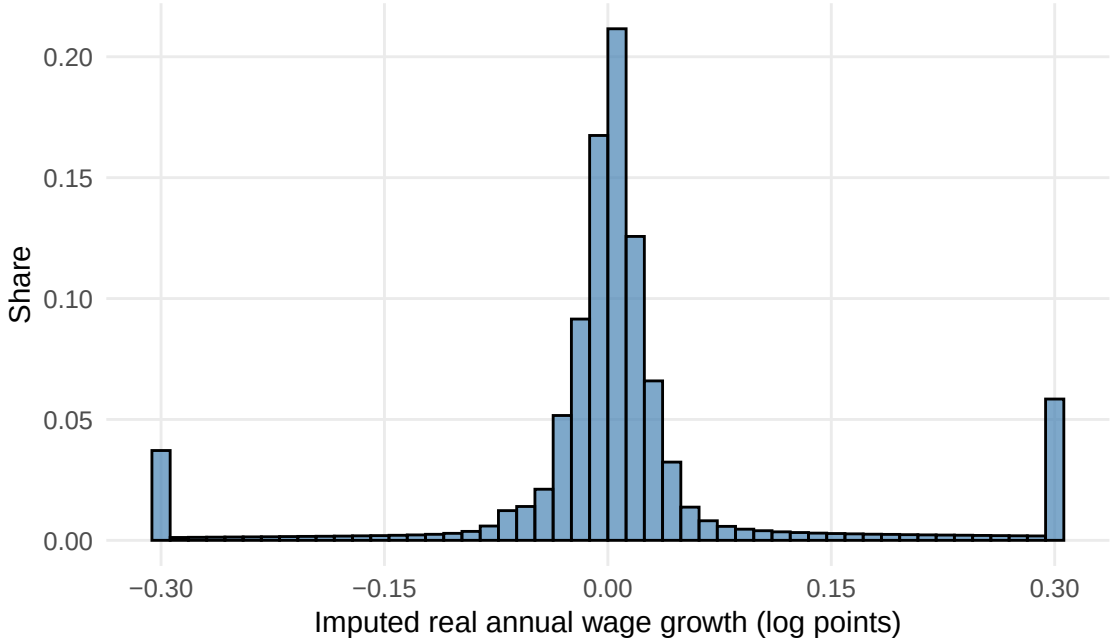



Figure A2: Imputed Real Annual Wage Growth Distribution

# Appendix B: Event-Study Specification

To assess the plausibility of the parallel trends assumption, I estimate an event-study version of equation (1), where the expected graduation year of the first freshman cohort affected by the introduction of Facebook at a given college is referred to as the *event*. The event-study model in a two-way fixed effects (TWFE) setting can be specified as follows:

$$Y_{fs,c,t} = \sum_{k=-5}^{0} \mathbf{1}\{t_{fs,c} - t_c = k\}\, \beta_k + \delta_{fs,\text{tier,region}} + \phi_c + \epsilon_{fs,c,t}, \tag{2}$$

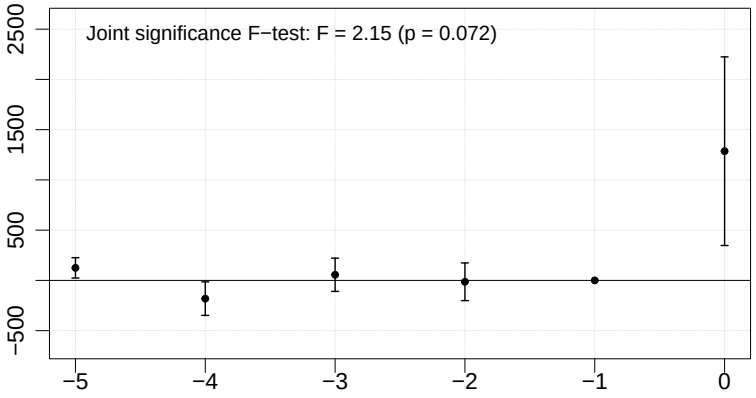
where $t_c$ corresponds to the *event* in college $c$ and $t_{fs,c}$ to the expected graduation year of freshman cohort $fs$ in college $c$. The remaining components of equation (2) are the same as in the baseline model. The event-study specification allows me to investigate nonlinear pre-trends in the career outcome variables between treatment and control cohorts. To allow for separate identification of the dynamic treatment effect coefficients $\beta_k$ and the cohort fixed effects in a setup that uses only not-yet-treated cohorts as comparison groups, I follow the recommendation by Schmidheiny and Siegloch (2023) and bin both endpoints of the event-time window. To be precise, I bin the event-time coefficients for $k \leq -6$ with $k = -5$, and for $k = 1$ with $k = 0$. Estimating separate coefficients for $k = 0$ and $k = 1$ would show whether the career benefits of online social network access are stronger for students with longer exposure during college. However, $k = 1$ is only identified for the first 36 colleges, which received access in the academic year 2004 (see Appendix Figure A1), so that a separate estimate is very imprecise.[45] Hence, the event studies in this setting should be primarily seen as plausibility tests of the parallel trends assumption with $\beta_0$ in the event-study model capturing the same pooled ITT estimate as $\alpha_1$ in the baseline specification. Event time $k = -1$ is omitted as the reference category. To ensure that the coefficients are not biased by the strong assumption of treatment effect homogeneity, I estimate the event study using the two-stage heterogeneity-robust estimator proposed by Gardner (2022).

Appendix Figures B1 to B18 show the results. Across all outcome variables, the pre-treatment coefficients are close to zero, and almost all coefficients are individually and jointly insignificant at the 5% level. In the few cases where the null hypothesis of zero pre-treatment coefficients is rejected, the event study figures clearly indicate that the pre-treatment dynamics cannot explain the estimated treatment effects (e.g., Figure B6, Panel (c)). Overall, the results support the parallel-trends assumption and rule

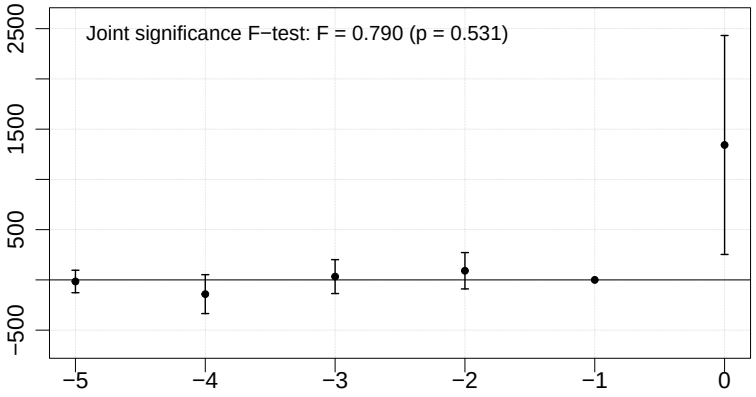
[45] Armona (2025) provides some evidence that longer exposure during college produces larger earnings effects, using a continuous treatment measure in a TWFE model. Consistent with this, estimating $k = 1$ separately generally yields slightly larger but noisy effects, which are statistically different neither from zero nor from the estimates for $k = 0$ (not shown here).

out concerns that the documented treatment effects simply reflect pre-existing differential trends in career outcomes between treated and control cohorts.

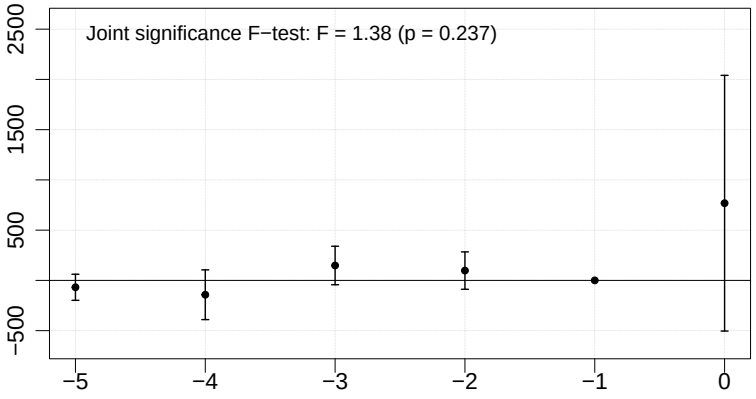
Figure B1: Facebook Access and Job-Based Real Annual Earnings – Event Studies




(a) 1–6 Years

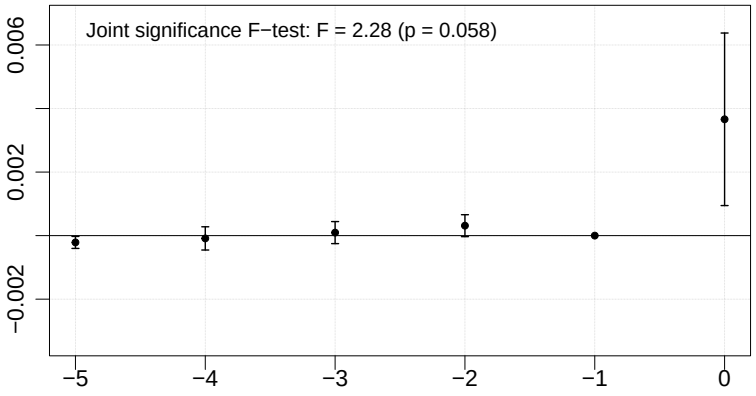



(b) 7–12 Years

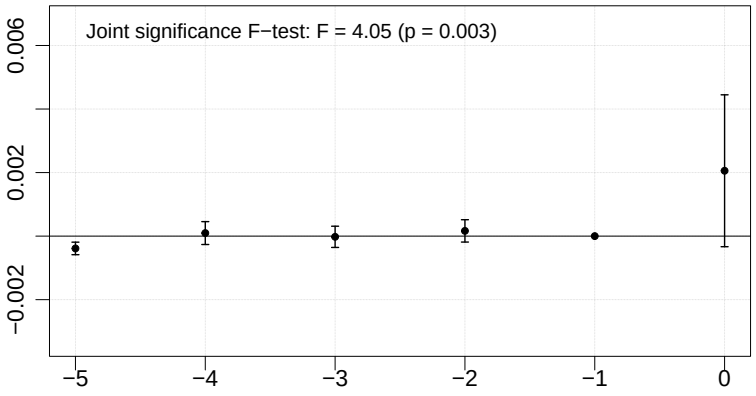



(c) 13–19 Years

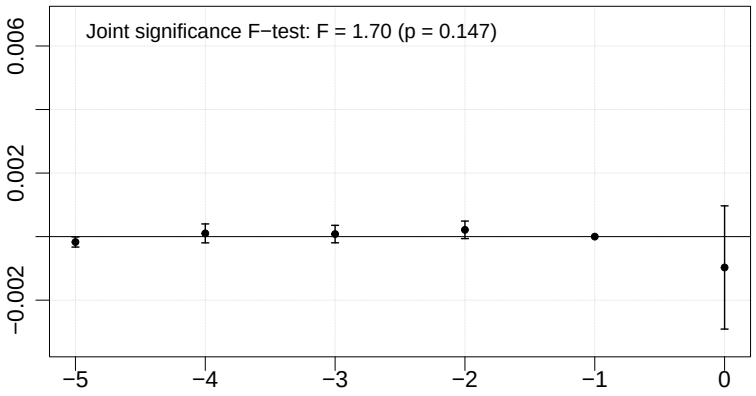
Figure B2: Facebook Access and Job-Based Promotions – Event Studies




(a) 2–6 Years

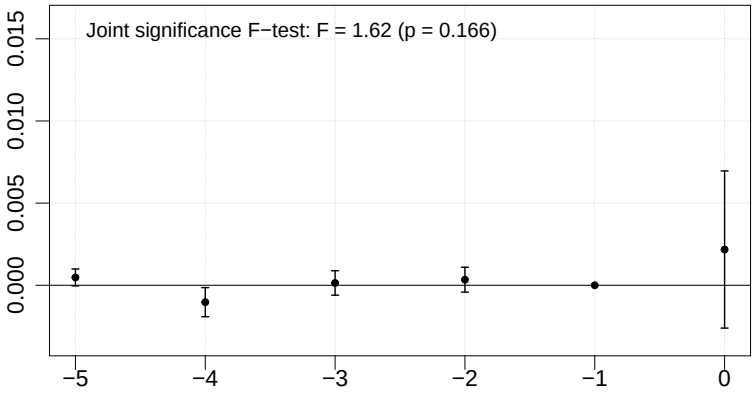



(b) 7–12 Years

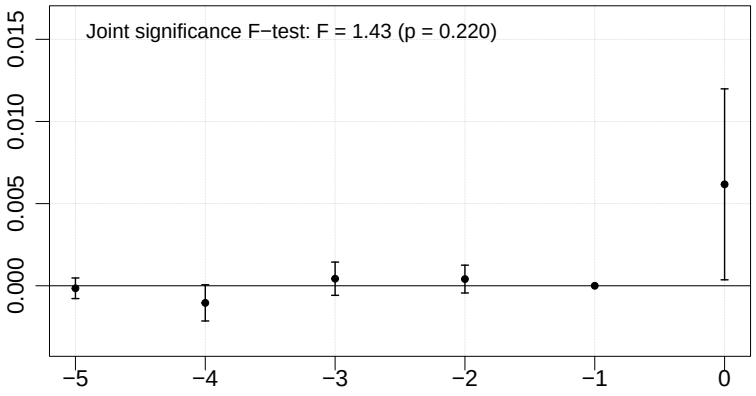



(c) 13–19 Years

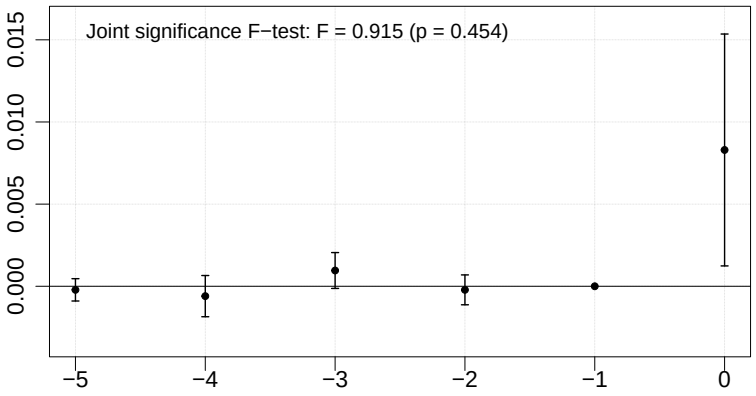
Figure B3: Facebook Access and Senior Leadership Positions – Event Studies




(a) 1–6 Years

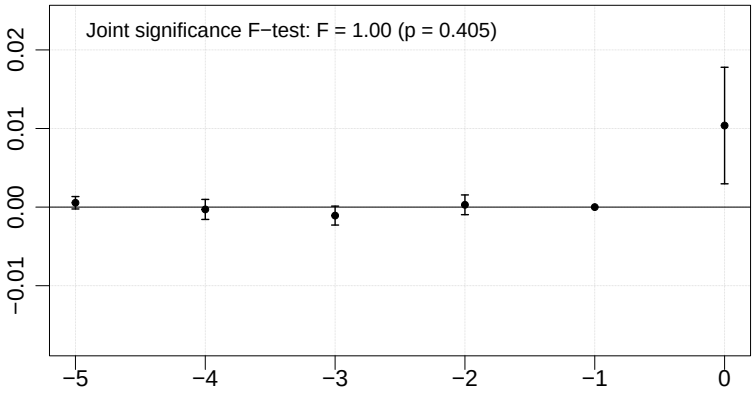



(b) 7–12 Years

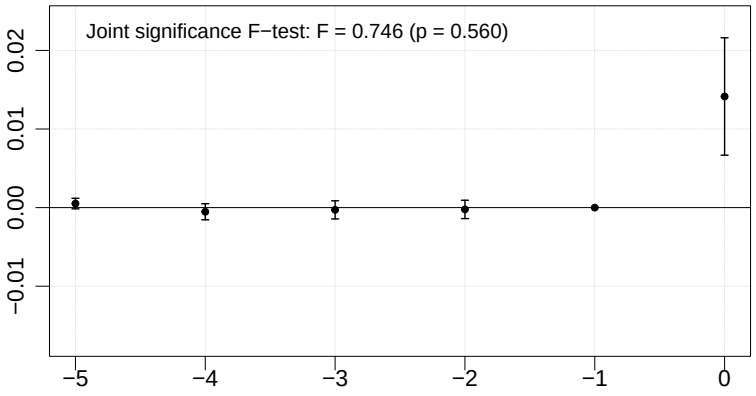



(c) 13–19 Years

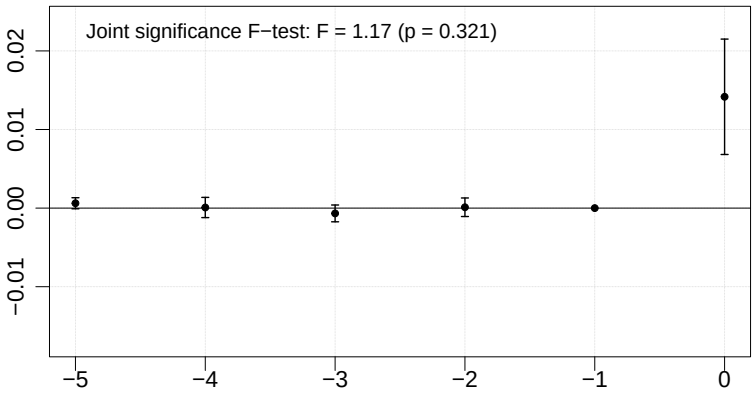
Figure B4: Facebook Access and the College-to-Work Transition (1) – Event Studies




(a) Employment

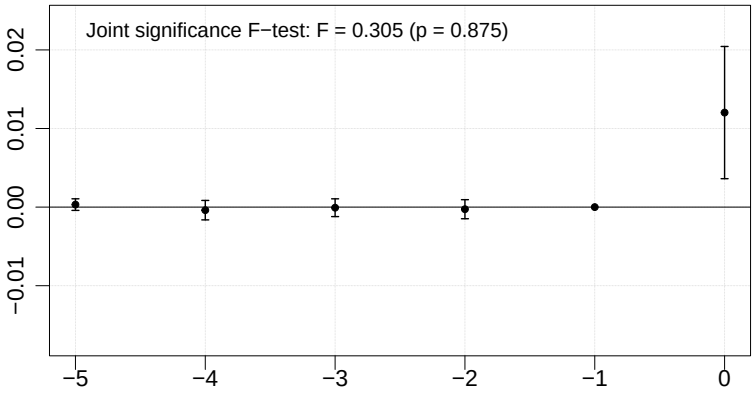



(b) High-Wage Firm

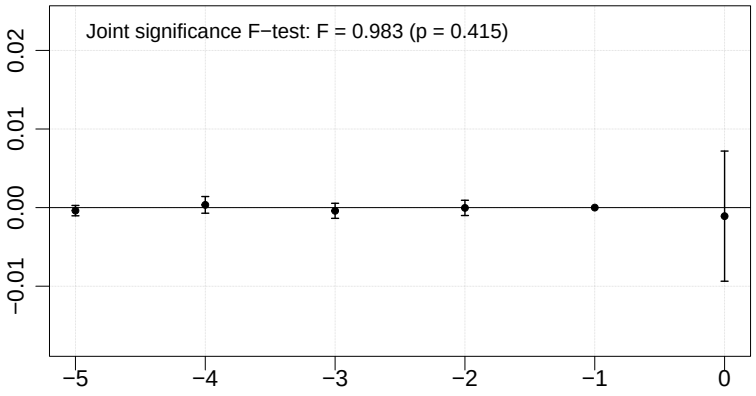



(c) High-Career-Support Firm

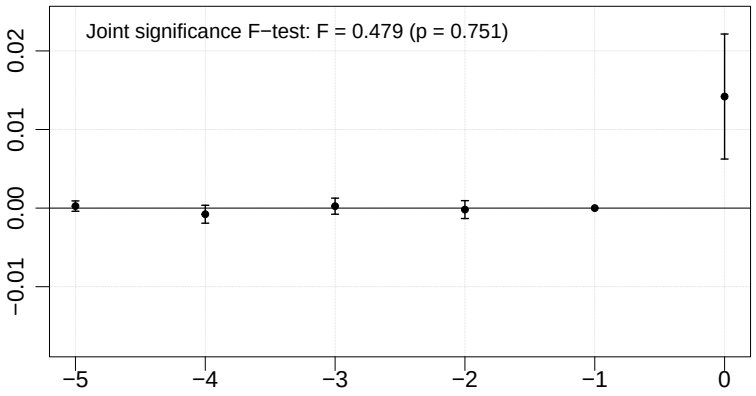
Figure B5: Facebook Access and the College-to-Work Transition (2) – Event Studies




(a) High-Wage Occupation

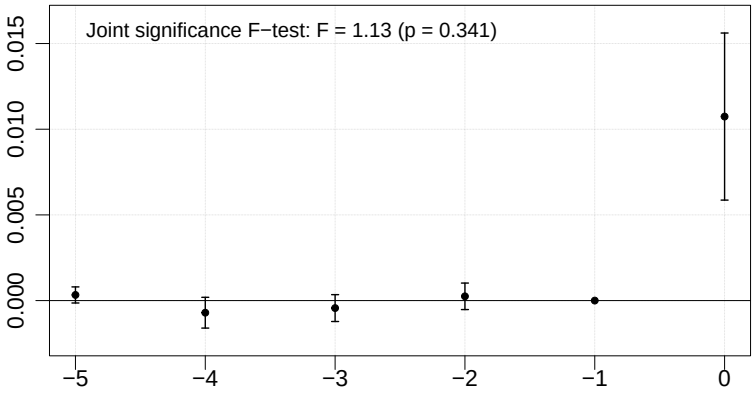



(b) Steep Wage Growth

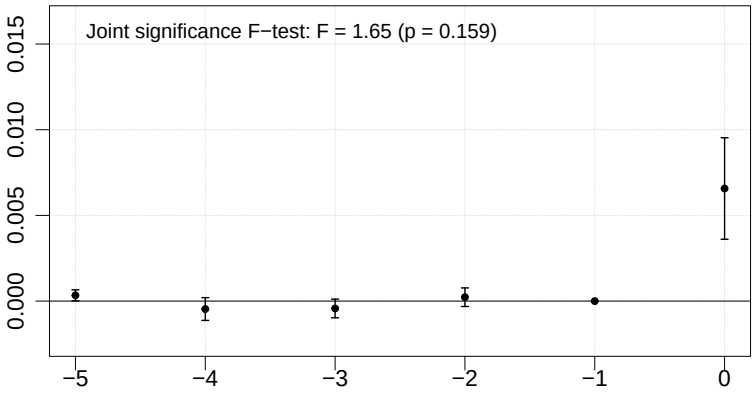



(c) Sr. Leadership Track

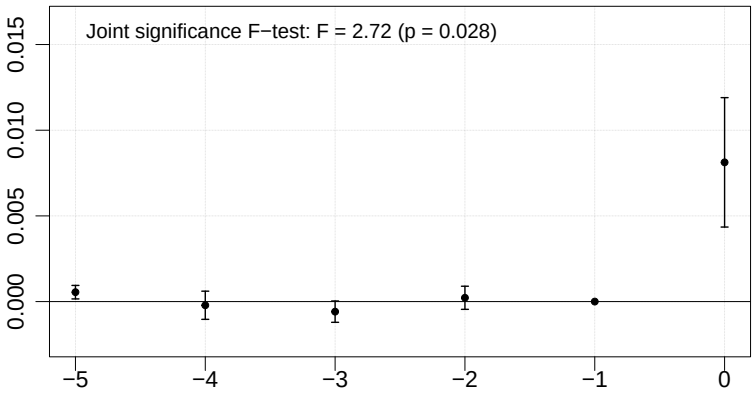
Figure B6: Facebook Access and Early-Career Job Help in the College Network (1) – Event Studies




(a) Job via College Network: Any Job

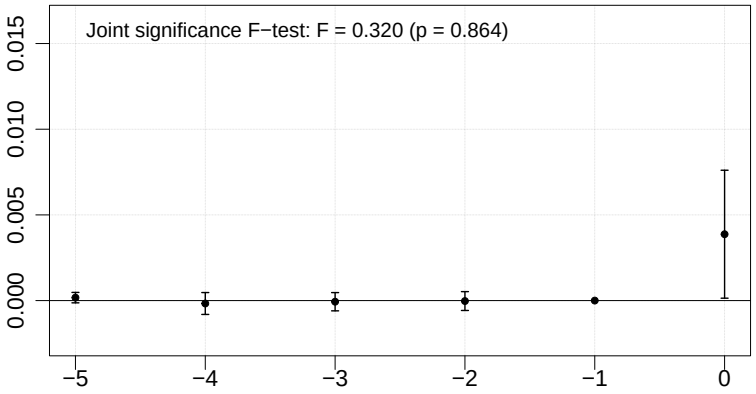



(b) Job via College Network: High-Wage Firm

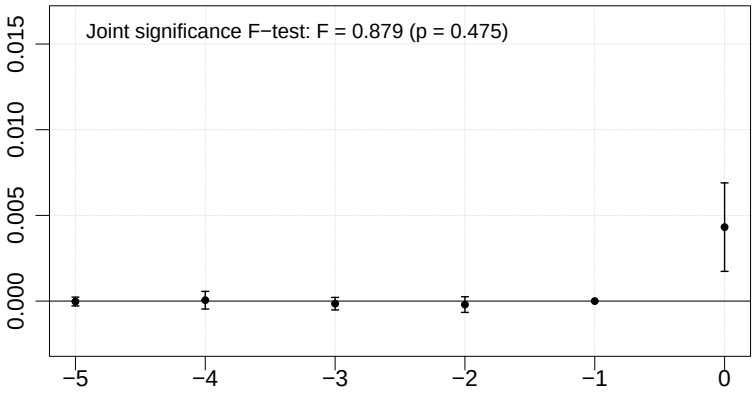



(c) Job via College Network: High-Career-Support Firm

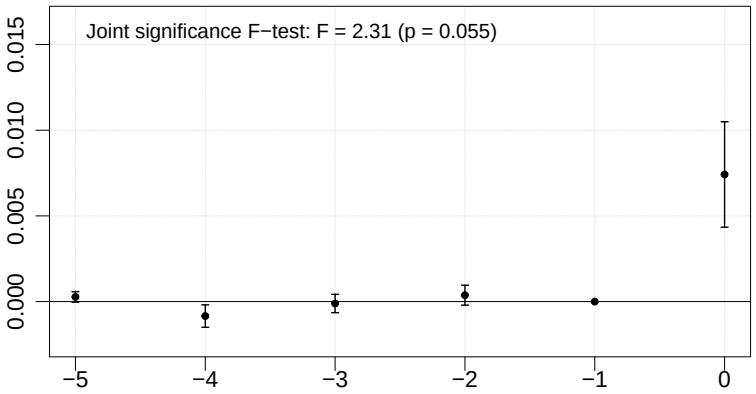
Figure B7: Facebook Access and Early-Career Job Help in the College Network (2) – Event Studies




(a) Job via College Network: High-Wage Occupation

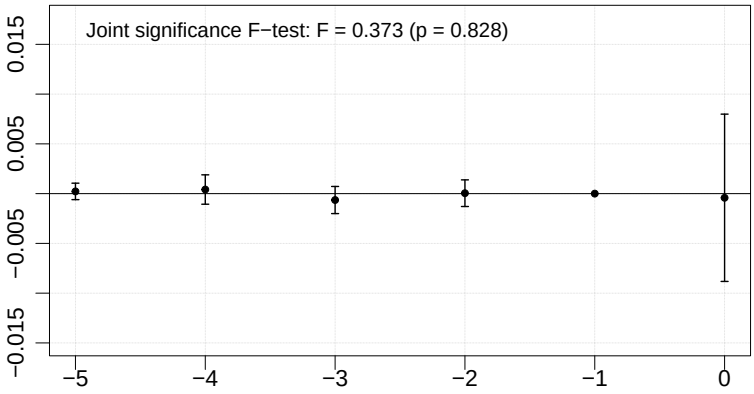



(b) Job via College Network: Steep Wage Growth

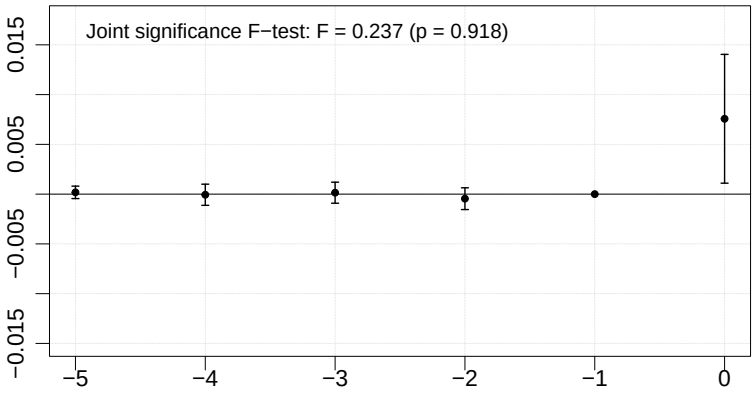



(c) Job via College Network: Sr. Leadership Track

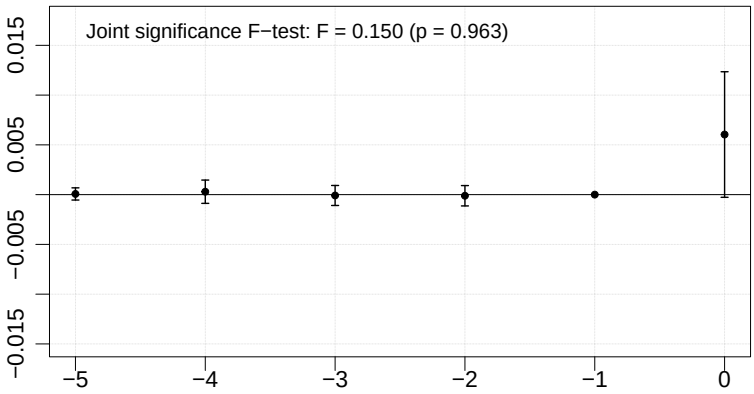
Figure B8: Facebook Access and Early-Career Jobs without College Network Help (1) – Event Studies




(a) Job without College Network: Any Job

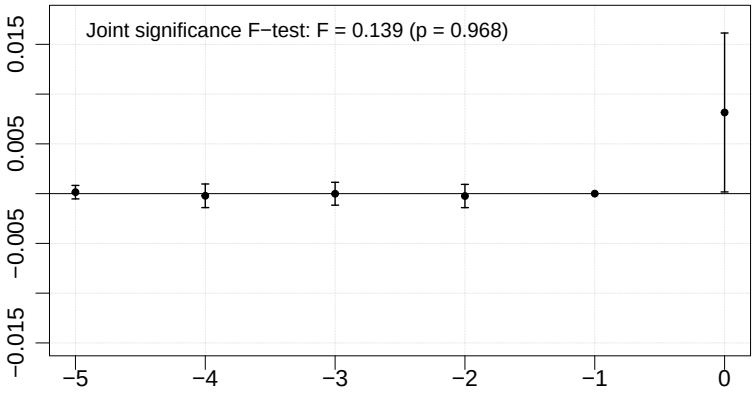



(b) Job without College Network: High-Wage Firm

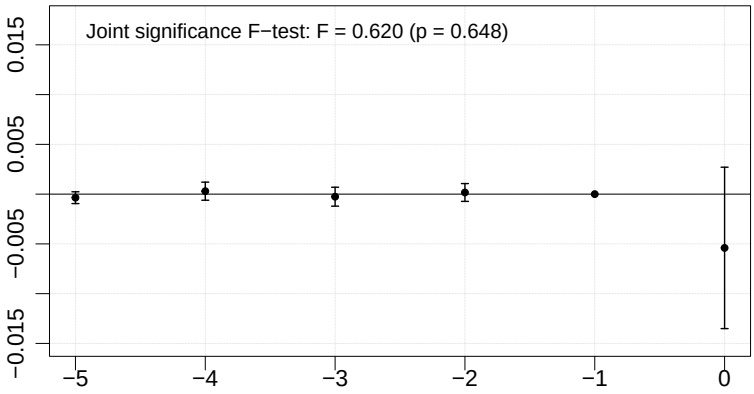



(c) Job without College Network: High-Career-Support Firm

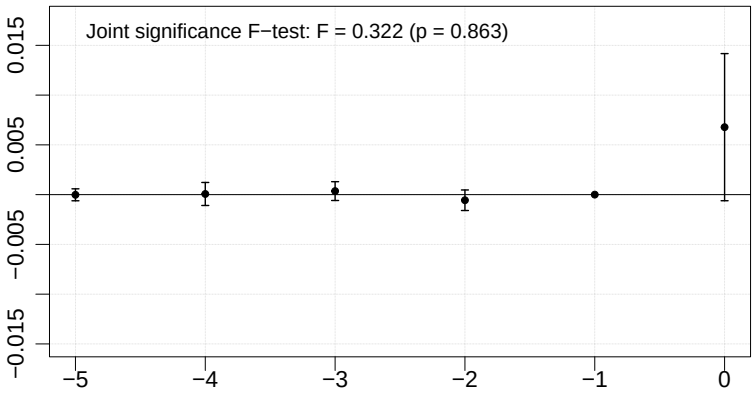
Figure B9: Facebook Access and Early-Career Jobs without College Network Help (2) – Event Studies




(a) Job without College Network: High-Wage Occupation

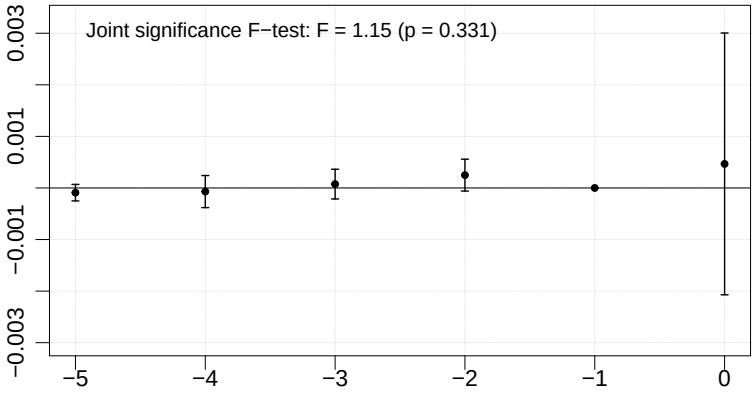



(b) Job without College Network: Steep Wage Growth

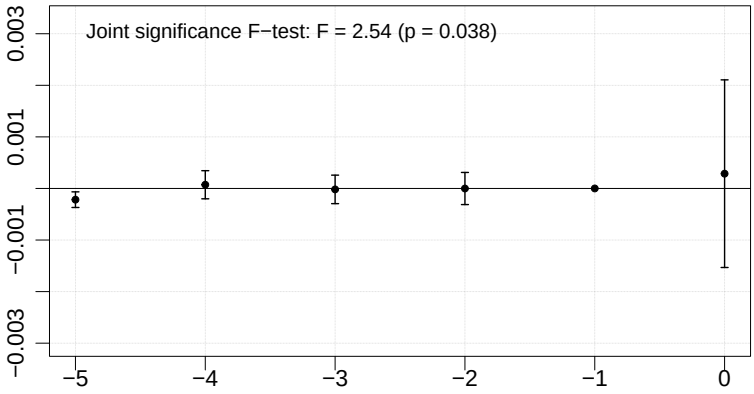



(c) Job without College Network: Sr. Leadership Track

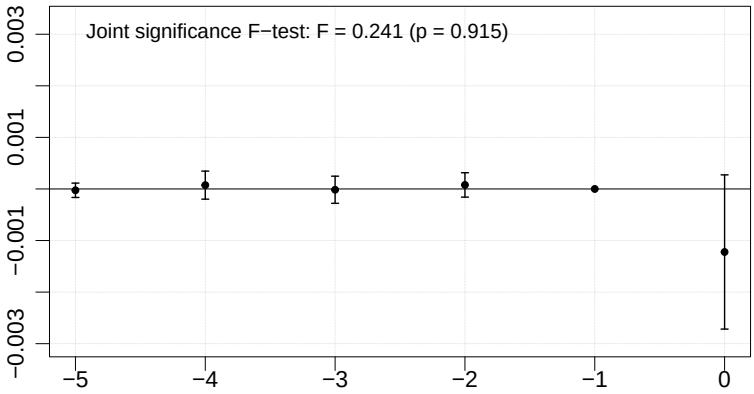
Figure B10: Facebook Access and Between-Firm Promotions – Event Studies




(a) 2–6 Years

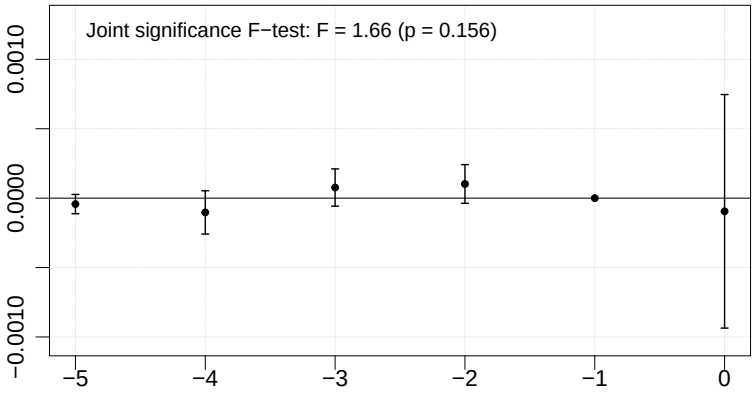



(b) 7–12 Years

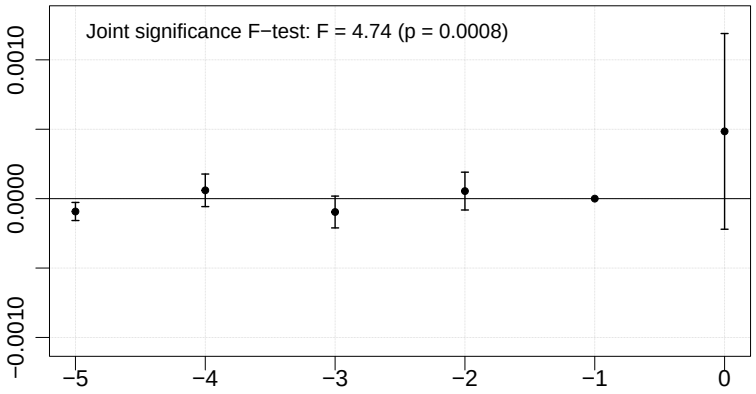



(c) 13–19 Years

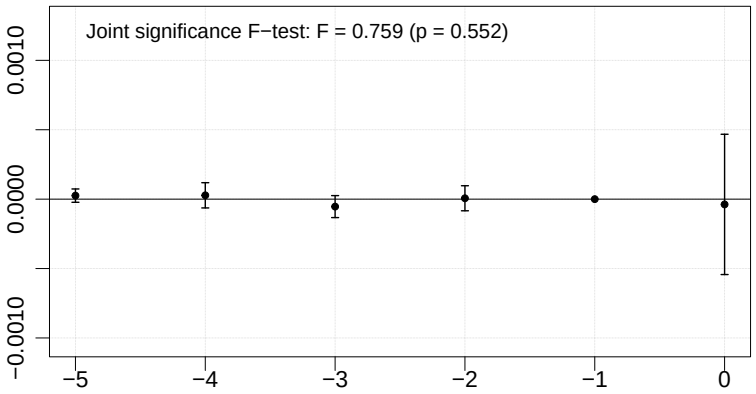
Figure B11: Facebook Access and Promotion via College Network – Event Studies




(a) 2–6 Years

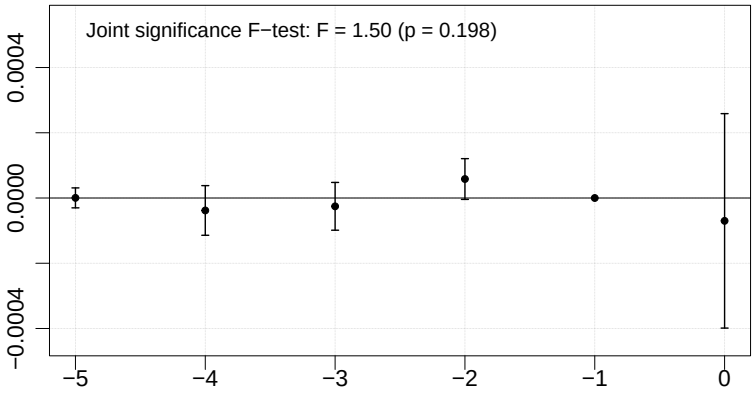



(b) 7–12 Years

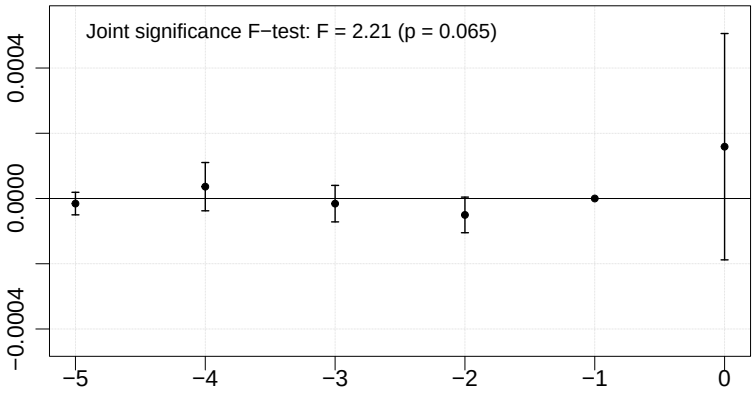



(c) 13–19 Years

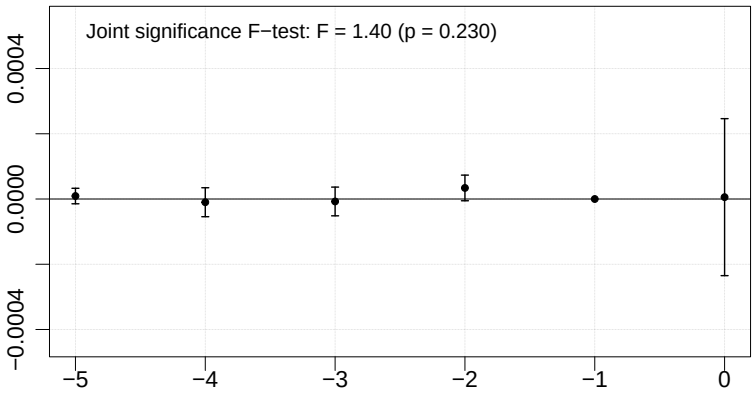
Figure B12: Facebook Access and Senior Leadership Transition via College Network – Event Studies




(a) 2–6 Years

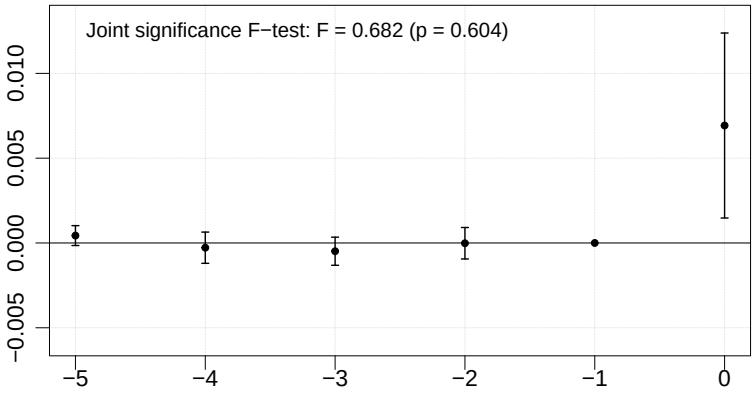



(b) 7–12 Years

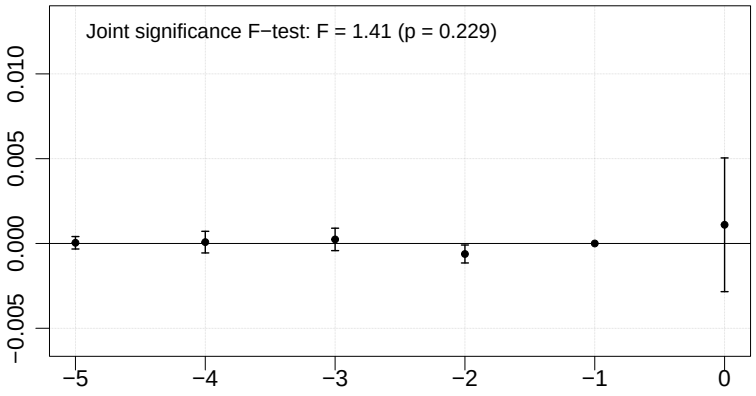



(c) 13–19 Years

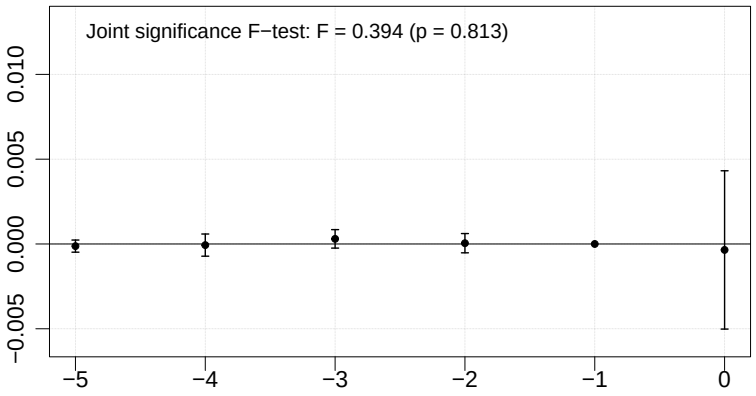
Figure B13: Facebook Access and Stable Worker-Firm Match – Event Studies




(a) 1–6 Years

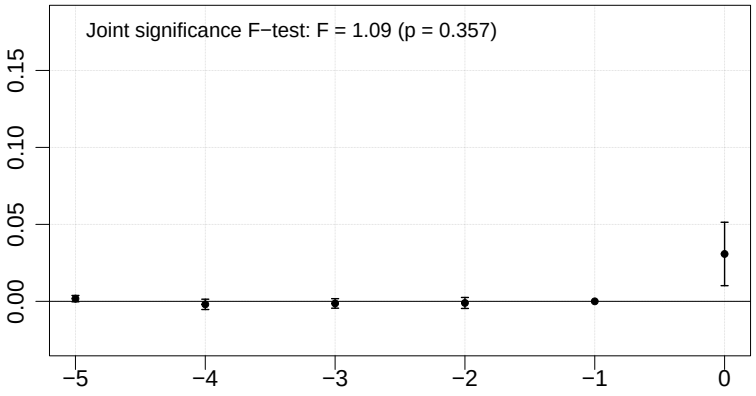



(b) 7–12 Years

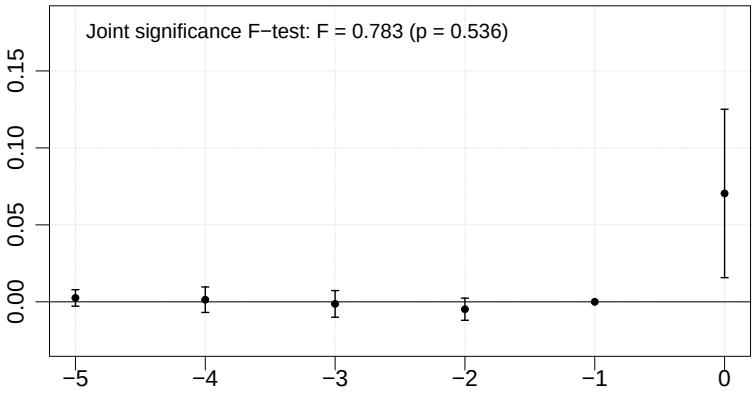



(c) 13–19 Years

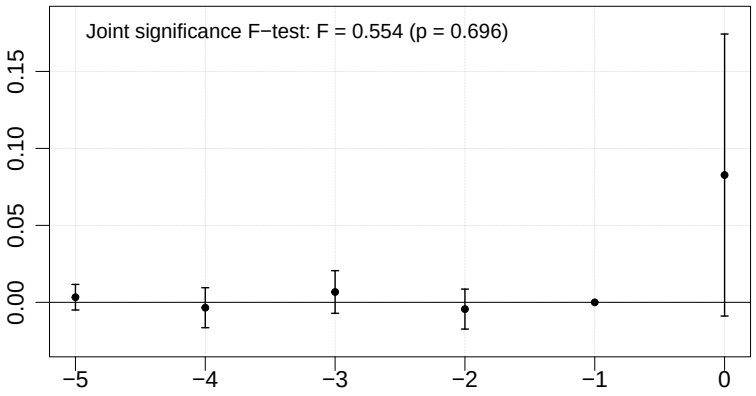
Figure B14: Facebook Access and Current Firm Tenure – Event Studies




(a) 1–6 Years

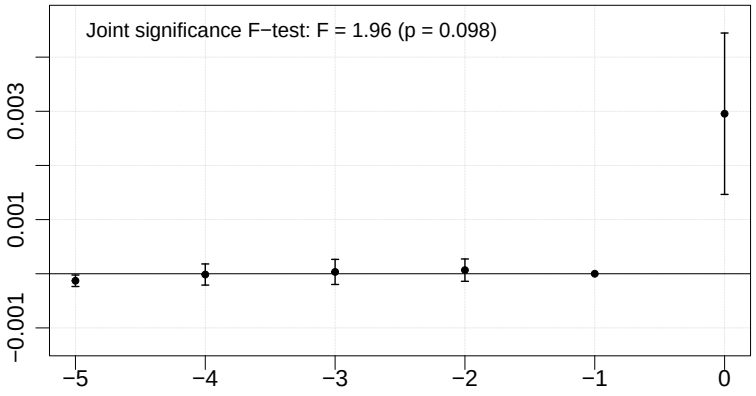



(b) 7–12 Years

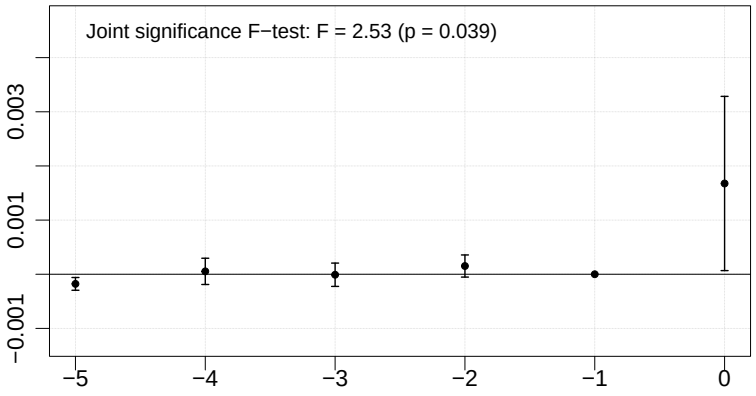



(c) 13–19 Years

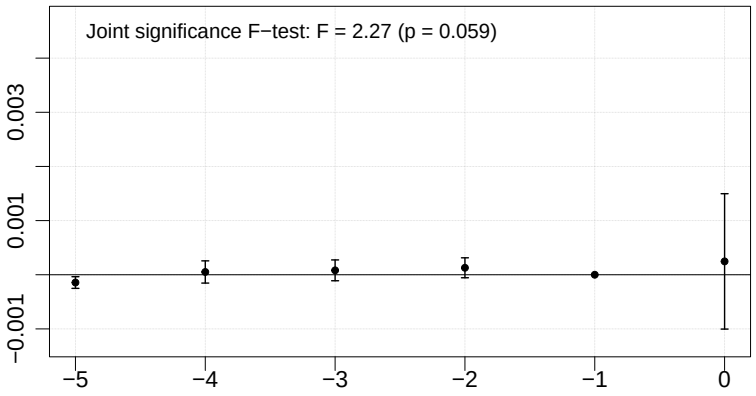
Figure B15: Facebook Access and Within-Firm Promotions – Event Studies




(a) 2–6 Years

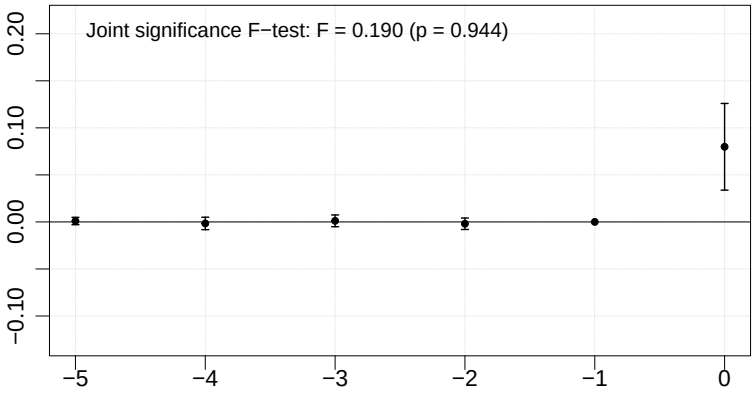



(b) 7–12 Years

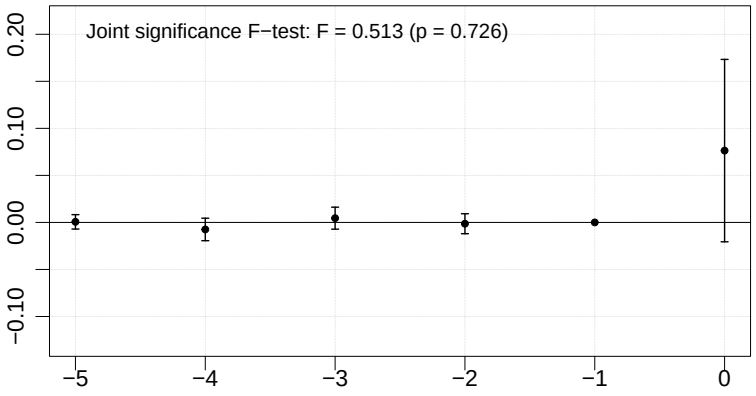



(c) 13–19 Years

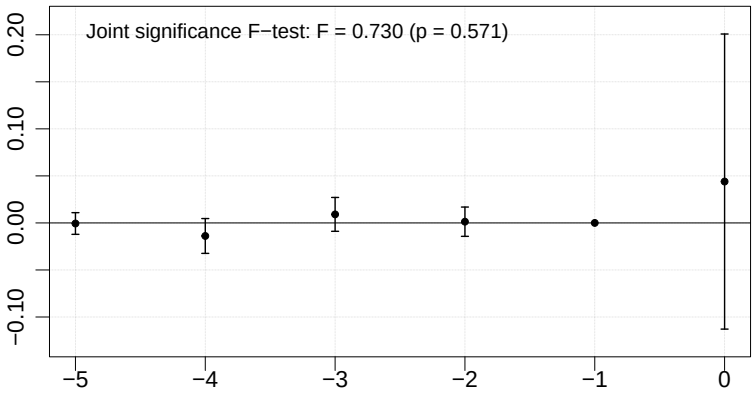
Figure B16: Facebook Access and Work Experience at High-Wage Firms – Event Studies




(a) Year 6

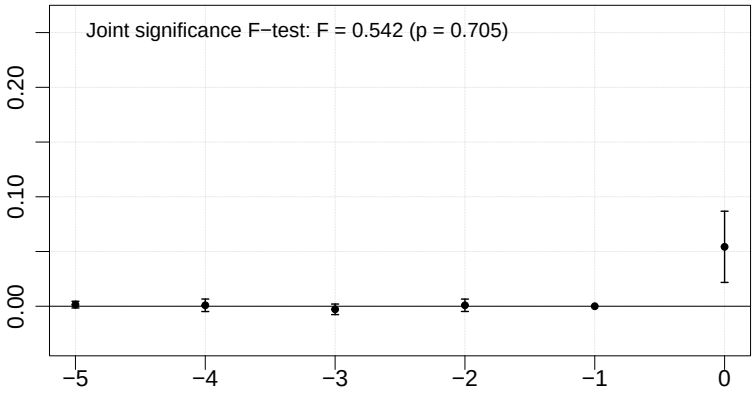



(b) Year 12

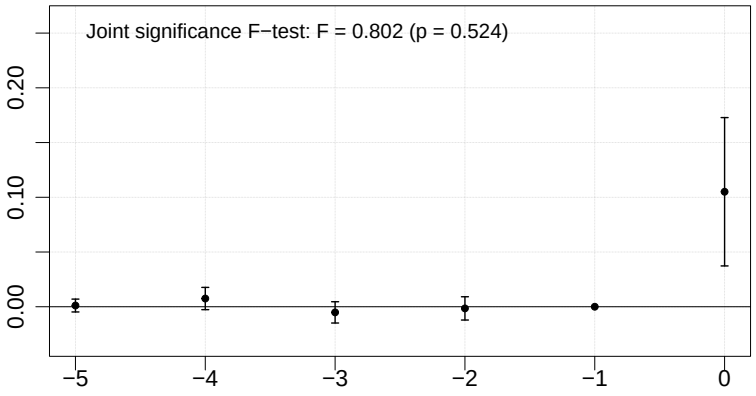



(c) Year 19

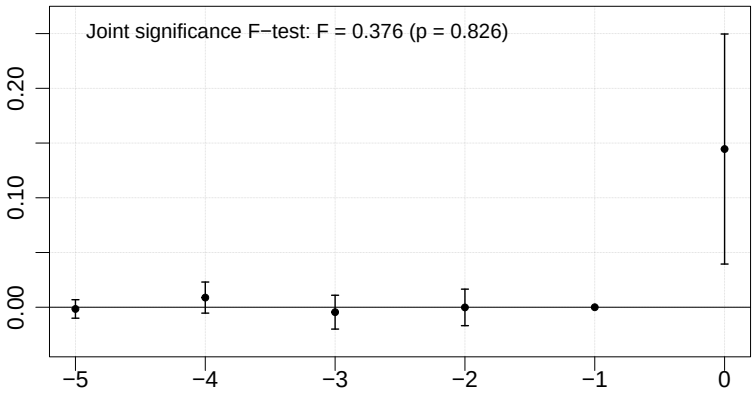
Figure B17: Facebook Access and Work Experience at High-Career-Support Firms – Event Studies




(a) Year 6

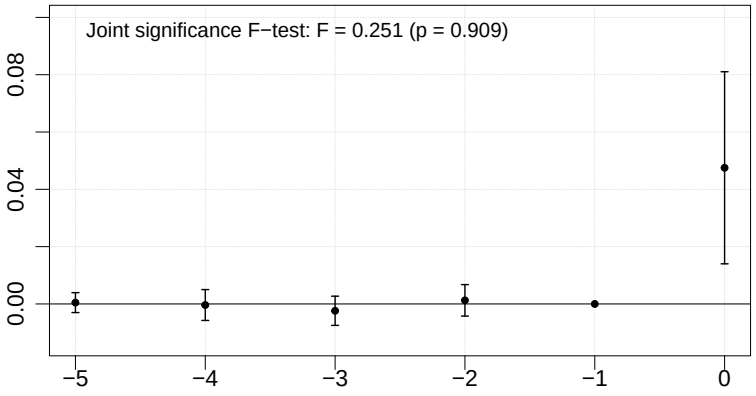



(b) Year 12

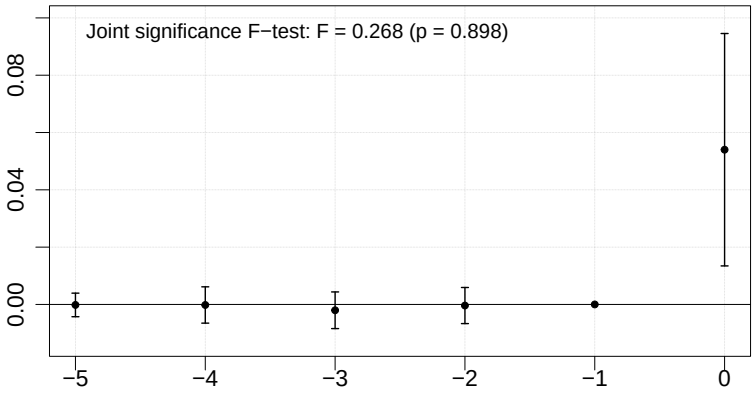



(c) Year 19

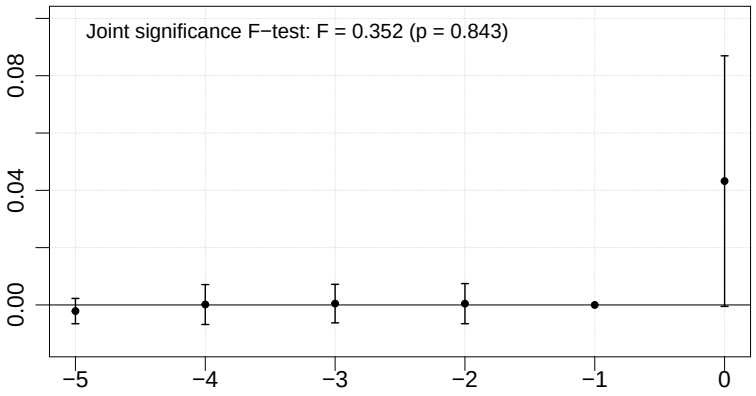
Figure B18: Facebook Access and Cumulative Work Experience – Event Studies




(a) Year 6




(b) Year 12




(c) Year 19

# Appendix C: Gender Heterogeneity

How does having access to online social networking platforms at labor market entry affect gender gaps in career outcomes? Despite decades of progress, women still work in lower-paying jobs and are systematically underrepresented at the top of the corporate world (e.g., Goldin 2014; Hindlian *et al.* 2018; Olivetti *et al.* 2024). While several studies highlight a range of explanations, including the role of labor supply (e.g., Bertrand *et al.* 2010), family responsibilities (e.g., Kleven *et al.* 2019), or preferences for risk and competition (Niederle and Vesterlund 2007; Buser *et al.* 2014; Wiswall and Zafar 2018; Mas and Pallais 2017), another part of the literature emphasizes the role of social networks, homophily, and referrals (e.g., Lalanne and Seabright 2016; Beaman *et al.* 2018; Zeltzer 2020; Mengel 2020; Cullen and Perez-Truglia 2023; von Essen and Smith 2023; Hampole *et al.* 2026). As Facebook reduces the costs of adding and maintaining social connections, this may have facilitated women's access to male-dominated networks, equalizing access to valuable labor market connections, and ultimately reducing gender gaps in career outcomes. Consistent with the first part of this hypothesis, the results by Armona and Bailey (2025) indicate that early Facebook adopters have online social networks with lower gender homophily.

This appendix investigates to what extent this observation translates into differential career effects for men and women. Appendix Table C1 starts by showing that the documented career benefits are homogeneously shared across gender. While the effects on job-based earnings are slightly larger for women in the medium and long run, consistent with the cross-sectional evidence in Armona (2025), the coefficients are not statistically distinguishable. Similarly, while single point estimates for the underlying mechanisms tend to differ across gender (see Appendix Table C2 to Table C6), the differences are, with few exceptions, not statistically significant. Nevertheless, consistent with the finding by Armona and Bailey (2025) of reduced gender homophily, Appendix Table C3 shows that during the college-to-work transition, students with early access to Facebook were more likely to receive job help from opposite-sex peers. Taken together, the results indicate that early access to Facebook did not meaningfully affect the gender gap in career outcomes, but may have facilitated women's access to otherwise male-dominated labor market networks (and vice versa).

Table C1: Facebook Access and Career Development – By Gender

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Real Annual Earnings (2024, USD)* | | | |
| FB Access*Male | 1,319* | 1,108 | 7.07 |
| | (669) | (843) | (981) |
| FB Access*Female | 1,131 | 1,639* | 1,806 |
| | (662) | (803) | (944) |
| Equality test (p-value) | 0.842 | 0.648 | 0.186 |
| Dep. Var. Mean (Male) | 79,840 | 113,823 | 131,041 |
| Dep. Var. Mean (Female) | 70,330 | 95,628 | 108,035 |
| Observations | 94,680 | 94,680 | 110,460 |
| *Panel B: Job-Based Promotion ($\geq$10 log pts.)* | | | |
| FB Access*Male | 0.003* | 0.002 | -0.00009 |
| | (0.002) | (0.002) | (0.001) |
| FB Access*Female | 0.004* | 0.002 | -0.002 |
| | (0.002) | (0.002) | (0.001) |
| Equality test (p-value) | 0.662 | 0.898 | 0.284 |
| Dep. Var. Mean (Male) | 0.078 | 0.087 | 0.076 |
| Dep. Var. Mean (Female) | 0.078 | 0.080 | 0.073 |
| Observations | 78,900 | 94,680 | 110,460 |
| *Panel C: Senior Leadership Positions* | | | |
| FB Access*Male | 0.001 | 0.007 | 0.009 |
| | (0.003) | (0.005) | (0.005) |
| FB Access*Female | 0.004 | 0.006 | 0.008 |
| | (0.003) | (0.004) | (0.005) |
| Equality test (p-value) | 0.623 | 0.889 | 0.896 |
| Dep. Var. Mean (Male) | 0.114 | 0.226 | 0.305 |
| Dep. Var. Mean (Female) | 0.084 | 0.161 | 0.221 |
| Observations | 94,680 | 94,680 | 110,460 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level separately for male and female graduates. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 Years after expected graduation. Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Equality test: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table C2: Facebook Access and the College-to-Work Transition – By Gender

| | | Indicators of High-Quality Early-Career Job Placement | | | | |
|---|---|---|---|---|---|---|
| | Employment | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
| FB Access*Male | 0.013* | 0.012* | 0.015** | 0.018** | -0.005 | 0.010 |
| | (0.006) | (0.005) | (0.005) | (0.006) | (0.005) | (0.005) |
| FB Access*Female | 0.008 | 0.016** | 0.013* | 0.004 | 0.005 | 0.020** |
| | (0.006) | (0.005) | (0.006) | (0.006) | (0.007) | (0.007) |
| Equality test (p-value) | 0.563 | 0.636 | 0.825 | 0.095 | 0.239 | 0.244 |
| Dep. Var. Mean (Male) | 0.645 | 0.216 | 0.165 | 0.282 | 0.134 | 0.201 |
| Dep. Var. Mean (Female) | 0.665 | 0.192 | 0.150 | 0.203 | 0.194 | 0.201 |
| Observations | 31,560 | 31,558 | 31,446 | 31,556 | 31,546 | 31,546 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level separately for male and female graduates. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Equality test: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table C3: Facebook Access and Early-Career Job Help in the College Network – By Gender

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Same-Sex Early-Career Job via College Network* | | | | | | |
| FB Access*Male | 0.009** | 0.005* | 0.008** | 0.004* | 0.004** | 0.006** |
| | (0.003) | (0.002) | (0.003) | (0.002) | (0.001) | (0.002) |
| FB Access*Female | 0.009** | 0.006*** | 0.004* | 0.003 | 0.004* | 0.005* |
| | (0.003) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| Equality test (p-value) | 0.998 | 0.536 | 0.251 | 0.716 | 0.978 | 0.641 |
| Dep. Var. Mean (Male) | 0.091 | 0.038 | 0.045 | 0.041 | 0.021 | 0.041 |
| Dep. Var. Mean (Female) | 0.079 | 0.028 | 0.037 | 0.025 | 0.023 | 0.035 |
| Observations | 31,560 | 31,558 | 31,446 | 31,556 | 31,546 | 31,546 |
| *Panel B: Opposite-Sex Early-Career Job via College Network* | | | | | | |
| FB Access*Male | 0.006* | 0.002 | 0.007** | 0.0004 | 0.004** | 0.005** |
| | (0.002) | (0.002) | (0.002) | (0.002) | (0.001) | (0.002) |
| FB Access*Female | 0.011** | 0.008*** | 0.006* | 0.003 | 0.004* | 0.007** |
| | (0.004) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| Equality test (p-value) | 0.260 | 0.020 | 0.676 | 0.433 | 0.889 | 0.575 |
| Dep. Var. Mean (Male) | 0.072 | 0.030 | 0.036 | 0.031 | 0.017 | 0.034 |
| Dep. Var. Mean (Female) | 0.082 | 0.031 | 0.040 | 0.028 | 0.024 | 0.037 |
| Observations | 31,560 | 31,558 | 31,446 | 31,556 | 31,546 | 31,546 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level separately for male and female graduates. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Equality test: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table C4: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – By Gender

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Between-Firm Promotions (≥10 log pts.)* | | | |
| FB Access*Male | 0.0005 | 0.0007 | -0.0001 |
| | (0.001) | (0.001) | (0.0010) |
| FB Access*Female | 0.0005 | 0.0002 | -0.003* |
| | (0.002) | (0.002) | (0.001) |
| Equality test (p-value) | 0.992 | 0.817 | 0.081 |
| Dep. Var. Mean (Male) | 0.056 | 0.057 | 0.048 |
| Dep. Var. Mean (Female) | 0.055 | 0.051 | 0.043 |
| Observations | 78,892 | 94,653 | 110,417 |
| *Panel B: Promotion via College Network (≥10 log pts.)* | | | |
| FB Access*Male | -0.00004 | 0.0009 | -0.00006 |
| | (0.0006) | (0.0005) | (0.0003) |
| FB Access*Female | -0.0002 | 0.00007 | -0.0000005 |
| | (0.0007) | (0.0006) | (0.0004) |
| Equality test (p-value) | 0.898 | 0.265 | 0.894 |
| Dep. Var. Mean (Male) | 0.012 | 0.011 | 0.007 |
| Dep. Var. Mean (Female) | 0.012 | 0.009 | 0.006 |
| Observations | 78,900 | 94,680 | 110,460 |
| *Panel C: Senior Leadership Transition via College Network* | | | |
| FB Access*Male | 0.0001 | 0.0003 | 0.00004 |
| | (0.0002) | (0.0002) | (0.0002) |
| FB Access*Female | -0.0003 | 0.00001 | -0.00002 |
| | (0.0003) | (0.0003) | (0.0002) |
| Equality test (p-value) | 0.195 | 0.369 | 0.832 |
| Dep. Var. Mean (Male) | 0.002 | 0.003 | 0.002 |
| Dep. Var. Mean (Female) | 0.002 | 0.002 | 0.002 |
| Observations | 78,900 | 94,680 | 110,460 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level separately for male and female graduates. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior Leadership Transition via College Network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Equality test: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table C5: Facebook Access and Worker-Firm Match Stability and Productivity – By Gender

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Stable Worker-Firm Match (≥6 Months)* | | | |
| FB Access*Male | 0.010* | 0.001 | -0.003 |
| | (0.004) | (0.002) | (0.002) |
| FB Access*Female | 0.003 | 0.0006 | 0.003 |
| | (0.004) | (0.003) | (0.004) |
| Equality test (p-value) | 0.210 | 0.853 | 0.199 |
| Dep. Var. Mean (Male) | 0.685 | 0.873 | 0.891 |
| Dep. Var. Mean (Female) | 0.693 | 0.856 | 0.867 |
| Observations | 94,680 | 94,680 | 110,460 |
| *Panel B: Current Firm Tenure (Years)* | | | |
| FB Access*Male | 0.045** | 0.119** | 0.105 |
| | (0.015) | (0.039) | (0.057) |
| FB Access*Female | 0.014 | 0.007 | 0.052 |
| | (0.017) | (0.041) | (0.062) |
| Equality test (p-value) | 0.169 | 0.046 | 0.533 |
| Dep. Var. Mean (Male) | 1.63 | 4.03 | 6.08 |
| Dep. Var. Mean (Female) | 1.66 | 4.01 | 5.99 |
| Observations | 94,680 | 94,680 | 110,460 |
| *Panel C: Within-Firm Promotions (≥10 log pts.)* | | | |
| FB Access*Male | 0.003* | 0.001 | 0.00003 |
| | (0.001) | (0.001) | (0.0009) |
| FB Access*Female | 0.004** | 0.002* | 0.0005 |
| | (0.001) | (0.001) | (0.0009) |
| Equality test (p-value) | 0.517 | 0.469 | 0.725 |
| Dep. Var. Mean (Male) | 0.022 | 0.030 | 0.029 |
| Dep. Var. Mean (Female) | 0.023 | 0.029 | 0.030 |
| Observations | 78,892 | 94,653 | 110,417 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level separately for male and female graduates. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 6 months. Current firm tenure: years at current employer. Within-firm promotions: indicator for upward title transition within the same firm. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Equality test: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table C6: Facebook Access, Firm Sorting, and Human Capital Accumulation – By Gender

| | 6Y | 12Y | 19Y |
|---|---|---|---|
| *Panel A: Work Experience at High-Wage Firms (Years)* | | | |
| FB Access*Male | 0.063 | 0.060 | 0.002 |
| | (0.033) | (0.066) | (0.100) |
| FB Access*Female | 0.097** | 0.088 | 0.086 |
| | (0.034) | (0.070) | (0.116) |
| Equality test (p-value) | 0.463 | 0.769 | 0.583 |
| Dep. Var. Mean (Male) | 1.56 | 3.71 | 6.35 |
| Dep. Var. Mean (Female) | 1.35 | 3.10 | 5.23 |
| Observations | 15,780 | 15,780 | 15,780 |
| *Panel B: Work Experience at High-Career-Support Firms (Years)* | | | |
| FB Access*Male | 0.062** | 0.140** | 0.195** |
| | (0.023) | (0.045) | (0.068) |
| FB Access*Female | 0.040 | 0.062 | 0.090 |
| | (0.026) | (0.051) | (0.088) |
| Equality test (p-value) | 0.519 | 0.249 | 0.345 |
| Dep. Var. Mean (Male) | 0.885 | 1.90 | 3.12 |
| Dep. Var. Mean (Female) | 0.792 | 1.68 | 2.77 |
| Observations | 15,692 | 15,674 | 15,677 |
| *Panel C: Cumulative Work Experience (Years)* | | | |
| FB Access*Male | 0.057* | 0.064* | 0.042 |
| | (0.027) | (0.031) | (0.033) |
| FB Access*Female | 0.036 | 0.044 | 0.049 |
| | (0.025) | (0.030) | (0.035) |
| Equality test (p-value) | 0.561 | 0.642 | 0.872 |
| Dep. Var. Mean (Male) | 4.46 | 10.13 | 16.86 |
| Dep. Var. Mean (Female) | 4.53 | 10.11 | 16.68 |
| Observations | 15,780 | 15,780 | 15,780 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level separately for male and female graduates. Each cell reports a coefficient from a separate DiD regression, estimated at 6, 12, and 19 years after expected graduation. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Work experience at high-career-support firms: cumulative years employed at a high-career-support firm. Cumulative work experience: total years of employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Equality test: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

# Appendix D: Randomization Inference

A key concern in the main empirical strategy is that even within the narrow selectivity-by-region comparisons, spurious correlations with the timing of the Facebook rollout might remain, biasing the main estimates. For example, this could be the case if Facebook (deliberately or not) expanded early to colleges *within the same selectivity and U.S. region* where the career outcomes of U.S. business and economics cohorts were on a differential trend. To formally test this, I implement a randomization inference test in which I randomly reshuffle Facebook's introduction year across colleges within each selectivity-by-region cell (N = 1,000). The test preserves the total number of treated colleges per cell and the overall distribution of treatment timing. Appendix Tables D1–D6 show the results. The placebo estimates have a dense distribution around zero and virtually all of the statistically significant effects from the main specification fall above the 95$^{th}$ percentile of the placebo distribution. The remaining variation in the Facebook rollout is therefore unlikely to reflect spurious correlations.

Table D1: Permutation Test: Career Development

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Real Annual Earnings (2024, USD)* | | | |
| Baseline | 1,286** | 1,343* | 768 |
| Mean (placebo) | 5.72 | 110 | 117 |
| SD (placebo) | 516 | 628 | 785 |
| Percentile rank | 99 | 97 | 80 |
| *Panel B: Job-Based Promotion (≥10 log pts.)* | | | |
| Baseline | 0.004** | 0.002 | -0.0010 |
| Mean (placebo) | -0.00007 | -0.00004 | -0.00008 |
| SD (placebo) | 0.001 | 0.001 | 0.0009 |
| Percentile rank | 100 | 98 | 15 |
| *Panel C: Senior Leadership Positions* | | | |
| Baseline | 0.002 | 0.006* | 0.008* |
| Mean (placebo) | 0.00006 | 0.0002 | 0.0001 |
| SD (placebo) | 0.002 | 0.003 | 0.004 |
| Percentile rank | 84 | 97 | 99 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* Each cell reports four quantities: (1) baseline Gardner (2022) two-stage DiD estimate (significance: $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$), (2) mean of the placebo distribution, (3) standard deviation of the placebo distribution, (4) percentile rank of the baseline estimate in the placebo distribution (share of permuted estimates ≤ the baseline). Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Permutations: $N = 1000$. Within each stratum (college selectivity * region), the vector of Facebook introduction years is randomly permuted across colleges; the number of treated colleges and the distribution of introduction timing per stratum are preserved. Standard errors clustered at the college level. *Data source:* Revelio Labs.

Table D2: Permutation Test: College-to-Work Transition

| | Employment | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| Baseline | 0.010** | 0.014*** | 0.014*** | 0.012** | -0.001 | 0.014*** |
| Mean (placebo) | -0.0002 | 0.0004 | 0.0004 | 0.0001 | -0.00002 | 0.0005 |
| SD (placebo) | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 | 0.005 |
| Percentile rank | 100 | 100 | 100 | 100 | 38 | 100 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* Each cell reports four quantities: (1) baseline Gardner (2022) two-stage DiD estimate (significance: $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$), (2) mean of the placebo distribution, (3) standard deviation of the placebo distribution, (4) percentile rank of the baseline estimate in the placebo distribution (share of permuted estimates ≤ the baseline). Permutations: $N = 1000$. Within each stratum (college selectivity * region), the vector of Facebook introduction years is randomly permuted across colleges; the number of treated colleges and the distribution of introduction timing per stratum are preserved. Standard errors clustered at the college level. *Data source:* Revelio Labs.

Table D3: Permutation Test: Early-Career Job Help in the College Network

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Early-Career Job via College Network* | | | | | | |
| Baseline | 0.011*** | 0.007*** | 0.008*** | 0.004* | 0.004** | 0.007*** |
| Mean (placebo) | 0.0004 | 0.0005 | 0.0003 | 0.0001 | 0.0001 | 0.0006 |
| SD (placebo) | 0.003 | 0.002 | 0.003 | 0.002 | 0.002 | 0.003 |
| Percentile rank | 100 | 100 | 100 | 96 | 100 | 100 |
| *Panel B: Early-Career Job without College Network* | | | | | | |
| Baseline | -0.0004 | 0.008* | 0.006 | 0.008* | -0.005 | 0.007 |
| Mean (placebo) | -0.0005 | -0.00002 | 0.00004 | -0.0002 | 0.00001 | -0.0002 |
| SD (placebo) | 0.004 | 0.004 | 0.003 | 0.004 | 0.003 | 0.004 |
| Percentile rank | 50 | 98 | 97 | 100 | 6 | 98 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* Panel A: early-career job at a firm with a college peer already working there. Panel B: early-career job at a firm without a college peer already working there. Each cell reports four quantities: (1) baseline Gardner (2022) two-stage DiD estimate (significance: $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$), (2) mean of the placebo distribution, (3) standard deviation of the placebo distribution, (4) percentile rank of the baseline estimate in the placebo distribution (share of permuted estimates $\leq$ the baseline). Permutations: $N = 1000$. Within each stratum (college selectivity * region), the vector of Facebook introduction years is randomly permuted across colleges; the number of treated colleges and the distribution of introduction timing per stratum are preserved. Standard errors clustered at the college level. *Data source:* Revelio Labs.

Table D4: Permutation Test: Promotion Ties

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Between-Firm Promotions (≥10 log pts.)* | | | |
| Baseline | 0.0005 | 0.0003 | -0.001 |
| Mean (placebo) | -0.00004 | 0.00004 | 0.00002 |
| SD (placebo) | 0.001 | 0.0008 | 0.0007 |
| Percentile rank | 68 | 62 | 3 |
| *Panel B: Promotion via College Network (≥10 log pts.)* | | | |
| Baseline | -0.00009 | 0.0005 | -0.00004 |
| Mean (placebo) | 0.00004 | 0.00005 | -0.0000009 |
| SD (placebo) | 0.0005 | 0.0004 | 0.0003 |
| Percentile rank | 39 | 86 | 45 |
| *Panel C: Senior Leadership Transition via College Network* | | | |
| Baseline | -0.00007 | 0.0002 | 0.000006 |
| Mean (placebo) | 0.000008 | -0.00001 | -0.000004 |
| SD (placebo) | 0.0002 | 0.0002 | 0.0001 |
| Percentile rank | 33 | 82 | 54 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* Each cell reports four quantities: (1) baseline Gardner (2022) two-stage DiD estimate (significance: $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$), (2) mean of the placebo distribution, (3) standard deviation of the placebo distribution, (4) percentile rank of the baseline estimate in the placebo distribution (share of permuted estimates ≤ the baseline). Permutations: $N = 1000$. Within each stratum (college selectivity * region), the vector of Facebook introduction years is randomly permuted across colleges; the number of treated colleges and the distribution of introduction timing per stratum are preserved. Standard errors clustered at the college level. *Data source:* Revelio Labs.

Table D5: Permutation Test: Worker-Firm Match Stability and Productivity

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Stable Worker-Firm Match (≥6 Months)* | | | |
| Baseline | 0.007* | 0.001 | -0.0004 |
| Mean (placebo) | -0.0001 | 0.0002 | 0.00002 |
| SD (placebo) | 0.003 | 0.002 | 0.002 |
| Percentile rank | 99 | 72 | 42 |
| *Panel B: Current Firm Tenure (Years)* | | | |
| Baseline | 0.031** | 0.070* | 0.083 |
| Mean (placebo) | -0.0002 | 0.001 | 0.001 |
| SD (placebo) | 0.010 | 0.021 | 0.040 |
| Percentile rank | 100 | 100 | 98 |
| *Panel C: Within-Firm Promotions (≥10 log pts.)* | | | |
| Baseline | 0.003*** | 0.002* | 0.0002 |
| Mean (placebo) | -0.00002 | -0.00008 | -0.0001 |
| SD (placebo) | 0.0008 | 0.0009 | 0.0006 |
| Percentile rank | 100 | 98 | 72 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* Each cell reports four quantities: (1) baseline Gardner (2022) two-stage DiD estimate (significance: $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$), (2) mean of the placebo distribution, (3) standard deviation of the placebo distribution, (4) percentile rank of the baseline estimate in the placebo distribution (share of permuted estimates ≤ the baseline). Permutations: $N = 1000$. Within each stratum (college selectivity * region), the vector of Facebook introduction years is randomly permuted across colleges; the number of treated colleges and the distribution of introduction timing per stratum are preserved. Standard errors clustered at the college level. *Data source:* Revelio Labs.

Table D6: Permutation Test: Firm Sorting and Human Capital Accumulation

| | 6Y | 12Y | 19Y |
|---|---|---|---|
| *Panel A: Work Exp. at High-Wage Firms (Years)* | | | |
| Baseline | 0.080*** | 0.076 | 0.044 |
| Mean (placebo) | 0.002 | 0.005 | 0.008 |
| SD (placebo) | 0.021 | 0.042 | 0.063 |
| Percentile rank | 100 | 97 | 71 |
| *Panel B: Work Exp. at High-Career-Support Firms (Years)* | | | |
| Baseline | 0.054** | 0.105** | 0.145** |
| Mean (placebo) | 0.002 | 0.003 | 0.003 |
| SD (placebo) | 0.017 | 0.031 | 0.047 |
| Percentile rank | 100 | 100 | 100 |
| *Panel C: Cumulative Work Experience (Years)* | | | |
| Baseline | 0.048** | 0.054** | 0.043 |
| Mean (placebo) | -0.0004 | -0.000005 | -0.0003 |
| SD (placebo) | 0.016 | 0.019 | 0.020 |
| Percentile rank | 100 | 100 | 99 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* Each cell reports four quantities: (1) baseline Gardner (2022) two-stage DiD estimate (significance: $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$), (2) mean of the placebo distribution, (3) standard deviation of the placebo distribution, (4) percentile rank of the baseline estimate in the placebo distribution (share of permuted estimates $\leq$ the baseline). Permutations: $N = 1000$. Within each stratum (college selectivity * region), the vector of Facebook introduction years is randomly permuted across colleges; the number of treated colleges and the distribution of introduction timing per stratum are preserved. Standard errors clustered at the college level. *Data source:* Revelio Labs.

# Appendix E: Spillovers

One key concern with the main specification is that the results are driven by spillovers. In particular, treated students might take away jobs from students in the control group, upward biasing the treatment effect and violating the stable-unit treatment value assumption (SUTVA). To assess the seriousness of this issue, I begin by predicting each potential control cohort's exposure to spillovers using the actual career overlap of U.S. business and economics students across colleges *before* the Facebook rollout. The main motivation behind this measure is that graduate placements are highly persistent over time (e.g., Hampole *et al.* 2025) and therefore well-suited to predicting future labor market competition across college cohorts. To account for the changing set of treated colleges over time, I define the predicted spillover exposure measure for each potential control cohort as follows:

$$J_{a,T_t} = \frac{\left| C_a^{1999\text{–}2003} \cap \left( \bigcup_{b \in T_t} C_b^{1999\text{–}2003} \right) \right|}{|C_a^{1999\text{–}2003}|}$$

where $J_{a,T_t}$ measures the share of graduates from college $a$ working in the same job as graduates from the set of colleges $T$ that already had access to Facebook in the academic year $t$. To ensure that the prediction is not contaminated by potential treatment effects, I use the cohorts from 1999–2003 *before* the Facebook rollout for computation. To closely reflect career overlap and labor market competition between college graduates, I define the same job as working in the same firm*2-digit SOC*year cell.[46]

Table E1 investigates the predicted spillover exposure measure for the potential control group cohorts in detail. In 2004, the average predicted fraction of control-group cohorts competing for the same jobs as graduates of the treated colleges was around 16%, consistent with the small number of highly selective institutions that had early access to Facebook. By 2005, this share increased to around 33% as Facebook expanded to a broader set of colleges. Table E1 regresses the exposure score on college characteristics to identify which control cohorts are most susceptible to spillovers. In 2004, institutions in the highest selectivity tiers and those in the Northeast had substantially higher predicted exposure scores, consistent with the idea that these colleges are the most likely to compete for the same jobs as the elite colleges Facebook targeted first. By 2005, as more colleges across the U.S. gained access to Facebook, differences in selectivity and region narrowed considerably, with colleges in the West and those with the lowest selectivity being slightly less exposed. Table E2 shows the twenty college cohorts

[46] Note that the ideal labor market competition measure would use information on job applications instead of realized outcomes. While this information is not available, the derived measure from LinkedIn profile data can be seen as a meaningful but lower-bound proxy for it.

with the highest predicted exposure scores. All twenty come from the 2005 graduation cohort, consistent with spillovers being more severe at the later stage of Facebook's expansion. The most-exposed college cohorts come predominantly from HBCUs (e.g., Johnson C. Smith University, Central State University, Cheyney University of Pennsylvania) and Catholic colleges (e.g., St. Catherine University, Mount St. Joseph University, College of Saint Benedict).

To probe the robustness of the main results to potential spillovers, I then drop every college that has at least one potential control cohort with a predicted exposure score above the population-weighted upper quartile of the 2005 exposure distribution ($Q^{\text{pop}}_{0.75}(J_{a,2005}) = 0.369$).[47] This removes 97 colleges, or 25% of the potential control student body graduating in 2005. Tables E3 to E8 present the results. Reassuringly, the results are very similar to the baseline estimates. Combined with the baseline findings, the results suggest that the treatment effects do not simply reflect a reallocation of jobs within the economy. They are rather consistent with the idea that access to Facebook reduces information frictions about job opportunities.

Table E1: Predicted Spillover Exposure Score and Observable College Characteristics

| | Predicted Spillover Exposure Score ($J_{a,T_t}$) | |
|---|---|---|
| | $J_{a,2004}$ | $J_{a,2005}$ |
| *U.S. Region (ref.: West)* | | |
| Midwest | 0.009* | 0.049*** |
| | (0.004) | (0.011) |
| Northeast | 0.028*** | 0.046*** |
| | (0.004) | (0.012) |
| South | 0.005 | 0.045*** |
| | (0.004) | (0.010) |
| *Selectivity (ref.: Less competitive)* | | |
| Competitive | 0.009* | 0.004 |
| | (0.003) | (0.007) |
| Very competitive | 0.030*** | 0.019 |
| | (0.004) | (0.010) |
| Highly competitive | 0.056*** | 0.032 |
| | (0.005) | (0.017) |
| Elite | 0.118*** | 0.178** |
| | (0.006) | (0.063) |
| Dep. Var. Mean (pop.-wtd.) | 0.157 | 0.332 |
| Upper Quartile (pop.-wtd.) | 0.177 | 0.369 |
| Observations | 1,118 | 422 |
| $R^2$ | 0.360 | 0.086 |

*Notes:* OLS regressions at the college-cohort level. Sample: all potential control college-cohorts during the Facebook expansion period by academic year. Dependent variable: predicted spillover exposure score in each academic year. *p<0.05; **p<0.01; ***p<0.001. *Data source:* Revelio Labs.

[47] The results are robust to using different thresholds.

Table E2: Top Twenty Control College-Cohorts with Highest Predicted Spillover Exposure

| College | Cohort | Exposure score |
|---|---|---|
| johnson c smith university | 2005 | 0.523 |
| barnard college | 2005 | 0.500 |
| lane college | 2005 | 0.488 |
| st catherine university | 2005 | 0.485 |
| eureka college | 2005 | 0.470 |
| central state university | 2005 | 0.469 |
| williams baptist college | 2005 | 0.461 |
| gwynedd mercy university | 2005 | 0.460 |
| college of saint benedict | 2005 | 0.452 |
| westminster college | 2005 | 0.448 |
| mount saint joseph university | 2005 | 0.448 |
| missouri university of science and technology | 2005 | 0.446 |
| cheyney university of pennsylvania | 2005 | 0.439 |
| university of arkansas at pine bluff | 2005 | 0.438 |
| university of michigan-dearborn | 2005 | 0.436 |
| fisk university | 2005 | 0.434 |
| saint johns university | 2005 | 0.434 |
| union college | 2005 | 0.434 |
| southern university and a & m college | 2005 | 0.434 |
| university of michigan-flint | 2005 | 0.434 |

*Notes:* Potential control college-cohorts with the twenty highest predicted spillover exposure scores.
*Data source:* Revelio Labs.

Table E3: Facebook Access and Career Development – Excluding Spillover-Exposed Colleges

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Real Annual Earnings (2024, USD)* | | | |
| FB Access | 1,537* | 1,657* | 1,033 |
| | (673) | (688) | (886) |
| Dep. Var. Mean | 76,108 | 106,652 | 122,017 |
| Observations | 44,178 | 44,178 | 51,541 |
| *Panel B: Job-Based Promotion (≥10 log pts.)* | | | |
| FB Access | 0.003* | 0.002 | -0.002* |
| | (0.002) | (0.002) | (0.001) |
| Dep. Var. Mean | 0.078 | 0.085 | 0.075 |
| Observations | 36,815 | 44,178 | 51,541 |
| *Panel C: Senior Leadership Positions* | | | |
| FB Access | 0.004 | 0.008* | 0.012** |
| | (0.004) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.102 | 0.201 | 0.272 |
| Observations | 44,178 | 44,178 | 51,541 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 Years after expected graduation. Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Colleges with any control cohort whose predicted spillover exposure score exceeds the population-weighted upper quartile of the 2005 distribution (0.369) are excluded entirely. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table E4: Facebook Access and the College-to-Work Transition – Excluding Spillover-Exposed Colleges

| | | Indicators of High-Quality Early-Career Job Placement | | | | |
|---|---|---|---|---|---|---|
| | Employment | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
| FB Access | 0.013* | 0.012* | 0.010* | 0.015* | 0.0008 | 0.016** |
| | (0.006) | (0.005) | (0.005) | (0.006) | (0.005) | (0.005) |
| Dep. Var. Mean | 0.653 | 0.207 | 0.160 | 0.250 | 0.159 | 0.202 |
| Observations | 14,726 | 14,726 | 14,711 | 14,724 | 14,724 | 14,724 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Colleges with any control cohort whose predicted spillover exposure score exceeds the population-weighted upper quartile of the 2005 distribution (0.369) are excluded entirely. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table E5: Facebook Access and Early-Career Job Help in the College Network – Excluding Spillover-Exposed Colleges

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Early-Career Job via College Network* | | | | | | |
| FB Access | 0.011** | 0.006*** | 0.007*** | 0.002 | 0.006*** | 0.008*** |
| | (0.003) | (0.002) | (0.002) | (0.003) | (0.002) | (0.002) |
| Dep. Var. Mean | 0.108 | 0.042 | 0.050 | 0.043 | 0.027 | 0.047 |
| Observations | 14,726 | 14,726 | 14,711 | 14,724 | 14,724 | 14,724 |
| *Panel B: Early-Career Job without College Network* | | | | | | |
| FB Access | 0.002 | 0.007 | 0.003 | 0.014* | -0.005 | 0.009 |
| | (0.006) | (0.005) | (0.004) | (0.006) | (0.006) | (0.005) |
| Dep. Var. Mean | 0.545 | 0.165 | 0.110 | 0.207 | 0.131 | 0.155 |
| Observations | 14,726 | 14,726 | 14,711 | 14,724 | 14,724 | 14,724 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Panel A: early-career job at a firm with a college peer already working there. Panel B: early-career job at a firm without a college peer already working there. The two panels partition the corresponding outcome in Table E4, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. Column 1 (Any Job) is the partition of the Employment outcome shown in Column 1 of that table. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Colleges with any control cohort whose predicted spillover exposure score exceeds the population-weighted upper quartile of the 2005 distribution (0.369) are excluded entirely. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table E6: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – Excluding Spillover-Exposed Colleges

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Between-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | -0.0009 | 0.0009 | -0.002* |
| | (0.002) | (0.001) | (0.0009) |
| Dep. Var. Mean | 0.056 | 0.055 | 0.046 |
| Observations | 36,815 | 44,175 | 51,541 |
| *Panel B: Promotion via College Network (≥10 log pts.)* | | | |
| FB Access | -0.0003 | 0.001** | -0.00002 |
| | (0.0005) | (0.0004) | (0.0003) |
| Dep. Var. Mean | 0.012 | 0.010 | 0.007 |
| Observations | 36,815 | 44,178 | 51,541 |
| *Panel C: Senior Leadership Transition via College Network* | | | |
| FB Access | -0.00003 | 0.0001 | -0.0002 |
| | (0.0002) | (0.0002) | (0.0002) |
| Dep. Var. Mean | 0.002 | 0.002 | 0.002 |
| Observations | 36,815 | 44,178 | 51,541 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior Leadership Transition via College Network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Colleges with any control cohort whose predicted spillover exposure score exceeds the population-weighted upper quartile of the 2005 distribution (0.369) are excluded entirely. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table E7: Facebook Access and Worker-Firm Match Stability and Productivity – Excluding Spillover-Exposed Colleges

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Stable Worker-Firm Match (≥6 Months)* | | | |
| FB Access | 0.010* | 0.003 | -0.002 |
| | (0.004) | (0.002) | (0.003) |
| Dep. Var. Mean | 0.688 | 0.866 | 0.881 |
| Observations | 44,178 | 44,178 | 51,541 |
| *Panel B: Current Firm Tenure (Years)* | | | |
| FB Access | 0.043** | 0.090** | 0.060 |
| | (0.013) | (0.035) | (0.059) |
| Dep. Var. Mean | 1.64 | 4.01 | 6.03 |
| Observations | 44,178 | 44,178 | 51,541 |
| *Panel C: Within-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.004*** | 0.0008 | -0.0003 |
| | (0.001) | (0.001) | (0.0009) |
| Dep. Var. Mean | 0.022 | 0.030 | 0.029 |
| Observations | 36,815 | 44,175 | 51,541 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 6 months. Current firm tenure: years at current employer. Within-firm promotions: indicator for upward title transition within the same firm. Colleges with any control cohort whose predicted spillover exposure score exceeds the population-weighted upper quartile of the 2005 distribution (0.369) are excluded entirely. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table E8: Facebook Access, Firm Sorting, and Human Capital Accumulation – Excluding Spillover-Exposed Colleges

| | 6Y | 12Y | 19Y |
|---|---|---|---|
| *Panel A: Work Experience at High-Wage Firms (Years)* | | | |
| FB Access | 0.076** | 0.060 | 0.004 |
| | (0.027) | (0.055) | (0.086) |
| Dep. Var. Mean | 1.48 | 3.47 | 5.92 |
| Observations | 7,363 | 7,363 | 7,363 |
| *Panel B: Work Exp. at High-Career-Support Firms (Years)* | | | |
| FB Access | 0.024 | 0.056 | 0.080 |
| | (0.021) | (0.048) | (0.080) |
| Dep. Var. Mean | 0.851 | 1.81 | 2.99 |
| Observations | 7,351 | 7,349 | 7,351 |
| *Panel C: Cumulative Work Experience (Years)* | | | |
| FB Access | 0.059* | 0.067* | 0.051 |
| | (0.025) | (0.030) | (0.030) |
| Dep. Var. Mean | 4.48 | 10.12 | 16.78 |
| Observations | 7,363 | 7,363 | 7,363 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, estimated at 6, 12, and 19 years after expected graduation. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. Cumulative work experience: total years of employment. Colleges with any control cohort whose predicted spillover exposure score exceeds the population-weighted upper quartile of the 2005 distribution (0.369) are excluded entirely. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

# Appendix F: Alternative Heterogeneity-Robust DiD Estimators

Table F1: Facebook Access and Career Development – Callaway and Sant'Anna (2021)

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Real Annual Earnings (2024, USD)* | | | |
| FB Access | 1,480** | 655 | 128 |
| | (554) | (590) | (741) |
| Dep. Var. Mean | 75,876 | 106,240 | 121,452 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel B: Job-Based Promotion (≥10 log pts.)* | | | |
| FB Access | 0.004** | -0.002 | -0.001 |
| | (0.001) | (0.002) | (0.001) |
| Dep. Var. Mean | 0.078 | 0.085 | 0.075 |
| Observations | 40,210 | 48,252 | 56,294 |
| *Panel C: Senior Leadership Positions* | | | |
| FB Access | 0.002 | 0.001 | 0.004 |
| | (0.003) | (0.003) | (0.004) |
| Dep. Var. Mean | 0.101 | 0.199 | 0.270 |
| Observations | 48,252 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Callaway and Sant'Anna (2021), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 Years after expected graduation. Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F2: Facebook Access and the College-to-Work Transition – Callaway and Sant'Anna (2021)

| | | Indicators of High-Quality Early-Career Job Placement | | | | |
|---|---|---|---|---|---|---|
| | Employment | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
| FB Access | 0.009* | 0.013** | 0.015*** | 0.008 | -0.003 | 0.010* |
| | (0.005) | (0.004) | (0.004) | (0.005) | (0.005) | (0.005) |
| Dep. Var. Mean | 0.653 | 0.206 | 0.159 | 0.249 | 0.159 | 0.201 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Callaway and Sant'Anna (2021), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F3: Facebook Access and Early-Career Job Help in the College Network – Callaway and Sant'Anna (2021)

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Early-Career Job via College Network* | | | | | | |
| FB Access | 0.011*** | 0.006*** | 0.009*** | 0.003 | 0.004* | 0.008*** |
| | (0.003) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| Dep. Var. Mean | 0.107 | 0.041 | 0.049 | 0.042 | 0.027 | 0.047 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| *Panel B: Early-Career Job without College Network* | | | | | | |
| FB Access | -0.002 | 0.007 | 0.006 | 0.005 | -0.007 | 0.003 |
| | (0.005) | (0.004) | (0.004) | (0.005) | (0.005) | (0.005) |
| Dep. Var. Mean | 0.546 | 0.165 | 0.110 | 0.207 | 0.132 | 0.154 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Callaway and Sant'Anna (2021), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Panel A: early-career job at a firm with a college peer already working there. Panel B: early-career job at a firm without a college peer already working there. The two panels partition the corresponding outcome in Table F2, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. Column 1 (Any Job) is the partition of the Employment outcome shown in Column 1 of that table. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F4: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – Callaway and Sant'Anna (2021)

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Between-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.001 | -0.002 | -0.002 |
| | (0.001) | (0.001) | (0.0009) |
| Dep. Var. Mean | 0.055 | 0.055 | 0.045 |
| Observations | 40,210 | 48,249 | 56,294 |
| *Panel B: Promotion via College Network (≥10 log pts.)* | | | |
| FB Access | 0.0004 | -0.0003 | -0.0001 |
| | (0.0005) | (0.0004) | (0.0003) |
| Dep. Var. Mean | 0.012 | 0.010 | 0.007 |
| Observations | 40,210 | 48,252 | 56,294 |
| *Panel C: Senior Leadership Transition via College Network* | | | |
| FB Access | 0.00004 | -0.00003 | 0.00010 |
| | (0.0002) | (0.0002) | (0.0002) |
| Dep. Var. Mean | 0.002 | 0.002 | 0.002 |
| Observations | 40,210 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Callaway and Sant'Anna (2021), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior Leadership Transition via College Network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F5: Facebook Access and Worker-Firm Match Stability and Productivity – Callaway and Sant'Anna (2021)

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Stable Worker-Firm Match (≥6 Months)* | | | |
| FB Access | 0.006 | -0.0004 | 0.0007 |
| | (0.003) | (0.002) | (0.003) |
| Dep. Var. Mean | 0.688 | 0.866 | 0.881 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel B: Current Firm Tenure (Years)* | | | |
| FB Access | 0.021 | 0.074** | 0.154** |
| | (0.012) | (0.029) | (0.054) |
| Dep. Var. Mean | 1.64 | 4.02 | 6.04 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel C: Within-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.002* | 0.0005 | 0.0004 |
| | (0.0008) | (0.0009) | (0.0008) |
| Dep. Var. Mean | 0.022 | 0.030 | 0.029 |
| Observations | 40,210 | 48,249 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Callaway–Sant'Anna (2021) estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 6 months. Current firm tenure: years at current employer. Within-firm promotions: indicator for upward title transition within the same firm. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F6: Facebook Access, Firm Sorting, and Human Capital Accumulation – Callaway and Sant'Anna (2021)

| | 6Y | 12Y | 19Y |
|---|---|---|---|
| *Panel A: Work Experience at High-Wage Firms (Years)* | | | |
| FB Access | 0.070* | 0.046 | 0.0009 |
| | (0.030) | (0.064) | (0.096) |
| Dep. Var. Mean | 1.48 | 3.45 | 5.88 |
| Observations | 8,042 | 8,042 | 8,042 |
| *Panel B: Work Exp. at High-Career-Support Firms (Years)* | | | |
| FB Access | 0.051** | 0.096* | 0.121* |
| | (0.019) | (0.041) | (0.061) |
| Dep. Var. Mean | 0.846 | 1.80 | 2.97 |
| Observations | 8,029 | 8,027 | 8,029 |
| *Panel C: Cumulative Work Experience (Years)* | | | |
| FB Access | 0.041* | 0.033 | 0.026 |
| | (0.019) | (0.024) | (0.029) |
| Dep. Var. Mean | 4.49 | 10.12 | 16.78 |
| Observations | 8,042 | 8,042 | 8,042 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Callaway–Sant'Anna (2021) estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, estimated at 6, 12, and 19 years after expected graduation. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. Cumulative work experience: total years of employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F7: Facebook Access and Career Development – de Chaisemartin and D'Haultfœuille (2020)

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Real Annual Earnings (2024, USD)* | | | |
| FB Access | 1,362* | 719 | 474 |
| | (557) | (602) | (685) |
| Dep. Var. Mean | 75,876 | 106,240 | 121,452 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel B: Job-Based Promotion (≥10 log pts.)* | | | |
| FB Access | 0.003* | -0.001 | -0.0006 |
| | (0.001) | (0.002) | (0.001) |
| Dep. Var. Mean | 0.078 | 0.085 | 0.075 |
| Observations | 40,210 | 48,252 | 56,294 |
| *Panel C: Senior Leadership Positions* | | | |
| FB Access | 0.002 | 0.0008 | 0.005 |
| | (0.003) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.101 | 0.199 | 0.270 |
| Observations | 48,252 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. de Chaisemartin and D'Haultfœuille (2020) estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 Years after expected graduation. Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Non-parametric region-by-selectivity strata trends are included. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F8: Facebook Access and the College-to-Work Transition – de Chaisemartin and D'Haultfœuille (2020)

| | | Indicators of High-Quality Early-Career Job Placement | | | | |
|---|---|---|---|---|---|---|
| | Employment | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
| FB Access | 0.009 | 0.011* | 0.012** | 0.006 | -0.003 | 0.009 |
| | (0.004) | (0.004) | (0.004) | (0.005) | (0.005) | (0.005) |
| Dep. Var. Mean | 0.653 | 0.206 | 0.159 | 0.249 | 0.159 | 0.201 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. de Chaisemartin and D'Haultfœuille (2020) estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Non-parametric region-by-selectivity strata trends are included. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F9: Facebook Access and Early-Career Job Help in the College Network – de Chaisemartin and D'Haultfœuille (2020)

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Early-Career Job via College Network* | | | | | | |
| FB Access | 0.010*** | 0.006** | 0.007*** | 0.003 | 0.003* | 0.007*** |
| | (0.003) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| Dep. Var. Mean | 0.107 | 0.041 | 0.049 | 0.042 | 0.027 | 0.047 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| *Panel B: Early-Career Job without College Network* | | | | | | |
| FB Access | -0.002 | 0.005 | 0.005 | 0.003 | -0.006 | 0.002 |
| | (0.005) | (0.004) | (0.004) | (0.005) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.546 | 0.165 | 0.110 | 0.207 | 0.132 | 0.154 |
| Observations | 16,084 | 16,084 | 16,066 | 16,082 | 16,082 | 16,082 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. de Chaisemartin and D'Haultfœuille (2020) estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Non-parametric region-by-selectivity strata trends are included. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Panel A: early-career job at a firm with a college peer already working there. Panel B: early-career job at a firm without a college peer already working there. The two panels partition the corresponding outcome in Table F8, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. Column 1 (Any Job) is the partition of the Employment outcome shown in Column 1 of that table. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F10: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – de Chaisemartin and D'Haultfœuille (2020)

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Between-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.001 | -0.002 | -0.0010 |
| | (0.001) | (0.001) | (0.0008) |
| Dep. Var. Mean | 0.055 | 0.055 | 0.045 |
| Observations | 40,210 | 48,249 | 56,294 |
| *Panel B: Promotion via College Network (≥10 log pts.)* | | | |
| FB Access | 0.0005 | -0.0004 | 0.00002 |
| | (0.0005) | (0.0005) | (0.0003) |
| Dep. Var. Mean | 0.012 | 0.010 | 0.007 |
| Observations | 40,210 | 48,252 | 56,294 |
| *Panel C: Senior Leadership Transition via College Network* | | | |
| FB Access | 0.00006 | -0.0001 | 0.0001 |
| | (0.0002) | (0.0002) | (0.0001) |
| Dep. Var. Mean | 0.002 | 0.002 | 0.002 |
| Observations | 40,210 | 48,252 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. de Chaisemartin and D'Haultfœuille (2020) estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. Non-parametric region-by-selectivity strata trends are included. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior Leadership Transition via College Network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F11: Facebook Access and Worker-Firm Match Stability and Productivity – de Chaisemartin and D'Haultfœuille (2020)

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Stable Worker-Firm Match (≥6 Months)* | | | |
| FB Access | 0.006 | -0.001 | 0.0008 |
| | (0.003) | (0.002) | (0.003) |
| Dep. Var. Mean | 0.688 | 0.866 | 0.881 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel B: Current Firm Tenure (Years)* | | | |
| FB Access | 0.018 | 0.046 | 0.123* |
| | (0.012) | (0.031) | (0.052) |
| Dep. Var. Mean | 1.64 | 4.02 | 6.04 |
| Observations | 48,252 | 48,252 | 56,294 |
| *Panel C: Within-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.002* | 0.0001 | 0.0005 |
| | (0.0008) | (0.001) | (0.0007) |
| Dep. Var. Mean | 0.022 | 0.030 | 0.029 |
| Observations | 40,210 | 48,249 | 56,294 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. de Chaisemartin and D'Haultfœuille (2020) estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Non-parametric region-by-selectivity strata trends are included. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 6 months. Current firm tenure: years at current employer. Within-firm promotions: indicator for upward title transition within the same firm. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table F12: Facebook Access, Firm Sorting, and Human Capital Accumulation – de Chaisemartin and D'Haultfœuille (2020)

| | 6Y | 12Y | 19Y |
|---|---|---|---|
| *Panel A: Work Experience at High-Wage Firms (Years)* | | | |
| FB Access | 0.062* | 0.050 | 0.028 |
| | (0.025) | (0.053) | (0.080) |
| Dep. Var. Mean | 1.48 | 3.45 | 5.88 |
| Observations | 8,042 | 8,042 | 8,042 |
| *Panel B: Work Exp. at High-Career-Support Firms (Years)* | | | |
| FB Access | 0.043* | 0.079* | 0.084 |
| | (0.018) | (0.037) | (0.057) |
| Dep. Var. Mean | 0.846 | 1.80 | 2.97 |
| Observations | 8,029 | 8,027 | 8,029 |
| *Panel C: Cumulative Work Experience (Years)* | | | |
| FB Access | 0.037 | 0.028 | 0.026 |
| | (0.020) | (0.024) | (0.026) |
| Dep. Var. Mean | 4.49 | 10.12 | 16.78 |
| Observations | 8,042 | 8,042 | 8,042 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. de Chaisemartin and D'Haultfœuille (2020) estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, estimated at 6, 12, and 19 years after expected graduation. Non-parametric region-by-selectivity strata trends are included. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. Cumulative work experience: total years of employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

# Appendix G: IPEDS Weights

Table G1: Facebook Access and Career Development – IPEDS-Weighted

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Real Annual Earnings (2024, USD)* | | | |
| FB Access | 1,193* | 965 | 545 |
| | (484) | (568) | (666) |
| Dep. Var. Mean | 75,362 | 104,709 | 119,172 |
| Observations | 94,686 | 94,686 | 110,467 |
| *Panel B: Job-Based Promotion (≥10 log pts.)* | | | |
| FB Access | 0.004* | 0.003* | -0.0002 |
| | (0.001) | (0.001) | (0.0010) |
| Dep. Var. Mean | 0.076 | 0.082 | 0.073 |
| Observations | 78,905 | 94,686 | 110,467 |
| *Panel C: Senior Leadership Positions* | | | |
| FB Access | 0.002 | 0.006 | 0.009* |
| | (0.003) | (0.003) | (0.004) |
| Dep. Var. Mean | 0.101 | 0.194 | 0.261 |
| Observations | 94,686 | 94,686 | 110,467 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 Years after expected graduation. Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Observations are weighted by gender and cohort size using IPEDS enrollment data. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table G2: Facebook Access and the College-to-Work Transition – IPEDS-Weighted

| | | Indicators of High-Quality Early-Career Job Placement | | | | |
|---|---|---|---|---|---|---|
| | Employment | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
| FB Access | 0.011** | 0.014*** | 0.014*** | 0.010* | -0.002 | 0.014** |
| | (0.004) | (0.004) | (0.004) | (0.004) | (0.005) | (0.004) |
| Dep. Var. Mean | 0.654 | 0.204 | 0.156 | 0.243 | 0.160 | 0.197 |
| Observations | 31,562 | 31,560 | 31,448 | 31,558 | 31,548 | 31,548 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Observations are weighted by gender and cohort size using IPEDS enrollment data. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table G3: Facebook Access and Early-Career Job Help in the College Network – IPEDS-Weighted

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Early-Career Job via College Network* | | | | | | |
| FB Access | 0.011*** | 0.006*** | 0.007*** | 0.003 | 0.004** | 0.007*** |
| | (0.003) | (0.002) | (0.002) | (0.002) | (0.001) | (0.002) |
| Dep. Var. Mean | 0.102 | 0.039 | 0.046 | 0.040 | 0.026 | 0.044 |
| Observations | 31,562 | 31,560 | 31,448 | 31,558 | 31,548 | 31,548 |
| *Panel B: Early-Career Job without College Network* | | | | | | |
| FB Access | 0.0003 | 0.008* | 0.007* | 0.007 | -0.006 | 0.007 |
| | (0.004) | (0.003) | (0.003) | (0.004) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.552 | 0.165 | 0.109 | 0.204 | 0.134 | 0.154 |
| Observations | 31,562 | 31,560 | 31,448 | 31,558 | 31,548 | 31,548 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Panel A: early-career job at a firm with a college peer already working there. Panel B: early-career job at a firm without a college peer already working there. The two panels partition the corresponding outcome in Table G2, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. Column 1 (Any Job) is the partition of the Employment outcome shown in Column 1 of that table. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Observations are weighted by gender and cohort size using IPEDS enrollment data. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table G4: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – IPEDS-Weighted

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Between-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.00009 | 0.0006 | -0.0006 |
| | (0.001) | (0.001) | (0.0007) |
| Dep. Var. Mean | 0.054 | 0.054 | 0.045 |
| Observations | 78,897 | 94,659 | 110,417 |
| *Panel B: Promotion via College Network (≥10 log pts.)* | | | |
| FB Access | 0.0001 | 0.0007 | -0.00002 |
| | (0.0004) | (0.0004) | (0.0003) |
| Dep. Var. Mean | 0.012 | 0.010 | 0.007 |
| Observations | 78,905 | 94,686 | 110,467 |
| *Panel C: Senior Leadership Transition via College Network* | | | |
| FB Access | -0.0001 | 0.0001 | -0.00006 |
| | (0.0002) | (0.0002) | (0.0001) |
| Dep. Var. Mean | 0.002 | 0.002 | 0.002 |
| Observations | 78,905 | 94,686 | 110,467 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior Leadership Transition via College Network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Observations are weighted by gender and cohort size using IPEDS enrollment data. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table G5: Facebook Access and Worker-Firm Match Stability and Productivity – IPEDS-Weighted

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Stable Worker-Firm Match (≥6 Months)* | | | |
| FB Access | 0.008** | 0.001 | -0.001 |
| | (0.003) | (0.002) | (0.003) |
| Dep. Var. Mean | 0.689 | 0.865 | 0.879 |
| Observations | 94,686 | 94,686 | 110,467 |
| *Panel B: Current Firm Tenure (Years)* | | | |
| FB Access | 0.035*** | 0.078** | 0.069 |
| | (0.010) | (0.029) | (0.049) |
| Dep. Var. Mean | 1.66 | 4.04 | 6.05 |
| Observations | 94,686 | 94,686 | 110,467 |
| *Panel C: Within-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.003*** | 0.002* | 0.0004 |
| | (0.0008) | (0.0009) | (0.0006) |
| Dep. Var. Mean | 0.022 | 0.028 | 0.028 |
| Observations | 78,897 | 94,659 | 110,417 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 6 months. Current firm tenure: years at current employer. Within-firm promotions: indicator for upward title transition within the same firm. Observations are weighted by gender and cohort size using IPEDS enrollment data. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table G6: Facebook Access, Firm Sorting, and Human Capital Accumulation – IPEDS-Weighted

| | 6Y | 12Y | 19Y |
|---|---|---|---|
| *Panel A: Work Experience at High-Wage Firms (Years)* | | | |
| FB Access | 0.073** | 0.057 | 0.014 |
| | (0.023) | (0.047) | (0.076) |
| Dep. Var. Mean | 1.46 | 3.40 | 5.78 |
| Observations | 15,781 | 15,781 | 15,781 |
| *Panel B: Work Exp. at High-Career-Support Firms (Years)* | | | |
| FB Access | 0.053** | 0.115** | 0.176** |
| | (0.017) | (0.036) | (0.059) |
| Dep. Var. Mean | 0.826 | 1.76 | 2.91 |
| Observations | 15,692 | 15,675 | 15,677 |
| *Panel C: Cumulative Work Experience (Years)* | | | |
| FB Access | 0.054** | 0.060** | 0.047 |
| | (0.017) | (0.021) | (0.024) |
| Dep. Var. Mean | 4.49 | 10.11 | 16.76 |
| Observations | 15,781 | 15,781 | 15,781 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, estimated at 6, 12, and 19 years after expected graduation. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. Cumulative work experience: total years of employment. Observations are weighted by gender and cohort size using IPEDS enrollment data. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

# Appendix H: Excluding Elite Colleges

Table H1: Facebook Access and Career Development – Excluding Elite Colleges

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Job-Based Real Annual Earnings (2024, USD)* | | | |
| FB Access | 1,519** | 1,518** | 1,143 |
| | (487) | (556) | (635) |
| Dep. Var. Mean | 74,659 | 103,734 | 118,123 |
| Observations | 45,102 | 45,102 | 52,619 |
| *Panel B: Job-Based Promotion (≥10 log pts.)* | | | |
| FB Access | 0.004** | 0.002 | -0.001 |
| | (0.001) | (0.001) | (0.0010) |
| Dep. Var. Mean | 0.077 | 0.084 | 0.074 |
| Observations | 37,585 | 45,102 | 52,619 |
| *Panel C: Senior Leadership Positions* | | | |
| FB Access | 0.004 | 0.006* | 0.009** |
| | (0.002) | (0.003) | (0.003) |
| Dep. Var. Mean | 0.099 | 0.191 | 0.259 |
| Observations | 45,102 | 45,102 | 52,619 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 Years after expected graduation. Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Elite colleges are excluded from the sample. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table H2: Facebook Access and the College-to-Work Transition – Excluding Elite Colleges

| | | Indicators of High-Quality Early-Career Job Placement | | | | |
|---|---|---|---|---|---|---|
| | Employment | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
| FB Access | 0.013*** | 0.013*** | 0.014*** | 0.011** | -0.0007 | 0.015*** |
| | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.652 | 0.197 | 0.152 | 0.242 | 0.157 | 0.190 |
| Observations | 15,034 | 15,034 | 15,016 | 15,032 | 15,032 | 15,032 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Elite colleges are excluded. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table H3: Facebook Access and Early-Career Job Help in the College Network – Excluding Elite Colleges

| | Any Job | High-Wage Firm | High-Career-Support Firm | High-Wage Occupation | Steep Wage Growth | Sr. Leadership Track |
|---|---|---|---|---|---|---|
| *Panel A: Early-Career Job via College Network* | | | | | | |
| FB Access | 0.012*** | 0.007*** | 0.010*** | 0.003* | 0.005*** | 0.008*** |
| | (0.002) | (0.001) | (0.002) | (0.002) | (0.001) | (0.001) |
| Dep. Var. Mean | 0.103 | 0.038 | 0.045 | 0.039 | 0.026 | 0.042 |
| Observations | 15,034 | 15,034 | 15,016 | 15,032 | 15,032 | 15,032 |
| *Panel B: Early-Career Job without College Network* | | | | | | |
| FB Access | 0.0009 | 0.006 | 0.004 | 0.008* | -0.006 | 0.007* |
| | (0.004) | (0.003) | (0.003) | (0.004) | (0.004) | (0.004) |
| Dep. Var. Mean | 0.549 | 0.159 | 0.107 | 0.203 | 0.131 | 0.148 |
| Observations | 15,034 | 15,034 | 15,016 | 15,032 | 15,032 | 15,032 |
| College FE | X | X | X | X | X | X |
| Cohort*Tier*Region FE | X | X | X | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Panel A: early-career job at a firm with a college peer already working there. Panel B: early-career job at a firm without a college peer already working there. The two panels partition the corresponding outcome in Table H2, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. Column 1 (Any Job) is the partition of the Employment outcome shown in Column 1 of that table. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Elite colleges are excluded. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table H4: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – Excluding Elite Colleges

| | 2–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Between-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.001 | 0.0003 | -0.001 |
| | (0.001) | (0.0009) | (0.0007) |
| Dep. Var. Mean | 0.055 | 0.054 | 0.045 |
| Observations | 37,585 | 45,099 | 52,619 |
| *Panel B: Promotion via College Network (≥10 log pts.)* | | | |
| FB Access | 0.0002 | 0.0005 | -0.0001 |
| | (0.0004) | (0.0004) | (0.0002) |
| Dep. Var. Mean | 0.012 | 0.010 | 0.007 |
| Observations | 37,585 | 45,102 | 52,619 |
| *Panel C: Senior Leadership Transition via College Network* | | | |
| FB Access | -0.00007 | 0.00010 | 0.00006 |
| | (0.0001) | (0.0002) | (0.0001) |
| Dep. Var. Mean | 0.002 | 0.002 | 0.002 |
| Observations | 37,585 | 45,102 | 52,619 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* $^{*}$p<0.05; $^{**}$p<0.01; $^{***}$p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior Leadership Transition via College Network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Elite colleges are excluded. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table H5: Facebook Access and Worker-Firm Match Stability and Productivity – Excluding Elite Colleges

| | 1–6 Years | 7–12 Years | 13–19 Years |
|---|---|---|---|
| *Panel A: Stable Worker-Firm Match (≥6 Months)* | | | |
| FB Access | 0.008** | 0.0008 | 0.00002 |
| | (0.003) | (0.002) | (0.002) |
| Dep. Var. Mean | 0.690 | 0.867 | 0.881 |
| Observations | 45,102 | 45,102 | 52,619 |
| *Panel B: Current Firm Tenure (Years)* | | | |
| FB Access | 0.035*** | 0.062* | 0.091* |
| | (0.011) | (0.028) | (0.046) |
| Dep. Var. Mean | 1.66 | 4.06 | 6.08 |
| Observations | 45,102 | 45,102 | 52,619 |
| *Panel C: Within-Firm Promotions (≥10 log pts.)* | | | |
| FB Access | 0.003*** | 0.001 | 0.00006 |
| | (0.0007) | (0.0008) | (0.0006) |
| Dep. Var. Mean | 0.022 | 0.030 | 0.029 |
| Observations | 37,585 | 45,099 | 52,619 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 6 months. Current firm tenure: years at current employer. Within-firm promotions: indicator for upward title transition within the same firm. Elite colleges are excluded. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table H6: Facebook Access, Firm Sorting, and Human Capital Accumulation – Excluding Elite Colleges

| | 6Y | 12Y | 19Y |
|---|---|---|---|
| *Panel A: Work Experience at High-Wage Firms (Years)* | | | |
| FB Access | 0.082*** | 0.089* | 0.077 |
| | (0.022) | (0.045) | (0.071) |
| Dep. Var. Mean | 1.42 | 3.32 | 5.67 |
| Observations | 7,517 | 7,517 | 7,517 |
| *Panel B: Work Exp. at High-Career-Support Firms (Years)* | | | |
| FB Access | 0.059*** | 0.110*** | 0.152** |
| | (0.016) | (0.032) | (0.048) |
| Dep. Var. Mean | 0.815 | 1.75 | 2.89 |
| Observations | 7,505 | 7,504 | 7,504 |
| *Panel C: Cumulative Work Experience (Years)* | | | |
| FB Access | 0.059*** | 0.065** | 0.057** |
| | (0.017) | (0.020) | (0.021) |
| Dep. Var. Mean | 4.49 | 10.14 | 16.80 |
| Observations | 7,517 | 7,517 | 7,517 |
| College FE | X | X | X |
| Cohort*Tier*Region FE | X | X | X |

*Notes:* *p<0.05; **p<0.01; ***p<0.001. Two-stage DiD using Gardner (2022), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, estimated at 6, 12, and 19 years after expected graduation. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. Cumulative work experience: total years of employment. Elite colleges are excluded. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.